Lie Groups and Their Representation Theory

Noncompact representations, geometry and physics

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Studying this course

This is a self-reading course in two parts of fourteen lectures. The division into lectures is for convenience, so lectures differ in length. Study at your own pace. A lecture may need several passes. The exercise sheet is separate. Allow another two to four hours for its core problems.

The prerequisites are *Linear Algebra for Physics*, *Finite Groups and Their Representations*, *Lie Algebras and Their Representations*, and basic category language. We use their algebraic results throughout. Ordinary multivariable calculus and elementary ordinary differential equations are also assumed.

The course in one page

This course asks what a group determines that its Lie algebra does not. For example, $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ have isomorphic Lie algebras. The module V_1, of spin one half, is a representation of $\mathrm{SU}(2)$ and not of $\mathrm{SO}(3)$. The course also asks how the representations of a group describe the functions on it.

Part I treats compact groups. It can be read alone. Lectures 1 to 4 build Lie groups, the exponential map, quotients and covering groups. Lecture 4 decides which finite-dimensional representations of a Lie algebra come from a given group. Lectures 5 to 9 treat the structure of a compact group: the invariant integral, maximal tori and their lattices, and highest weights. Lecture 8 lists the irreducible representations of a compact connected group. Lectures 10 to 14 treat functions on a compact group. Lecture 10 proves the Peter–Weyl theorem, which contains Fourier series. Lectures 11 and 12 apply it to convolution and to functions on homogeneous spaces. Lecture 13 treats spin groups and projective representations. Lecture 14 recovers the group from its representations.

Part II removes compactness. Lectures 15 to 17 rebuild the tools of Part I: generators of unitary representations, Haar measure of infinite total mass, and induction. Lectures 18 and 19 treat the Heisenberg group and coadjoint orbits. Lectures 20 to 23 treat $\mathrm{SL}(2,\mathbb{R})$ and end with the list of its irreducible unitary representations. Lectures 24 and 25 treat the Lorentz and Poincaré groups. Lectures 26 and 27 return to compact groups and realise their representations on line bundles and on coadjoint orbits. Lecture 28 compares the function algebra of a group with its enveloping algebra, and it prepares for a later course on quantum groups.

Reading a lecture

Start with the objectives and outcomes. Work through the displayed calculations before reading the next paragraph. Then answer the self-checks without looking at their answers.

Attempt the core exercises before opening the worked solutions. Compare the reasoning as well as the final result. Some sheets include an optional extension. It is marked explicitly and is not needed for the next lecture.

The notes introduce the geometry and analysis needed at each stage. They do not replace courses on differential geometry, topology or functional analysis. Long existence theorems are stated with their hypotheses and a source. The main calculations are carried out here.

A cited theorem is marked as cited, by the word *cited* or a source in its title, or by a sentence next to it. Where a proof is only sketched, the text names the omitted step. The mark [C: source] gives the source of a cited statement, and the mark [I] means that a result comes from a prerequisite course. A cited theorem is an input to the argument that follows it. Longer checks that a source's hypotheses apply are gathered under *Sources and hypotheses* at the end of a lecture.

Conventions

Lie groups are finite-dimensional real Lie groups. Their manifolds are Hausdorff and second countable. Compactness and connectedness are stated when they are needed. Representations are continuous and act on complex vector spaces. A Hilbert-space unitary representation is strongly continuous. Inner products are antilinear in the first slot.

A real Lie algebra and its complexification are distinct objects. In particular, $\mathfrak{k} = \mathrm{Lie}(K)$ is real, while highest-weight theory applies to $\mathfrak{k}_{\mathbb{C}}$. For a torus with Lie algebra $\mathfrak{t}$, our period lattice is

$$\Lambda = \ker(\exp\colon \mathfrak{t} \to T).$$

We write a character as $\chi(\exp H) = e^{i\ell(H)}$. Thus its period condition is $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

The module V_m of $\mathrm{SU}(2)$ has dimension $m + 1$ and spin $m/2$. The operator of a group representation is $\Pi(g)$, and its differential is $d\Pi(X)$. When a basis is chosen, D denotes an operator matrix. Every change of convention is explained where it is used.

Books used in the notes

The reading lists give a route into the books. The following abbreviations are used in some lectures.

H Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, Springer, 2015. This is the main route for matrix groups, compact groups and integration of representations.

BD Theodor Bröcker and Tammo tom Dieck, *Representations of Compact Lie Groups*, Springer, 1985. This supplies the general compact-group theory and reconstruction.

S Mark R. Sepanski, *Compact Lie Groups*, Springer, 2007. This is a second route through compact structure and Peter–Weyl theory.

Knapp Beyond
Anthony W. Knapp, *Lie Groups Beyond an Introduction*, second edition. This supplies general Lie-group structure.

Knapp Representation Theory
Anthony W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. This supplies noncompact examples, smooth vectors and induced representations.

Kirillov Jr.
Alexander Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Cambridge University Press, 2008.

Kirillov Elements
A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.

Kirillov Orbit Method
A. A. Kirillov, *Lectures on the Orbit Method*, American Mathematical Society, 2004.

Warner
Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Springer edition, 1983.

Further primary sources are named in the relevant lectures. Knapp's two books serve different purposes; their titles are distinguished throughout.

Part II

Noncompact representations, geometry and physics

Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it. Prove that translations have no nonzero finite-dimensional invariant subspace.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. That theory rests on averaging. The Haar integral of a compact group has finite total mass, so Lecture 5 can average an inner product, and Lecture 10 can average an operator. A noncompact Lie group also has an invariant integral, but its total mass is infinite (Lecture 16). So it has no normalised finite Haar integral, and these averaging arguments are no longer available in general.

The theory changes in the following ways. First, a finite-dimensional representation need not preserve any inner product. For example, for $m \geq 1$ and $t \neq 0$, the matrix $\operatorname{diag}(e^t, e^{-t}) \in \mathrm{SL}(2,\mathbb{R})$ acts on the module V_m with the eigenvalue e^{mt}, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves more: every finite-dimensional unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. So a nontrivial unitary representation of this group is infinite-dimensional.

This is not a property of every noncompact group. The group $\mathbb{R}$ has the one-dimensional unitary characters $x \mapsto e^{icx}$. It shows another change. Its translation representation on $L^2(\mathbb{R})$ is unitary, and it has no nonzero finite-dimensional invariant subspace (Proposition 15.4 below). So a unitary representation need not contain any finite-dimensional piece.

Finally, on an infinite-dimensional space the derivative of a representation may be defined only on a proper subspace. We begin with this point, on translations. Lecture 16 and Lecture 17 then build Haar measure and induction without compactness. We apply these tools to the Heisenberg group, to $\mathrm{SL}(2,\mathbb{R})$, and to the Lorentz and Poincaré groups. Lecture 26 and Lecture 27 return to compact groups, and Lecture 28 returns to Lecture 14.

The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g \rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We want the indicator $1_{[0,1]}$ as a vector of $L^2(\mathbb{R})$, because it will be our example of a vector with no derivative. As in Lecture 10, a general element of a completion has no pointwise values. So we say which functions define vectors. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\, dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and for two such functions f and $\tilde{f}$,

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\, dx, \qquad \langle f, \tilde{f} \rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx.$$

Changing f at finitely many points does not change the vector. § 15.6 constructs this vector and proves these formulas.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t)f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t)f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1)f$ is the derivative of $t \mapsto \lambda(t)f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t)f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.16, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

By the Peter–Weyl theorem (Lecture 10), every unitary representation of a compact group splits into finite-dimensional pieces. The next proposition shows that the unitary representation λ of the noncompact group $\mathbb{R}$ contains no such piece at all.

Proposition 15.4. *The representation λ has no nonzero finite-dimensional invariant subspace.*

Proof. First we show that matrix coefficients decay. Let $f, g \in C_c(\mathbb{R})$. For large $|t|$ the supports of g and $\lambda(t)f$ are disjoint, so $\langle g, \lambda(t)f\rangle = 0$. Now let $f, g \in L^2(\mathbb{R})$, and choose $f_0, g_0 \in C_c(\mathbb{R})$ close to f and g in norm. Since $\lambda(t)$ is unitary,

$$|\langle g, \lambda(t)f\rangle - \langle g_0, \lambda(t)f_0\rangle| \le \|g - g_0\|\,\|f\| + \|g_0\|\,\|f - f_0\|.$$

Hence $\langle g, \lambda(t)f\rangle \to 0$ as $|t| \to \infty$, for all $f, g \in L^2(\mathbb{R})$.

Suppose that $W \neq 0$ is a finite-dimensional invariant subspace, with orthonormal basis $e_1, \dots, e_k$. Let $f \in W$ be a unit vector. Then $\lambda(t)f$ is a unit vector of W, so

$$\sum_{j=1}^{k} |\langle e_j, \lambda(t) f\rangle|^2 = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Each of the finitely many terms tends to 0 as $|t| \to \infty$. This is a contradiction. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

Figure 15.1 shows the difference between a fixed vector and a vector chosen after t. For the fixed step $1_{[0,1]}$ and small $t > 0$, the translate differs from the step only on two slivers of total length $2t$. So $\|\lambda(t)1_{[0,1]} - 1_{[0,1]}\| \to 0$ as $t \to 0$. The comb of Exercise 15.1 depends on t, since its teeth have width t. Translation by t reverses its sign on the overlap, which keeps the ratio $\|\lambda(t)f - f\|/\|f\|$ away from 0 for every t. This does not contradict strong continuity, because strong continuity fixes the vector first.

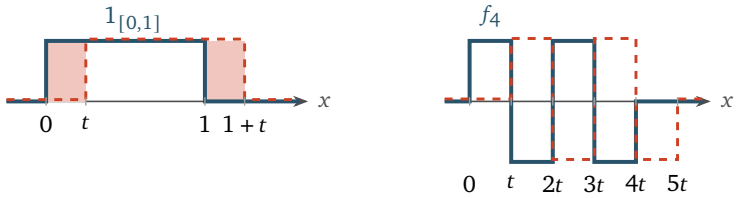


Figure 15.1: Two vectors, heavy, and their translates by t, dashed. The first panel shows the fixed step $1_{[0,1]}$, with the two slivers shaded. The second panel shows the comb f_4 of Exercise 15.1, whose teeth have width t.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.5. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.6. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.
(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.
(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.
(d) *For* $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v\, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v\, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v\, dg.$$

Now h appears only inside the smooth scalar function $(h, g) \mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h \mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n \geq 0$, $\int_G \varphi_n\, dg = 1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v - v\| \leq \sup_{g \in \operatorname{supp}\varphi_n} \|\Pi(g)v - v\| \longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G = \mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.7. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A) \to \mathcal{H}$ defined on a subspace $D(A) \subset \mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B \subset A$ if $D(B) \subset D(A)$ and $Bv = Av$ for $v \in D(B)$.

Definition 15.8. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w \in \mathcal{H}$ for which some $u \in \mathcal{H}$ satisfies

$$\langle w, Av\rangle = \langle u, v\rangle \qquad \text{for all } v \in D(A),$$

and then $A^*w = u$. The operator A is *skew-symmetric* if $\langle Av, w\rangle = -\langle v, Aw\rangle$ for all $v, w \in D(A)$. It is *skew-adjoint* if $A^* = -A$, including $D(A^*) = D(A)$, and *self-adjoint* if $A^* = A$, including $D(A^*) = D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A \subset A^*$.

Suppose $B \subset A$ and B is densely defined. Then $D(A) \supset D(B)$ is dense too, so both adjoints exist. In this case $A^* \subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v \in D(B)$.

Lemma 15.9. *Let B be densely defined and skew-symmetric. If $v_n \in D(B)$, $v_n \to 0$ and $Bv_n \to u$, then $u = 0$.*

Proof. For every $w \in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w, u\rangle = \lim_n \langle w, Bv_n\rangle = -\lim_n \langle Bw, v_n\rangle = 0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u = 0$. ∎

Definition 15.10. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar{B}$ is the set of $v \in \mathcal{H}$ for which there are $v_n \in D(B)$ with $v_n \to v$ and (Bv_n) convergent. Then $\bar{B}v = \lim_n Bv_n$.

The value $\bar{B}v$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.9.

Definition 15.11. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t \to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.12 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) *Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.*

(c) *Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.*

We cite most of this theorem and prove only the invariance statement in (a). The sources are Hall, *Quantum Theory for Mathematicians*, Section 10.2, and Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4. The section on sources and hypotheses at the end of this lecture matches their conventions with ours. For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$.

Corollary 15.13. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.6, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.12(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.12(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

The generator B_X can have a larger domain than $\mathcal{H}^\infty$. That domain can depend on X, and B_X need not preserve it. For example, Exercise 15.2 gives a vector g in the domain of the generator of translations whose image is outside that domain. Brackets $[d\Pi(X), d\Pi(Y)]$ need one subspace on which every $d\Pi(X)$ is defined and which every $d\Pi(X)$ preserves. By Proposition 15.6, the space $\mathcal{H}^\infty$ has both properties.

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, the generator B_1 is the operator that the next section calls A. The subscript matters: the physical generator $A_1 = iA$ is self-adjoint, while A is skew-adjoint. With $\hbar = 1$, the operator A_1 is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R} \to \mathbb{C}$ such that

$$\sup_{x \in \mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j, k \geq 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.14. *Let $f \in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)| \le C(1+y^2)^{-1}$ for all y. Then $f \in D(A)$ and $Af = -f'$.*

Proof. Fix $0 < |h| \le 1$. Taylor's formula with integral remainder, for $s \mapsto f(x - sh)$ on $[0,1]$, gives

$$f(x-h) = f(x) - hf'(x) + h^2 \int_0^1 (1-s) f''(x-sh)\, ds.$$

So the error function is

$$q_h(x) = \frac{f(x-h) - f(x)}{h} + f'(x) = h \int_0^1 (1-s) f''(x-sh)\, ds,$$

and $|q_h(x)| \le \frac{|h|}{2} \sup_{|y-x|\le 1} |f''(y)|$. If $|y - x| \le 1$, then $x^2 \le (|y|+1)^2 \le 2y^2 + 2$, so $1 + x^2 \le 3(1+y^2)$. Therefore

$$|q_h(x)| \le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2 \le \frac{9C^2h^2}{4} \int_{\mathbb{R}} \frac{dx}{(1+x^2)^2} = \frac{9\pi C^2 h^2}{8}.$$

Thus $\left\| \frac{\lambda(h)f - f}{h} - (-f') \right\| = \|q_h\| \to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)| \le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.15. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f \in \mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f = -f'$. The closure of $B = -d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y = x - t$. Expanding $x^k = (y+t)^k$ bounds $|x^k f^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^i f^{(j)}(y)|$.

Smoothness. Each $f \in \mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.14. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''| \le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})$. By the lemma, $t \mapsto \lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f - \lambda(t_0)f}{h} = \lambda(t_0)\, \frac{\lambda(h)f - f}{h} \longrightarrow \lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t \mapsto \lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f' \in \mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t \mapsto \lambda(t)\big((-1)^k f^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \ge 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R,R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \le \sup_{|t|\le 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.12(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise. By analogy with the compact groups of Lecture 11, it is the convolution $\varphi_n * f$.

Example 15.16. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2-2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \le 6(1+x^2)$, we get $|w''(x)| \le 6(1+x^2)^{-2} \le 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.14, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.15. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.6(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \le 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \le t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.13, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. All derivatives of w satisfy the hypotheses of the Taylor-remainder lemma, Lemma 15.14, so w is a smooth vector that is not in $\mathcal{S}(\mathbb{R})$. So both inclusions are strict. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

Remark 15.17. We cite Plancherel's theorem (Folland, *Real Analysis*, second edition, Chapter 8). The Fourier transform $\hat{f}(p) = (2\pi)^{-1/2} \int_{\mathbb{R}} f(x) e^{-ipx}\, dx$ extends to a unitary operator of $L^2(\mathbb{R})$, and it turns $\lambda(t)$ into multiplication by e^{-ipt}. As in Exercise 15.3, $D(A)$ is the set of f with $\int_{\mathbb{R}} p^2 |\hat{f}(p)|^2\, dp < \infty$, and P acts as multiplication by p. In quantum mechanics, $D(A)$ is the set of states for which P^2 has a finite expectation value.

This describes the three vectors above. The step $h = 1_{[0,1]}$ has $|\hat{h}(p)|^2 = 2\sin^2(p/2)/(\pi p^2)$, so $\int p^2 |\hat{h}|^2\, dp$ diverges. The tent g of Exercise 15.2 has $|\hat{g}(p)|^2 = 8\sin^4(p/2)/(\pi p^4)$, so $\int p^2 |\hat{g}|^2\, dp$ converges, while $\int p^4 |\hat{g}|^2\, dp$ diverges. The function w has $\hat{w}(p) = \sqrt{\pi/2}\, e^{-|p|}$, so all these integrals converge. Finally, $\|\lambda(t) - I\| = \sup_p |e^{-ipt} - 1| = 2$ for $t \neq 0$, which agrees with Exercise 15.1.

15.6 Piecewise continuous functions as vectors

We now construct the vector of a square integrable function, as stated in § 15.1. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Then f has finitely many jumps $x_1 < \cdots < x_J$, and the one-sided limits at these jumps make f bounded, say $|f| \le M$. Choose $\delta > 0$ so small that the intervals $[x_j - \delta, x_j + \delta]$ are disjoint. Let g_δ equal f outside these intervals, and let g_δ be linear on each of them, with the values of f at the two endpoints. Then $g_\delta \in C_c(\mathbb{R})$ and $|g_\delta| \le M$. So $|f - g_\delta| \le 2M$, and $f - g_\delta$ vanishes outside J intervals of length 2δ. Hence

$$\int_{\mathbb{R}} |f - g_\delta|^2\, dx \le 8JM^2\delta.$$

For piecewise continuous functions u and $\tilde{u}$ that vanish outside a bounded interval, write $\|u\|_2 = \big(\int_{\mathbb{R}} |u|^2\, dx\big)^{1/2}$. On $C_c(\mathbb{R})$ this is the norm of $L^2(\mathbb{R})$. The Cauchy–Schwarz inequality for Riemann integrals, $\int_{\mathbb{R}} |\bar{u}\tilde{u}|\, dx \le \|u\|_2 \|\tilde{u}\|_2$, gives the triangle inequality for $\|\cdot\|_2$.

Take $\delta_n \to 0$ and $g_n = g_{\delta_n}$, so that $\|f - g_n\|_2 \to 0$. Since $\|g_n - g_m\|_2 \le \|g_n - f\|_2 + \|f - g_m\|_2$, the sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. By the same inequality, every sequence in $C_c(\mathbb{R})$ that approximates f in $\|\cdot\|_2$ has this limit. Sums and scalar multiples of approximating sequences approximate sums and scalar multiples, so the vector depends linearly on f. By the triangle inequality, the vector f has norm $\|f\| = \lim_n \|g_n\|_2 = \|f\|_2$.

Let $\tilde{f}$ be a second such function, approximated by $\tilde{g}_n \in C_c(\mathbb{R})$. By continuity of the inner product, $\langle f, \tilde{f}\rangle = \lim_n \int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx$. Also

$$\int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx - \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx = \int_{\mathbb{R}} \overline{(g_n - f)}\, \tilde{g}_n\, dx + \int_{\mathbb{R}} \bar{f}\, (\tilde{g}_n - \tilde{f})\, dx.$$

By the Cauchy–Schwarz inequality, the right side has absolute value at most $\|g_n - f\|_2 \|\tilde{g}_n\|_2 + \|f\|_2 \|\tilde{g}_n - \tilde{f}\|_2$. This tends to 0, because $\|\tilde{g}_n\|_2$ stays bounded. So $\langle f, \tilde{f}\rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx$.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot$

$1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f\cdot 1_{[-m,m]} - f\cdot 1_{[-n,n]}\right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m>n\to\infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n\to\infty$ in the formulas for the truncations gives the two formulas of § 15.1. The second improper integral converges absolutely, since $\int_{-n}^{n}|\bar f\tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

15.7 Self-checks

1. Let $\dim\mathcal{H}<\infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G\to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x)=xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f\in\mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.15, gives $Af=-f'=(x^2-1)e^{-x^2/2}$. Then $\langle f,Af\rangle=\int_{\mathbb{R}} x(x^2-1)e^{-x^2}\,dx=0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f,Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0,\infty)$. Then $D\subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.12(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]}\neq 0$. It is not invariant either. Given $t<0$, choose a nonzero $f\in D$ supported in $(0,|t|)$. Then $\lambda(t)f$ is supported in $(t,0)$, so $\lambda(t)f\notin D$.
4. Let $D=C_c^\infty(\mathbb{R}\setminus\{0\})$. This space is dense in $L^2(\mathbb{R})$ and contained in $D(A)$, but it is not invariant. Is the closure of $A|_D$ equal to A?
   Answer. No. Let g be a C^1 function with g and g' square integrable and $g(x)\to 0$ as $x\to-\infty$. Then $|g(0)|^2=\int_{-\infty}^{0}(|g|^2)'\,dx\le 2\|g\|\,\|g'\|$. Suppose that $v_n\in D$ with $v_n\to w$ and $Av_n\to Aw$, where $w=(1+x^2)^{-1}$. Take $g=w-v_n$. Since $v_n(0)=0$ and $w(0)=1$, we get $1\le 2\|w-v_n\|\,\|w'-v_n'\|\to 0$, which is false. So $w\in D(A)$ is not in the domain of the closure. The invariance hypothesis in the criterion cannot be dropped.

15.8 Summary

Unitary representations are required to be strongly continuous, not norm continuous. The translation representation on $L^2(\mathbb{R})$ has no nonzero finite-dimensional invariant subspace. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

Sources and hypotheses

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\mathrm{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\mathrm{Hall}} = -iA$. In our physical convention $H = iA$, so $H_{\mathrm{Hall}} = -H$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in Theorem 15.12(b).

15.9 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
- Etingof, MIT 18.757 lecture notes, Lectures 1–7: an alternative introduction to continuous representations of Lie groups.

Lecture 16

Noncompact Haar Measure and the Modular Function

Objectives. *Lecture 5 integrated over compact groups. This lecture drops compactness. It asks which translation-invariant integrals a Lie group carries and how left and right integration differ. A noncompact Lie group has infinite Haar volume, so the normalised averaging of Lecture 5 is no longer available in general. Every Lie group has a left-invariant and a right-invariant density. With equal normalisations at the identity, the left density divided by the right density is the modular function. It controls right translation, inversion and the right regular representation.*

By the end of this lecture you will be able to:

- Construct the left and right Haar densities of a Lie group, prove that each is unique up to a positive factor, and prove that the total mass is infinite when the group is noncompact.
- Prove the right-translation and inversion identities for $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.
- Compute both Haar densities and Δ for the affine group $ax + b$.
- Construct the left and right regular representations as strongly continuous unitary representations on $L^2(G)$, and state the integration rules of Part II with their hypotheses.

Lecture 5 built a left-invariant density on a compact Lie group by translating a density at e. That construction did not use compactness. Compactness entered later. It made the integral finite on every continuous function, and it allowed normalisation to total mass 1. It made the image of $j(h) = |\det \mathrm{Ad}(h)|$ a compact subgroup of $(0, \infty)$, so $j \equiv 1$. It gave strict positivity through a finite cover. Without compactness we integrate only compactly supported functions. Strict positivity comes from charts, and j survives as a homomorphism. The left Haar density μ_L below is the density δ of Lecture 5, and the modular function below is $\Delta = j^{-1}$.

Throughout, G is a Lie group of dimension n with Lie algebra $\mathfrak{g} = T_eG$. We write left translation, right translation, conjugation and inversion as

$$L_g(x) = gx, \qquad R_g(x) = xg, \qquad C_g = L_g \circ R_{g^{-1}}, \qquad \iota(x) = x^{-1}.$$

Then $R_g \circ R_h = R_{hg}$ and $d(C_g)_e = \mathrm{Ad}(g)$.

16.1 Densities and integration

Under a linear map T of $\mathbb{R}^n$, Lebesgue measure scales by $|\det T|$. A density imposes this rule on each tangent space.

Definition 16.1. Let V be a real vector space of dimension n. A *positive density* on V is a nonzero function $\delta\colon V^n \to [0,\infty)$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \qquad (T \in \operatorname{End} V).$$

Let $e_1,\dots,e_n$ be a basis and $v_j = Te_j$. Then $\delta(v_1,\dots,v_n) = |\det T|\,\delta(e_1,\dots,e_n)$. So δ is fixed by the number $\delta(e_1,\dots,e_n)$, which is positive since $\delta \neq 0$. Two positive densities on V therefore differ by a positive factor.

Definition 16.2. A *smooth positive density* μ on G assigns a positive density μ_x on T_xG to each $x \in G$ such that, in every chart with coordinates y, the coefficient $m(y) = \mu_x(\partial_1,\dots,\partial_n)$ is smooth.

For a diffeomorphism F of G and a smooth function $\varphi > 0$ put

$$(F^*\mu)_x(v_1,\dots,v_n) = \mu_{F(x)}(dF_x v_1,\dots,dF_x v_n), \qquad (\varphi\mu)_x = \varphi(x)\,\mu_x.$$

The chain rule gives $(F\circ H)^* = H^*F^*$. The definitions give $F^*(\varphi\mu) = (\varphi\circ F)\,F^*\mu$.

Let $C_c(G)$ be the space of continuous functions $f\colon G \to \mathbb{C}$ with compact support.

If $\operatorname{supp} f$ lies in one chart, set $\int_G f\mu = \int_{\mathbb{R}^n} f(y)\,m(y)\,dy$. A change of chart multiplies the coefficient m by the absolute Jacobian determinant. Hence, by the change of variables formula, the integral is well defined.

For general $f \in C_c(G)$ we use a partition of unity, adapted from Lecture 5. Here a *partition of unity* on $\operatorname{supp} f$ is a finite family of smooth functions $\psi_i \geq 0$ on G, each with compact support in one chart, such that $\sum_i \psi_i = 1$ on $\operatorname{supp} f$. On a compact manifold, Lecture 5 requires the sum to equal 1 everywhere. Since f vanishes outside $\operatorname{supp} f$, we have $f = \sum_i \psi_i f$. We add the chart integrals of the functions $\psi_i f$.

We cite this construction and the following three facts. Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII, carries out the same construction with differential forms.

(I1) $f \mapsto \int_G f\mu$ is linear, and $\int_G f\mu > 0$ if $f \geq 0$ and $f \neq 0$.
(I2) $\int_G (f\circ F)\,F^*\mu = \int_G f\mu$ for every diffeomorphism F of G.
(I3) $\int_G f\,(\varphi\mu) = \int_G (f\varphi)\,\mu$.

In a chart, $F^*\mu$ has coefficient $m(F(y))\,|\det DF(y)|$. So (I2) is the change of variables formula.

16.2 Left and right Haar densities

Theorem 16.3 (Haar densities). *Fix a positive density μ_e on $\mathfrak{g}$.*

(a) There is exactly one smooth positive density μ_L with $(\mu_L)_e = \mu_e$ and $L_g^*\mu_L = \mu_L$ for all $g \in G$. Every left-invariant smooth positive density is a positive multiple of μ_L.

(b) There is exactly one smooth positive density μ_R with $(\mu_R)_e = \mu_e$ and $R_g^*\mu_R = \mu_R$ for all $g \in G$. Every right-invariant smooth positive density is a positive multiple of μ_R.

(c) Write $\int_G f(x)\,d_Lx$ for $\int_G f\mu_L$, and similarly d_Rx. For $f \in C_c(G)$ and $g \in G$,

$$\int_G f(gx)\,d_Lx = \int_G f(x)\,d_Lx, \qquad \int_G f(xg)\,d_Rx = \int_G f(x)\,d_Rx.$$

Proof. Uniqueness. Let ν be a left-invariant smooth positive density. Evaluate $L_{x^{-1}}^*\nu = \nu$ at x:

$$\nu_x(v_1,\dots,v_n) = \nu_e\big(d(L_{x^{-1}})_x v_1,\dots,d(L_{x^{-1}})_x v_n\big). \tag{16.1}$$

So ν is determined by ν_e. Two positive densities on $\mathfrak{g}$ differ by a positive factor. Thus $\nu_e = c\mu_e$ for some $c > 0$. Once μ_L is constructed below, the formula gives $\nu = c\mu_L$.

Existence. Define μ_L by (16.1) with $\nu_e = \mu_e$. Linear maps act entrywise on tuples $v = (v_1,\dots,v_n)$. Since $L_{(gx)^{-1}} \circ L_g = L_{x^{-1}}$, the chain rule gives

$$(L_g^*\mu_L)_x(v) = \mu_e\big(d(L_{(gx)^{-1}})_{gx}\,d(L_g)_x v\big) = \mu_e\big(d(L_{x^{-1}})_x v\big) = (\mu_L)_x(v).$$

Smoothness. This step uses the smooth group law. Choose a basis $X_1,\dots,X_n$ of $\mathfrak{g}$. The left-invariant fields $\tilde{X}_i(x) = d(L_x)_e X_i$ are smooth (Lecture 1). Since $L_{x^{-1}} \circ L_x = \mathrm{id}$, we have $d(L_{x^{-1}})_x \tilde{X}_i(x) = X_i$. Hence

$$(\mu_L)_x\big(\tilde{X}_1(x),\dots,\tilde{X}_n(x)\big) = \mu_e(X_1,\dots,X_n) =: c_0 > 0.$$

In a chart write $\tilde{X}_i = \sum_j P_{ji}(y)\,\partial_j$. The matrix $P(y)$ is smooth. It is invertible because $d(L_x)_e$ is an isomorphism. The density rule gives $c_0 = |\det P(y)|\,m(y)$. So $m = c_0\,|\det P|^{-1}$ is smooth.

Right translations. Use $\nu_x(v) = \nu_e(d(R_{x^{-1}})_x v)$. The key identity is $R_{(xg)^{-1}} \circ R_g = R_{x^{-1}}$, since $z\,g\,g^{-1}x^{-1} = zx^{-1}$. The fields $d(R_x)_e X_i$ replace $\tilde{X}_i$.

Integrals. Apply the change of variables (I2) with $F = L_g$ or $F = R_g$. ∎

On a compact group one normalises the total mass to 1, as in Lecture 5. For noncompact G this is impossible, as we now prove. Since the constant function 1 is not in $C_c(G)$, we state the result for compactly supported functions.

Proposition 16.4 (Infinite Haar volume). *Let G be a noncompact Lie group, and let $f \in C_c(G)$ with $f \geq 0$ and $f \neq 0$. There are elements $g_1, g_2, \dots$ of G such that the translates $f_k(x) = f(g_k^{-1}x)$ have pairwise disjoint supports. Hence $F_N = f_1 + \cdots + f_N$ lies in $C_c(G)$, it satisfies $0 \leq F_N \leq \sup f$, and*

$$\int_G F_N\,d_Lx = N\int_G f\,d_Lx, \qquad \int_G f\,d_Lx > 0.$$

Proof. We choose the elements one at a time. Let $S = \operatorname{supp} f$, a compact set, and take $g_1 = e$. Suppose $g_1, \dots, g_{m-1}$ are chosen. The set $\bigcup_{k<m} g_k S S^{-1}$ is compact, as a finite union of continuous images of $S \times S$. Since G is not compact, there is an element g_m outside it.

The support of f_k is $g_k S$. If $g_m S$ met $g_k S$ with $k < m$, then $g_m s = g_k s'$ for some $s, s' \in S$. That gives $g_m = g_k s' s^{-1} \in g_k S S^{-1}$, which contradicts the choice of g_m. So the supports are pairwise disjoint.

At each point at most one term of F_N is nonzero, so $0 \le F_N \le \sup f$. Left invariance in Theorem 16.3(c) gives $\int_G f_k \, d_L x = \int_G f \, d_L x$, and this value is positive by the positivity statement (I1). ∎

So the left Haar integral is unbounded on functions in $C_c(G)$ with values in $[0, \sup f]$, and the total mass of G is infinite. The right Haar integral behaves in the same way, with right translates. A noncompact Lie group therefore has no normalised finite Haar integral, and the averaging arguments of Lecture 5 are no longer available in general.

We use the same density μ_e for both μ_L and μ_R.

16.3 The modular function

Definition 16.5. The *modular function* of G is $\Delta(g) = |\det \operatorname{Ad}(g)|^{-1}$. The group G is *unimodular* if $\Delta \equiv 1$.

Since $\operatorname{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism (Lecture 2), Δ is a smooth homomorphism into the multiplicative group $(0, \infty)$.

Theorem 16.6 (Integration identities)**.** *Let $f \in C_c(G)$ and $g \in G$.*

(i) $R_g^* \mu_L = \Delta(g)\, \mu_L$. *Hence* $\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x$.

(ii) $\mu_R = \Delta^{-1} \mu_L$, *that is,* $\int_G f(x)\, d_R x = \int_G f(x)\, \Delta(x)^{-1}\, d_L x$.

(iii) $\int_G f(x^{-1})\, d_L x = \int_G f(x)\, d_R x$ *and* $\int_G f(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G f(x)\, d_L x$.

Proof. (i) Left and right translations commute. Hence

$$L_h^* R_g^* \mu_L = (R_g \circ L_h)^* \mu_L = (L_h \circ R_g)^* \mu_L = R_g^* L_h^* \mu_L = R_g^* \mu_L.$$

So $R_g^* \mu_L$ is left-invariant. By the uniqueness in Theorem 16.3(a), it equals $c(g) \mu_L$ for some $c(g) > 0$. We find $c(g)$ at e. Since $L_{g^{-1}} \circ R_g = C_{g^{-1}}$,

$$(R_g^* \mu_L)_e(v) = \mu_e\big(d(L_{g^{-1}})_g \, d(R_g)_e v\big) = \mu_e\big(\operatorname{Ad}(g^{-1}) v\big) = |\det \operatorname{Ad}(g)|^{-1} \mu_e(v).$$

Thus $c(g) = \Delta(g)$. Now the change of variables (I2) with $F = R_g$ and the multiplication rule (I3) give

$$\int_G f \mu_L = \int_G (f \circ R_g)\, R_g^* \mu_L = \Delta(g) \int_G f(xg)\, d_L x.$$

(ii) Use $F^*(\varphi \mu) = (\varphi \circ F) F^* \mu$, the right-translation identity (i) and the homomorphism property of Δ:

$$\big(R_g^* (\Delta^{-1} \mu_L)\big)_x = \Delta(xg)^{-1} \Delta(g)\, (\mu_L)_x = \Delta(x)^{-1} (\mu_L)_x.$$

So $\Delta^{-1}\mu_L$ is right-invariant with value μ_e at e. By uniqueness in Theorem 16.3(b) it equals μ_R. Then the multiplication rule (I3) gives the integral form.

(iii) First, $d\iota_e = -\mathrm{id}$. Indeed, the multiplication map $\mathrm{mult}\colon G\times G\to G$ satisfies $\mathrm{mult}(x,e)=x$ and $\mathrm{mult}(e,x)=x$ for $x\in G$. So $d(\mathrm{mult})_{(e,e)}(v,w)=v+w$ for $v,w\in\mathfrak{g}$. Differentiating $x\,\iota(x)=e$ at e gives $v+d\iota_e v=0$. Therefore

$$(\iota^*\mu_L)_e(v_1,\dots,v_n)=\mu_e(-v_1,\dots,-v_n)=|\det(-\mathrm{id})|\,\mu_e(v_1,\dots,v_n)=\mu_e(v_1,\dots,v_n).$$

Next, $(xg)^{-1}=g^{-1}x^{-1}$ says $\iota\circ R_g=L_{g^{-1}}\circ\iota$. Hence

$$R_g^*\iota^*\mu_L=(\iota\circ R_g)^*\mu_L=(L_{g^{-1}}\circ\iota)^*\mu_L=\iota^*L_{g^{-1}}^*\mu_L=\iota^*\mu_L.$$

By the uniqueness in Theorem 16.3(b), $\iota^*\mu_L=\mu_R$. Apply the change of variables (I2) with $F=\iota$ to $f\circ\iota$. This gives $\int_G f\,d_Rx=\int_G f(x^{-1})\,d_Lx$. For the second identity apply the first to $h=f\Delta\in C_c(G)$ and use the density relation (ii):

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\int_G h(x)\,d_Rx=\int_G f(x)\Delta(x)\Delta(x)^{-1}\,d_Lx.$$ ∎

Remark 16.7. The right-translation identity (i) is equivalent to $\int_G f(xg^{-1})\,d_Lx=\Delta(g)\int_G f\,d_Lx$. For Borel sets E, using the Radon measure of § 16.6, it reads $\mu_L(Eg)=\Delta(g)\mu_L(E)$. This is Folland's convention. Some texts call $|\det \mathrm{Ad}(g)|$ the modular function. When comparing sources, check which defining identity is used.

Corollary 16.8. *(a) Every compact Lie group is unimodular.*

(b) If G is connected, then $\Delta(\exp X)=e^{-\operatorname{tr}\operatorname{ad}X}$. So G is unimodular if and only if $\operatorname{tr}\operatorname{ad}X=0$ for all $X\in\mathfrak{g}$. This holds when $\mathfrak{g}$ is abelian, nilpotent or semisimple.

Proof. (a) Since G is compact and Δ is continuous, the image $\Delta(G)$ is a compact subgroup of $(0,\infty)$. If $t\neq 1$ lies in it, then so does the unbounded set $\{t^k:k\in\mathbb{Z}\}$. So $\Delta(G)=\{1\}$.

(b) We have $\mathrm{Ad}(\exp X)=e^{\operatorname{ad}X}$ (Lecture 2) and $\det e^{A}=e^{\operatorname{tr}A}$. This gives the formula. If all traces vanish, then $\Delta=1$ on $\exp\mathfrak{g}$. This set contains a neighbourhood U of e (Lecture 2). The subgroup H generated by U is open, since $hU\subset H$ for $h\in H$. It is closed, since its complement is a union of open cosets. By connectedness $H=G$, so $\Delta\equiv 1$. Conversely, $\Delta\equiv 1$ gives $e^{-t\operatorname{tr}\operatorname{ad}X}=1$ for all t, so $\operatorname{tr}\operatorname{ad}X=0$. For nilpotent $\mathfrak{g}$ each $\operatorname{ad}X$ is nilpotent. For semisimple $\mathfrak{g}$ we have $\mathfrak{g}=[\mathfrak{g},\mathfrak{g}]$ and $\operatorname{tr}\operatorname{ad}[X,Y]=\operatorname{tr}[\operatorname{ad}X,\operatorname{ad}Y]=0$. ∎

16.4 The affine group

The *$ax+b$ group* is

$$G=\left\{g(a,b)=\begin{pmatrix}a & b\\ 0 & 1\end{pmatrix}:\ a>0,\ b\in\mathbb{R}\right\}.$$

It acts on $\mathbb{R}$ by $x\mapsto ax+b$. The pair (a,b) is a global chart onto $(0,\infty)\times\mathbb{R}$. So integrals over G are ordinary double integrals. Matrix multiplication gives

$$g(a,b)\,g(a',b')=g(aa',\ ab'+b),\qquad g(a,b)^{-1}=g(a^{-1},-a^{-1}b).$$

A tangent vector at g is a matrix $\begin{pmatrix}\alpha & \beta\\ 0 & 0\end{pmatrix}$, which is $\alpha\partial_a+\beta\partial_b$ in the chart. The Lie algebra has basis

$$X=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix},\qquad Y=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix},\qquad [X,Y]=Y.$$

Translations are restrictions of linear maps on matrices. Hence $d(L_g)_eZ=gZ$ and $d(R_g)_eZ=Zg$. We fix the density μ_e on $\mathfrak{g}$ by $\mu_e(X,Y)=1$.

Left density. For $g=g(a,b)$,

$$gX=\begin{pmatrix}a & 0\\ 0 & 0\end{pmatrix}=a\,\partial_a,\qquad gY=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=a\,\partial_b.$$

So the matrix P from the smoothness step in the proof of Theorem 16.3 is

$$P=\begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}.$$

Then $m_L=|\det P|^{-1}=a^{-2}$.

Right density. Similarly,

$$Xg=\begin{pmatrix}a & b\\ 0 & 0\end{pmatrix}=a\,\partial_a+b\,\partial_b,\qquad Yg=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}=\partial_b,\qquad P=\begin{pmatrix}a & 0\\ b & 1\end{pmatrix}.$$

Then $m_R=a^{-1}$. In summary,

$$d_Lx=a^{-2}\,da\,db,\qquad d_Rx=a^{-1}\,da\,db.$$

Modular function. With $g^{-1}=\begin{pmatrix}a^{-1} & -a^{-1}b\\ 0 & 1\end{pmatrix}$ we compute

$$gXg^{-1}=\begin{pmatrix}1 & -b\\ 0 & 0\end{pmatrix}=X-bY,\qquad gYg^{-1}=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=aY.$$

In the basis (X,Y),

$$\mathrm{Ad}(g)=\begin{pmatrix}1 & 0\\ -b & a\end{pmatrix},\qquad \det\mathrm{Ad}(g)=a,\qquad \Delta(g(a,b))=a^{-1}.$$

This matches the density relation in Theorem 16.6(ii): $\Delta^{-1}m_L=a\cdot a^{-2}=a^{-1}=m_R$. It also matches the trace formula in Corollary 16.8(b). Indeed $\operatorname{ad}X$ fixes Y and kills X, so $\operatorname{tr}\operatorname{ad}X=1$. And $\exp tX=g(e^t,0)$ has $\Delta=e^{-t}$. Hence the $ax+b$ group is not unimodular.

The connectedness hypothesis in Corollary 16.8(b) is needed. Let H be the subgroup of elements $g(a,b)$ with $a\in 2^{\mathbb{Z}}$ and $b\in\mathbb{R}$. It is closed, since $2^{\mathbb{Z}}$ is closed in $(0,\infty)$. Its Lie algebra $\mathbb{R}Y$ is abelian. Yet $\mathrm{Ad}(g)Y=gYg^{-1}=aY$ for $g=g(a,b)\in H$. So the modular function Δ_H of H satisfies $\Delta_H(g(a,b))=a^{-1}$, which is not identically 1.

Direct check of the right-translation identity in Theorem 16.6(i). Let $x=g(a',b')$ and $g=g(a,b)$. Then $xg=g(a'a,\ a'b+b')$. Substitute $u=a'a$, $v=a'b+b'$. Then $du\,dv=a\,da'\,db'$ and $a'^{-2}=a^2u^{-2}$. So

$$\int_G f(xg)\,d_Lx=\iint f(a'a,\ a'b+b')\,\frac{da'\,db'}{a'^2}=\iint f(u,v)\,\frac{a^2}{u^2}\,\frac{du\,dv}{a}=a\int_G f\,d_Lx.$$

This is $\Delta(g)^{-1}\int_G f\,d_Lx$.

Direct check of the second inversion identity in Theorem 16.6(iii). Here $x^{-1}=g(1/a,-b/a)$ and $\Delta(x^{-1})=a$. Substitute $u=1/a$, $v=-b/a$. The Jacobian determinant of $(a,b)\mapsto(u,v)$ is

$$\det\begin{pmatrix}-a^{-2} & 0\\ ba^{-2} & -a^{-1}\end{pmatrix}=a^{-3},\qquad\text{so}\qquad da\,db=u^{-3}\,du\,dv.$$

Since $a\cdot a^{-2}=a^{-1}=u$,

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\iint f\big(\tfrac{1}{a},-\tfrac{b}{a}\big)\,\frac{a\,da\,db}{a^2}=\iint f(u,v)\,u\,u^{-3}\,du\,dv=\int_G f\,d_Lx.$$

Translating a rectangle

In the $ax+b$ group the modular function can be computed from weighted areas, without differentiating a matrix. For a region E in the (a,b)-chart we write

$$\mu_L(E)=\iint_E a^{-2}\,da\,db$$

and call it the *left Haar measure* of E. Now let

$$E=\{(a,b):1\le a\le 2,\ 0\le b\le 1\}.$$

Its left Haar measure is

$$\mu_L(E)=\int_1^2\int_0^1 a^{-2}\,db\,da=\big[-a^{-1}\big]_1^2=\frac{1}{2}.$$

For $h=g(s,t)$, right translation sends (a,b) to $(as,b+at)$. Its Jacobian in (a,b) is s, while the new density coefficient is $(as)^{-2}$. Hence

$$\mu_L(Eh)=\int_E (as)^{-2}s\,da\,db=s^{-1}\mu_L(E)=\Delta(h)\mu_L(E).$$

The shear parameter t changes the shape of the region but not this scale factor. Left translation by h has Jacobian s^2 and density coefficient $(sa)^{-2}$, so it preserves $\mu_L(E)$.

An integrand $f(xh)$ involves a third region. The function $x\mapsto \mathbf{1}_E(xh)$ equals 1 exactly when $xh\in E$, so it is the indicator of Eh^{-1}. Figure 16.1 draws the three regions for $h=g(2,1)$. Here $hE=[2,4]\times[1,3]$, and Eh is the parallelogram with vertices $(2,1)$, $(4,2)$, $(4,3)$ and $(2,2)$. The region Eh^{-1} is the set of (a,b) with $\tfrac12\le a\le 1$ and $-a\le b\le 1-a$.

The weighted masses are

$$\mu_L(hE)=\tfrac12=\mu_L(E),\qquad \mu_L(Eh)=\tfrac14=\Delta(h)\mu_L(E),\qquad \mu_L(Eh^{-1})=1=\Delta(h)^{-1}\mu_L(E).$$

The inverse in the region Eh^{-1} explains why the integral identity in Theorem 16.6(i) has $\Delta(h)^{-1}$, whereas the region identity has $\Delta(h)$.

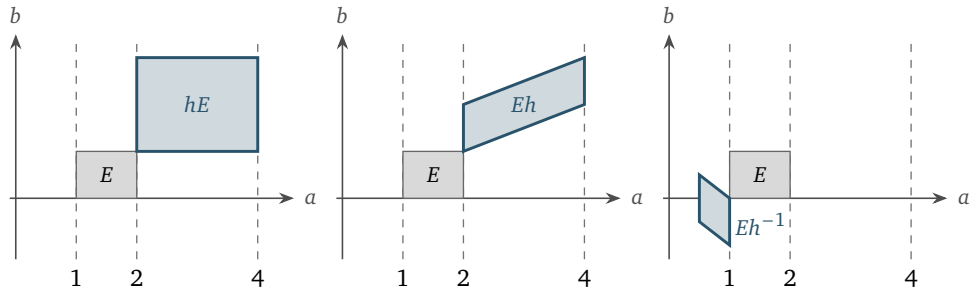


Figure 16.1: The rectangle E in the (a,b) half-plane, in grey, with the left translate hE, the right translate Eh and the region Eh^{-1} for $h = g(2,1)$. The measure of a region is the integral of $a^{-2}\,da\,db$, not its Euclidean area.

16.5 Regular representations

Definition 16.9. The space $L^2(G)$ is the Hilbert completion of $C_c(G)$ for the inner product

$$\langle f, h\rangle = \int_G \overline{f(x)}\, h(x)\, d_L x.$$

By the positivity statement (I1), this inner product is positive definite. For $f \in C_c(G)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

Lemma 16.10. *Let $f \in C_c(G)$ and $\varepsilon > 0$. There is a neighbourhood $V = V^{-1}$ of e with $|f(yx) - f(x)| < \varepsilon$ and $|f(xy) - f(x)| < \varepsilon$ for all $x \in G$ and $y \in V$.*

Proof. We treat $f(yx)$. The other case is symmetric. If U_L works for $f(yx)$ and U_R for $f(xy)$, put $V' = U_L \cap U_R$ and $V = V' \cap V'^{-1}$.

Fix a compact neighbourhood W of e. Choose a decreasing neighbourhood basis $V_1 \supset V_2 \supset \cdots$ of e inside W. Suppose no V_k works. Then there are $y_k \in V_k$ and x_k with $|f(y_k x_k) - f(x_k)| \geq \varepsilon$. One of x_k and $y_k x_k$ lies in $K = \operatorname{supp} f$. So x_k lies in the compact set $K \cup W^{-1}K$. Since G is second countable and Hausdorff, compact subsets are sequentially compact. So a subsequence has $x_k \to x$. As $y_k \to e$, also $y_k x_k \to x$. Continuity gives $f(y_k x_k) - f(x_k) \to 0$, which is a contradiction. ∎

Proposition 16.11. *The maps λ and ρ extend to strongly continuous unitary representations of G on $L^2(G)$, and $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.*

Proof. Both are homomorphisms on $C_c(G)$. For example,

$$\rho(g)\rho(h)f(x) = \Delta(g)^{1/2}\Delta(h)^{1/2} f(xgh) = \rho(gh)f(x).$$

Both operators preserve norms. By the left invariance in Theorem 16.3(c) and the right-translation identity in Theorem 16.6(i), applied to $|f|^2$,

$$\|\lambda(g)f\|^2 = \int_G |f(g^{-1}x)|^2\, d_L x = \|f\|^2, \qquad \|\rho(g)f\|^2 = \Delta(g)\int_G |f(xg)|^2\, d_L x = \|f\|^2.$$

Each operator is a bijection of $C_c(G)$, with inverse $\lambda(g^{-1})$ or $\rho(g^{-1})$. An isometric bijection of a dense subspace extends to an isometry with closed, dense range, hence to a unitary operator. The homomorphism property passes to the extensions by continuity. The operators commute, by the computation

$$\lambda(g)\rho(h)f(x) = \Delta(h)^{1/2} f(g^{-1}xh) = \rho(h)\lambda(g)f(x).$$

Strong continuity. Fix $f \in C_c(G)$ with support K, and a compact neighbourhood $W = W^{-1}$ of e. For $g \in W$, the functions $\lambda(g)f - f$ and $f(\cdot\, g) - f$ vanish outside $WK \cup KW$. A finite sum of chart bump functions gives $\varphi \in C_c(G)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on $WK \cup KW$. Fix $\varepsilon > 0$. Take the neighbourhood supplied by the uniform-continuity statement of Lemma 16.10 and intersect it with W to obtain $V \subset W$. For $g \in V$,

$$\|\lambda(g)f - f\|^2 \leq \varepsilon^2 \int_G \varphi \, d_L x.$$

The function φ was fixed before ε, and $\varepsilon > 0$ is arbitrary. Hence $\lambda(g)f \to f$ as $g \to e$. For ρ use

$$\|\rho(g)f - f\| \leq \Delta(g)^{1/2} \, \|f(\cdot\, g) - f\| + |\Delta(g)^{1/2} - 1| \, \|f\|$$

and the continuity of Δ. The squared norm $\|f(\cdot\, g) - f\|^2$ obeys the same bound as $\|\lambda(g)f - f\|^2$, by the second estimate of Lemma 16.10. Hence $\rho(g)f \to f$ as $g \to e$.

Thus λ and ρ are continuous at e on the dense subspace $C_c(G)$. The dense-set criterion for strong continuity in Lecture 15 extends this to every vector of $L^2(G)$ and every point of G. ∎

Without the factor $\Delta(g)^{1/2}$, right translation multiplies squared norms by $\Delta(g)^{-1}$. On the $ax + b$ group the factor $\Delta(g)^{-1}$ equals a for $g = g(a, b)$. Exercise 16.5 shows that λ and ρ are unitarily equivalent.

We test this on the rectangle E of the $ax + b$ group, with $h = g(s, t)$ as in § 16.4. For $f \in C_c(G)$, the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, gives $\|f(\cdot\, h)\|_2^2 = \Delta(h)^{-1}\|f\|_2^2 = s\,\|f\|_2^2$.

The indicator $\mathbf{1}_E$ of E is not continuous, so it is not in $C_c(G)$. It defines a vector of $L^2(G)$, as follows. For $0 < \delta \leq \frac{1}{2}$ let $\varphi_\delta(a, b) = u_\delta(a)v_\delta(b)$ be a product of trapezoid functions. The function u_δ equals 1 on $[1, 2]$, vanishes outside $[1 - \delta, 2 + \delta]$, and is linear in between. The function v_δ is built in the same way on $[0, 1]$. Then $\varphi_\delta \in C_c(G)$ and $0 \leq \varphi_\delta \leq 1$. It equals $\mathbf{1}_E$ except on a region of Euclidean area $4\delta + 4\delta^2$, and $a^{-2} \leq 4$ on this region. Hence $\int_G |\mathbf{1}_E - \varphi_\delta|^2 \, d_L x \leq 4(4\delta + 4\delta^2) \to 0$. So the functions φ_δ form a Cauchy family in $L^2(G)$, whose limit is the vector $\mathbf{1}_E$, with $\|\mathbf{1}_E\|_2^2 = \mu_L(E)$. Applying the identity above to φ_δ and passing to the limit gives

$$\|\mathbf{1}_E(\,\cdot\, h)\|_2^2 = \mu_L(Eh^{-1}) = s\,\mu_L(E).$$

Multiplication by $\Delta(h)^{1/2} = s^{-1/2}$ restores the norm in both cases.

16.6 Locally compact groups

A topological group is *locally compact* if e has a compact neighbourhood. We cite the following result (Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2). Let G be a locally compact Hausdorff group. A linear functional I on $C_c(G)$ is *positive* if $I(f) \geq 0$ whenever $f \geq 0$. There is a nonzero positive linear functional I on $C_c(G)$ with $I(f(g\,\cdot)) = I(f)$ for all $f \in C_c(G)$ and $g \in G$. It is unique up to a positive factor. There is a unique continuous homomorphism $\Delta\colon G \to (0,\infty)$ satisfying $I(f(\cdot\, g^{-1})) = \Delta(g) I(f)$.

A general locally compact group has no normalisation at e that compares left and right integrals. The cited result instead contains two identities. They replace the density relation in Theorem 16.6(ii) and the inversion identities in part (iii). First, the functional $f \mapsto I(f\Delta^{-1})$ is right-invariant. Second, I takes the same value on $f \in C_c(G)$ and on the function $x \mapsto f(x^{-1})\Delta(x^{-1})$.

The Riesz representation theorem realises I as a Radon measure. We cite Folland, *Real Analysis*, second edition, Theorem 7.2 in §7.1.

For a Lie group, uniqueness makes I a multiple of $f \mapsto \int_G f\, d_L x$. So the two modular functions agree. Since $C_c(G)$ is dense in L^2 of a Radon measure (Folland, *Real Analysis*, Proposition 7.9 in §7.2), our $L^2(G)$ is the usual one, in the sense made precise below.

Integration rules for Part II

We now collect the measure theory used in the rest of Part II. We cite it from Folland, *Real Analysis*, second edition, Chapters 2, 6 and 7, and we do not prove it. Let X be a Lie group, a quotient G/H of a Lie group by a closed subgroup, or an open subset of $\mathbb{R}^n$. Such a space is locally compact and second countable, and it is a countable union of compact sets. We state the rules only for these spaces. General locally compact groups need further hypotheses in the product theorems.

A *Borel set* of X is a member of the σ-algebra generated by the open sets. A function $f\colon X \to \mathbb{C}$ is *measurable* if the preimage of every open set is a Borel set. A *positive Radon measure* μ on X assigns a value in $[0,\infty]$ to each Borel set. It is countably additive and finite on compact sets. On our spaces $\mu(E)$ is the infimum of $\mu(U)$ over open $U \supset E$, and the supremum of $\mu(K)$ over compact $K \subset E$. A *null set* is a Borel set of measure 0, or a subset of one. A property holds *almost everywhere* if it fails only on a null set.

The word "positive" says only that the values are nonnegative. The measure μ has *full support* if $\mu(U) > 0$ for every nonempty open set U. Equivalently, $\int_X \varphi\, d\mu > 0$ for every nonzero $\varphi \in C_c(X)$ with $\varphi \geq 0$. For example, the point mass at e is a positive Radon measure without full support. Haar measures have full support by the positivity statement (I1), and so do the quotient measures of Lecture 17. Lecture 25 also uses positive measures without full support.

Suppose that μ has full support. Then $\int_X \bar{\varphi}\psi\, d\mu$ is an inner product on $C_c(X)$, and its Hilbert completion is the usual space $L^2(X,\mu)$. Its vectors are the measurable f with $\int_X |f|^2\, d\mu < \infty$, and two functions define one vector when they agree almost everywhere. On $\mathbb{R}$, the elements of the completion $L^2(\mathbb{R})$ of Lecture 15 are therefore classes of functions of this kind.

A measurable f is *integrable* if $\int_X |f|\,d\mu < \infty$. Let μ and ν be positive Radon measures on such spaces X and Y. We use the following rules.

- *Tonelli.* If $F\colon X \times Y \to [0,\infty]$ is measurable, then the two iterated integrals of F are equal. Their common value may be infinite.
- *Fubini.* If $F\colon X \times Y \to \mathbb{C}$ is measurable and one iterated integral of $|F|$ is finite, then the two iterated integrals of F exist, are finite and are equal. We use Tonelli to check the hypothesis.
- *Dominated convergence.* Let f_j be measurable with $f_j \to f$ almost everywhere. Suppose there is one integrable function g with $|f_j| \le g$ almost everywhere for all j. Then f is integrable and $\int_X f_j\,d\mu \to \int_X f\,d\mu$.
- *Differentiation under the integral.* Let $I \subset \mathbb{R}$ be an open interval. Let $F(s,\cdot)$ be integrable for each $s \in I$, and let $s \mapsto F(s,x)$ be differentiable for each $x \in X$. Suppose there is one integrable function g with $|\partial_s F(s,x)| \le g(x)$ for all $s \in I$ and $x \in X$. Then $\frac{d}{ds}\int_X F(s,x)\,d\mu(x) = \int_X \partial_s F(s,x)\,d\mu(x)$. Indeed, the mean value inequality bounds the difference quotients by g, and dominated convergence applies to them.
- *Change of variables.* Let $\Phi\colon U \to V$ be a C^1 diffeomorphism of open subsets of $\mathbb{R}^n$, and let $f\colon V \to \mathbb{C}$ be integrable, or measurable with values in $[0,\infty]$. Then $\int_V f(y)\,dy = \int_U f(\Phi(x))\,|\det D\Phi(x)|\,dx$.

The common bound in dominated convergence cannot be dropped. On $\mathbb{R}$ the functions $j\,\mathbf{1}_{(0,1/j)}$ tend to 0 everywhere, but each has integral 1. For the differentiation rule, take $F(s,t) = e^{-t^2+ist}$ on $\mathbb{R}$. Then $|\partial_s F(s,t)| = |t|e^{-t^2}$, which is integrable and does not depend on s. So $\int_{\mathbb{R}} e^{-t^2+ist}\,dt$ may be differentiated under the integral.

For Tonelli and Fubini, here is the bound that Lecture 18 uses. Let $f, h \in L^2(\mathbb{R})$ and $\lambda \neq 0$. Tonelli lets us integrate over t first. Cauchy–Schwarz and translation invariance bound the inner integral by $\|h\|\,\|f\|$. The two Gaussian integrals are $2\sqrt{\pi}$ and $2\sqrt{\pi}/|\lambda|$. Hence

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\,|f(t+a)|\,dt\,da\,db \le \frac{4\pi}{|\lambda|}\,\|h\|\,\|f\|.$$

This finite bound is the hypothesis of Fubini for the same integrand without absolute values.

Self-checks

1. In the $ax+b$ group let $g = g(2,5)$. Suppose $\int_G f\,d_L x = 7$. Find $\Delta(g)$ and $\int_G f(xg)\,d_L x$.

   Answer. $\Delta(g) = 1/2$. By the right-translation identity in Theorem 16.6(i), the integral is $\Delta(g)^{-1} \cdot 7 = 14$.

2. Suppose some left-invariant smooth positive density on G is also right-invariant. Must G be unimodular?

   Answer. Yes. By the uniqueness in Theorem 16.3(a), the density is $c\mu_L$ for some $c > 0$, so $R_g^*\mu_L = \mu_L$ for all $g \in G$. By the right-translation identity in Theorem 16.6(i), this means $\Delta(g) = 1$ for all $g \in G$.

3. On the $ax+b$ group, by what factor does $f \mapsto f(\cdot\, g(3,1))$ multiply $\|f\|^2$ in $L^2(G)$?

   Answer. By the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, the factor is $\Delta(g(3,1))^{-1} = 3$. Hence the map is not unitary without the factor $\Delta^{1/2}$.

Summary

Translating a fixed positive density on $\mathfrak{g}$ gives the left and right Haar densities. Each is unique up to a positive factor. On a noncompact Lie group the total mass is infinite, so no normalised finite Haar integral exists. Right translation scales μ_L by $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. When μ_L and μ_R have the same value at e, we have $\mu_R = \Delta^{-1}\mu_L$, and inversion exchanges μ_L and μ_R. For the $ax + b$ group, $d_L x = a^{-2}\,da\,db$, $d_R x = a^{-1}\,da\,db$ and $\Delta = a^{-1}$. The factor $\Delta^{1/2}$ makes right translation unitary on $L^2(G)$. Haar measures and quotient measures are positive Radon measures with full support, and the cited integration rules of § 16.6 hold in the second-countable spaces of Part II.

The main groups of Part II are unimodular: $\mathbb{R}^n$, the Heisenberg group, $\mathrm{SL}(2,\mathbb{R})$, and the Lorentz and Poincaré groups. Their closed subgroups often are not. For example, the upper triangular subgroup of $\mathrm{SL}(2,\mathbb{R})$ with positive diagonal is an $ax + b$ group (Exercise 20.6). Lecture 17 compares Δ_G with Δ_H to build unitary representations.

Reading

- Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII “Integration”, sections “Differential Forms and Measure Zero” and “Haar Measure for Lie Groups”. Compare its normalisation of the modular function with ours.
- Kirillov, *Elements of the Theory of Representations*, §9, from the course reference route. Check its modular-function convention.
- Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, sections “Haar Measure” and “The Modular Function”, for general locally compact groups.

Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

We check that $\mathcal{F}_c(\sigma)$ is not zero when $V \neq 0$. Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h)F(xh)\,dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1}\operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h')F(xh')\,dh' = \sigma(h_0)^{-1}P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

This element can be nonzero. Let $v \in V$ be nonzero. By strong continuity of σ, we can choose $\beta \in C_c(G)$ with $\beta \geq 0$ and $\beta(e) > 0$, supported so near e that $\operatorname{Re}\langle v, \sigma(h)v\rangle > 0$ for $h \in H \cap \operatorname{supp}\beta$. For $F(x) = \beta(x)v$ the number $\langle v, P_\sigma F(e)\rangle = \int_H \beta(h)\langle v, \sigma(h)v\rangle\,dh$ has positive real part. So $\mathcal{F}_c(\sigma) \neq 0$.

17.2 Quotient measures

Rescale a function on the line: $u \mapsto u((t - b_0)/a_0)$ with $a_0 > 0$. This multiplies $\|u\|^2$ by a_0, because dt is not invariant under dilation. Dividing by $a_0^{1/2}$ repairs the norm. This lecture carries out that repair for any quotient G/H.

In this lecture a *measure* on a locally compact space X means a positive Radon measure with full support, in the terms of Lecture 16. Equivalently, it is a positive linear functional $\varphi \mapsto \int_X \varphi\,d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Haar measures have this property. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi\rangle = \int_X \bar{\varphi}\psi\,d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\,dx = \Delta_G(g)^{-1}\int_G F(x)\,dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\,dh. \tag{17.2}$$

It is well defined by left invariance of dh.

Lemma 17.2. *If a measure μ on G/H satisfies $\int_G F\,dx = \int_{G/H} F^H\,d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1}\int_G F\,dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\,dh = \Delta_H(h_0)^{-1}F^H(xH).$$

Hence $\Delta_G(h_0)^{-1}\int_G F\,dx = \Delta_H(h_0)^{-1}\int_G F\,dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\,dx > 0$. Dividing both sides by $\int_G F\,dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax+b$ group and its scaling subgroup A, we will see $\Delta_G(a,0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

The weight must transform in a definite way. Suppose that a continuous function $\rho > 0$ and a measure μ satisfy $\int_G F\rho\,dx = \int_{G/H} F^H\,d\mu$ for all $F \in C_c(G)$. Apply this identity to $F_{h_0}(x) = F(xh_0)$. As in the proof of Lemma 17.2, the right side is multiplied by $\Delta_H(h_0)^{-1}$. On the left, the modular identity applied to $x \mapsto F(x)\rho(xh_0^{-1})$ gives

$$\int_G F(xh_0)\rho(x)\,dx = \Delta_G(h_0)^{-1}\int_G F(x)\rho(xh_0^{-1})\,dx.$$

Since F is arbitrary, $\Delta_G(h_0)^{-1}\rho(xh_0^{-1}) = \Delta_H(h_0)^{-1}\rho(x)$ for all $x \in G$. Replacing x by xh_0 gives the transformation law in part (a) of the next theorem.

Theorem 17.3 (Quotient measures; cited)**.** *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, “Measures on Homogeneous Spaces”, cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$. In this lecture ρ denotes a rho-function, and λ below denotes its cocycle. They are not the regular representations of Lecture 16.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1 g_2, p) = \lambda(g_1, g_2 p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1 g_2, p) = \frac{\rho(g_1 g_2 x)}{\rho(g_2 x)} \cdot \frac{\rho(g_2 x)}{\rho(x)} = \lambda(g_1, g_2 p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\,\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g, \cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

The substitution $p = gq$ in the quasi-invariance identity (17.6) gives $d\mu(gq) = \lambda(g, q)\,d\mu(q)$. So $\lambda(g, \cdot)$ is the Jacobian of $q \mapsto gq$ relative to μ, and the cocycle identity (17.5) is the chain rule. As an analogy, think of $\|f\|^2\,d\mu$ as a probability density. Then an amplitude carries the square root of the Jacobian.

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle\,d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f\,d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1}f(g^{-1}x)$. Also $q(\operatorname{supp}\Pi(g)f) = g\cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1},p)^{1/2}\,\lambda(g_2^{-1},g_1^{-1}p)^{1/2}\,f(g_2^{-1}g_1^{-1}x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1g_2)^{-1},p)^{1/2}$. Also $\lambda(e,p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g,\cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1},p)\,n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\,\lambda(g^{-1},gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\,d\mu = \int n_f(p)\,\lambda(g^{-1},gp)\,\lambda(g,p)\,d\mu(p) = \int n_f\,d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G\sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation* $\operatorname{Ind}_H^G\sigma$ *is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B\cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1},p)^{1/2}f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g,p)\mapsto\lambda(g^{-1},p)$ is jointly continuous and equals one at $g = e$. Also $(g,p)\mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B\times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p\in S_i$ as $g\to e$. There are only finitely many S_i, so the same holds on S.

Choose $\varphi\in C_c(G/H)$ with $\varphi\ge 0$ and $\varphi\ge 1$ on S. If the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \le \varepsilon^2\int_{G/H}\varphi\,d\mu.$$

This proves that $\Pi(g)f\to f$ in the norm of the completion as $g\to e$, for each $f\in\mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\,\lambda(g,\cdot)\,d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim. For example, the representation Π_0 of § 17.5 is induced from the trivial character of the translation subgroup N. A character of the $ax+b$ group that is trivial on N would have a nonzero N-map to that trivial character, yet no such character occurs in Π_0. We do not use reciprocity.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a,b)(a',b') = (aa', b + ab'), \qquad (a,b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $t \mapsto at + b$. By Lecture 16, $dx = a^{-2}\,da\,db$ is a left Haar measure and $\Delta_G(a,b) = 1/a$. Let $N = \{n_c = (1,c)\}$ be the translations and $A = \{(a,0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

The quotient G/N carries an invariant measure, as we see below. We treat it first because its points carry the orbit of the frequency. The affine line G/A of the opening preview has no invariant measure, and it comes at the end of § 17.5.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b + ac)$, the coset $(a,b)N$ is $\{a\} \times \mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a, ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_{\mathrm{Leb}}(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_{\mathrm{Leb}}} = da.$$

For the Lebesgue choice, take $g = (a_0, b_0)$ and $x = (a, 0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_{\mathrm{Leb}}(g, a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\, da = a_0 \int_0^\infty \varphi(u)\, du$. The cocycle identity reads $a_1 a_2 = a_1 \cdot a_2$, because $g_1 g_2$ has first entry $a_1 a_2$.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a, b + ac) = e^{-i\xi c} f(a, b)$. Take $b = 0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f \circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c} e^{-i\xi b/a}$. So $f \mapsto \varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2 \rho(a)\, da/a$.

Figure 17.1 draws such a function. Left translation by n_c sends (a, b) to $(a, b + c)$. So on the coset over a, the function f oscillates in b with frequency ξ/a. In the coset parameter $c = b/a$ the frequency is ξ.

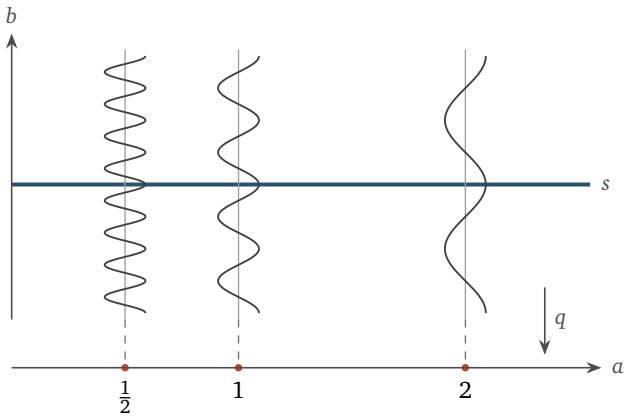


Figure 17.1: The real part of an equivariant function $f(a,b) = e^{-i\xi b/a}\varphi(a)$ with $\xi = 2\pi$, drawn as a horizontal offset on the three cosets of N over $a = \frac{1}{2}, 1, 2$. The heavy line is the section s, and q sends each vertical coset to its point of the base $(0,\infty)$.

Let $g = (a_0, b_0)$. Then $g^{-1}(a, 0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x = (a, 0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty), da/a), \tag{17.9}$$

$$(\Pi^{\mathrm{Leb}}_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2} e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty), da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g = (a_0, b_0)$, $g_j = (a_j, b_j)$ and $\varphi, \psi \in C_c((0,\infty))$.*

(a) $\|\Pi^{\mathrm{Leb}}_\xi(g)\psi\| = \|\psi\|$, *while* $\psi \mapsto e^{i\xi b_0/a}\psi(a/a_0)$ *multiplies* $\|\psi\|^2$ *by* a_0.

(b) $\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2) = \Pi^{\mathrm{Leb}}_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^{\mathrm{Leb}}_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a) &= a_1^{-1/2}e^{i\xi b_1/a}\,(\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big)\\ &= (a_1 a_2)^{-1/2}e^{i\xi(b_1+a_1 b_2)/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big) = (\Pi^{\mathrm{Leb}}_\xi(g_1 g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_{Leb} is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^{\mathrm{Leb}}_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_{\mathrm{Leb}})^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a, 0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let $\kappa > 0$. The operator $(U\varphi)(a) = \varphi(a/\kappa)$ is unitary on $L^2((0,\infty), da/a)$, and $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ for all $g \in G$.*

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$(U\Pi_\xi(g)\varphi)(a) = (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),$$
$$(\Pi_{\kappa\xi}(g)U\varphi)(a) = e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. ∎

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0,\infty) \cong G/N$, composed with the quotient map $G \to G/N$.

We do not prove irreducibility here. Mackey's theorem, cited in Lecture 25, applies with $n = 1$ and $H = (0,\infty)$. It shows that Π_+ and Π_- are irreducible. It also shows that, together with the characters $(a, b) \mapsto a^{is}$ with $s \in \mathbb{R}$, they are all the irreducible unitary representations of G, up to unitary equivalence. The representation Π_0 is not irreducible. By Exercise 17.4(a) it is the translation representation on $L^2(\mathbb{R})$, composed with $(a_0, b_0) \mapsto \log a_0$. By the proposition on finite-dimensional invariant subspaces in Lecture 15, it has no invariant line. An irreducible unitary representation of the abelian group $(0,\infty)$ is one-dimensional, by the Schur lemma that Lecture 18 cites. So Π_0 is reducible.

In physics this is the pattern of Wigner's classification. Translations act by phases, the remaining group moves the frequency along an orbit, and the representation lives on that orbit. Lecture 25 carries this out for the Poincaré group. Nothing in this lecture depends on it.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a,b)\rho(a,b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}}\left(\int_0^\infty F(a,t)\,\frac{da}{a}\right)dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1,t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0}\right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\, dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer. Exercise 17.6 shows that this representation is unitarily equivalent to $\Pi_1 \oplus \Pi_{-1}$.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.
2. Is $\rho(a, b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a, 0)n_1) = \rho(a, a) = a + a^2 \neq a = \rho(a, 0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

   $$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. By Mackey's theorem, cited in Lecture 25, Π_+ and Π_- are irreducible, while Π_0 is not.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, "Measures on Homogeneous Spaces": quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements,

Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.

Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. On this representation, the bracket $[X, Y] = Z$ becomes the commutation relation $[Q, P] = i\hbar$ of quantum mechanics. We then treat the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. We prove the irreducibility of the Schrödinger representation from an explicit rank-one operator, and we cite the uniqueness. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient, use it to prove that Π_λ is irreducible for $\lambda \neq 0$, and contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a, b, c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a, b, c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a, b, c)(a', b', c') = (a + a',\ b + b',\ c + c' + ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H) = \{(0,0,c) : c\in\mathbb{R}\} \cong \mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z, \qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ) = I+W+\tfrac12 W^2 = \big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1} = (a+s,b,c)(-a,-b,-c+ab) = (s,0,-sb) = \exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X = X-bZ$. In the same way $\mathrm{Ad}(g)Y = Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & a & 1 \end{pmatrix}, \qquad \det D_{\mathrm{Ad}}(g) = 1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle = \int_{\mathbb{R}} \overline{f(t)}\,h(t)\,dt$. We use the integration rules stated at the end of Lecture 16, in its section on locally compact groups: the identification of $L^2(\mathbb{R})$ with classes of functions, Tonelli and Fubini, dominated convergence and change of variables. To differentiate under an integral, we apply dominated convergence to the difference

quotients, using a common integrable bound for the derivative. We also use two standard facts. First, Lebesgue integrals are invariant under $t \mapsto t + a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a, b, c)f)(t) = e^{i\lambda(c+bt)} f(t + a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t + a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a, b, c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a, b, c) = (0, 0, c)(0, b, 0)(a, 0, 0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15, and $V_1(b)$ is the modulation $M(b)$ of Exercise 15.5. So at $\lambda = 1$ the Weyl relations below are the relation of Exercise 15.5(d).

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0, 0, c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a, b, c)$ and $g' = (a', b', c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t + a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t + a + a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t + a + a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0, 0, 0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t + a) = e^{i\lambda b(t+a)} f(t + a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R + 1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s)-f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R+2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f-h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g\to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab'-a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

This answers the question left open by the Heisenberg multiplier example of Lecture 13. There $\omega(x,y) = x_1y_2 - x_2y_1$ on $\mathbb{R}^2$ and $c(x,y) = e^{i\omega(x,y)/2}$. For $x \in \mathbb{R}^2$ put $U_x = \Pi_1(\exp(x_1X + x_2Y))$. The exponential formula (18.3) and the group law give

$$U_xU_y = e^{i\omega(x,y)/2}\,U_{x+y}.$$

So the multiplier c is realised on $L^2(\mathbb{R})$. By Exercise 18.4(d), Π_1 has no nonzero finite-dimensional invariant subspace, as that example requires. The extension $E = \mathbb{R}^2 \times \mathbb{T}$ of Exercise 13.5 is the quotient H/Γ with $\Gamma = \{(0,0,2\pi k) : k \in \mathbb{Z}\}$, through the map $(a,b,c) \mapsto ((a,b), e^{i(c-ab/2)})$.

The exponent $ab' - a'b$ is a signed area in the plane $H/Z(H)$, as Figure 18.1 shows. Multiply on the right by $(a,0,0)$ and then by $(0,b,0)$. This route ends at (a,b,ab). The other route ends at $(a,b,0)$. The central coordinates differ by the area ab of the rectangle, and this is the phase in the Weyl relations. Similarly, the four steps g, g', g^{-1}, g'^{-1} form a closed loop in the plane, and by (18.2) they end at $(0,0,ab'-a'b)$. The loop runs counterclockwise when $ab' - a'b > 0$. The operator products in the figure name these group products. As operators they act on a vector from the right, so the order of action is the reverse of the order of the route.

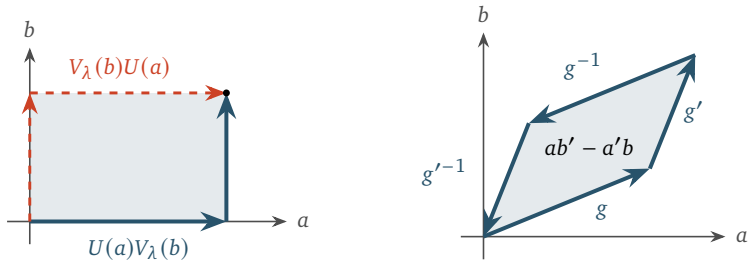


Figure 18.1: Routes in the plane $H/Z(H)$, with each step a multiplication on the right. In the first panel, two routes join e to the same point of the plane, and the rectangle lies between them. In the second panel, the loop of a commutator encloses the parallelogram on (a,b) and (a',b'), and its signed area is $ab' - a'b$.

Remark 18.4. Let $A = \{(0,b,c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0,b,c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s\to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X,Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R,R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left|\frac{f(t+s)-f(t)}{s} - f'(t)\right| \le \tfrac{1}{2}|s|\,\|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left|\frac{e^{i\lambda st}-1}{s} - i\lambda t\right| |f(t)| \le \tfrac{1}{2}\lambda^2|s|R^2|f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s}-1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

Since $\mathcal{D}$ is dense and invariant under each $\Pi_\lambda(\exp sW)$, the dense invariant-domain criterion of Stone's theorem in Lecture 15 shows that B_W is the closure of $d\Pi_\lambda(W)$ on $\mathcal{D}$.

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). The minus sign in P is there because $U(a)$ is the translation by $-a$. On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q,P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. A commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. For example, on $L^2(0,1)$ take $Qf = tf$ and $P = -i\hbar\, d/dt$ with periodic boundary conditions. Both operators are self-adjoint, and $[Q,P] = i\hbar$ on $C_c^\infty(0,1)$. Here $\exp(iP/\hbar) = I$, since P has the eigenvectors $e^{2\pi i n t}$ with eigenvalues $2\pi\hbar n$. The Weyl relation at $a = 1$ would then give $I = e^{ib/\hbar} I$ for every $b \in \mathbb{R}$, which is false (Hall, *Quantum Theory for Mathematicians*, Section 14.1). Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c} I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U, V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann)**.** *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We prove part (a) in § 18.4, after a kernel computation that does not use this theorem. We cite part (b), the uniqueness statement, from Kirillov, *Lectures on the Orbit Method*, Chapter 2. The proof of (a) does not prove (b).

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The uniqueness statement in the Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

In quantum mechanics, the first two consequences say that every irreducible Weyl pair with parameter $1/\hbar$ is the Schrödinger pair, which is the precise form of the statement that matrix mechanics and wave mechanics agree.

A standard proof of (b), going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \, \big\langle f, \rho(\exp(aX + bY))h \big\rangle \, da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a, b, c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on $|v| \leq V$

by the common integrable bound $2(|t+u|+V)e^{V^2}e^{-(t+u)^2}$ for the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2-(t+a)^2/2+i\lambda b t}\, dt.$$

Complete the square with $w = \frac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

Compare Schur orthogonality (Lecture 5). For a unit vector v in an irreducible unitary representation of dimension d of a compact group, $\int |\langle v, \Pi(g)v\rangle|^2\, dg = 1/d$, and $d \int \overline{\langle v, \Pi(g)v\rangle}\, \Pi(g)\, dg$ is the orthogonal projection onto $\mathbb{C}v$. The form B has the same shape on $H/Z(H)$, with φ in place of v. The number $|\lambda|/(2\pi)$ takes the place of d. Only the shape can be compared, because the measure here is $da\, db$ and not a normalised integral.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model, for $\lambda \neq 0$. The computation does not use that theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Tonelli, Cauchy–Schwarz in t and translation invariance give, as at the end of Lecture 16,

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus Fubini's theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4-(t+a/2)^2} f(t+a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u-t)^2}{4} - \frac{(t+u)^2}{4} = -\frac{t^2+u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t)\langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof of the uniqueness statement. Here its rank-one integral kernel is visible. In quantum mechanics, with $\lambda = \hbar = 1$, the function φ is the ground state of the oscillator $\frac{1}{2}(P^2 + Q^2)$, the vector $\Pi_1(g)\varphi$ is a coherent state, and E_1 is the projection on the ground state.

We can now prove part (a) of the theorem. The rank of E_λ forces every closed invariant subspace to contain φ or to be orthogonal to it, and the orbit of φ spans a dense subspace.

Proof of Theorem 18.9(a). Fix $\lambda \neq 0$, and let $W \subset L^2(\mathbb{R})$ be a closed invariant subspace.

The orbit of φ spans a dense subspace. Let $f \in L^2(\mathbb{R})$ be orthogonal to $\Pi_\lambda(g)\varphi$ for all $g \in H$. Taking $g = (0, b, 0)$ gives

$$\int_{\mathbb{R}} \overline{f(t)}\, \varphi(t)\, e^{i\lambda b t}\, dt = 0 \qquad \text{for every } b \in \mathbb{R}.$$

By the Cauchy–Schwarz inequality, $u = \bar{f}\varphi$ is integrable. We cite the injectivity of the Fourier transform on $L^1(\mathbb{R})$: if u is integrable and $\int_{\mathbb{R}} u(t)e^{ist}\, dt = 0$ for every $s \in \mathbb{R}$, then $u = 0$ almost everywhere (Folland, *Real Analysis*, second edition, Chapter 8). Since $\lambda \neq 0$, the number $s = \lambda b$ runs over $\mathbb{R}$. So $\bar{f}\varphi = 0$ almost everywhere. Since $\varphi > 0$ everywhere, $f = 0$.

The operator E_λ commutes with the projection onto W. Since $\Pi_\lambda(g)^* = \Pi_\lambda(g^{-1})$, the subspace $W^\perp$ is also invariant. Let $f \in W$ and $h \in W^\perp$. Then $\langle h, \Pi_\lambda(g)f\rangle = 0$ for all $g \in H$. The form defining E_λ therefore gives $\langle h, E_\lambda f\rangle = 0$. Hence $E_\lambda W \subset W$. The same argument with W and $W^\perp$ exchanged gives $E_\lambda W^\perp \subset W^\perp$. So E_λ commutes with the orthogonal projection P_W onto W.

Conclusion. Since $E_\lambda \varphi = \varphi$, we get $E_\lambda P_W \varphi = P_W E_\lambda \varphi = P_W \varphi$. So $P_W\varphi$ lies in the range $\mathbb{C}\varphi$ of E_λ, and $P_W\varphi = \kappa\varphi$ for some $\kappa \in \mathbb{C}$. Since $P_W^2 = P_W$, we have $\kappa = 0$ or $\kappa = 1$. If $\kappa = 1$, then $\varphi \in W$, and the first step gives $W = L^2(\mathbb{R})$. If $\kappa = 0$, then $\varphi \in W^\perp$, and the first step gives $W^\perp = L^2(\mathbb{R})$, so $W = 0$. ∎

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a,b,c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The pieces one expects in Π_0 are the plane waves $e^{i\alpha t}$, on which $U(a)$ acts by $e^{i\alpha a}$. They are not square integrable. The Fourier transform of Lecture 15 turns $U(a)$ into multiplication by $e^{i\alpha a}$ on $L^2(\mathbb{R}, d\alpha)$. So Π_0 is an integral of the characters $\chi_{\alpha,0}$ rather than a sum. This is called a *direct integral*, and we do not develop it.

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.
3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi \rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi \rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. For $\lambda \neq 0$ the Gaussian form computes an explicit rank-one projection, and this proves that Π_λ is irreducible. The uniqueness statement of the Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.

Lecture 19

Coadjoint orbits and the nilpotent orbit method

Objectives. *A representation acts on vectors. The coadjoint action of a Lie group instead acts on linear functionals on its Lie algebra. Each orbit of this action has a natural symplectic form. For connected simply connected nilpotent groups, these orbits label all irreducible strongly continuous unitary representations up to unitary equivalence. We prove the geometric construction and work through the Heisenberg example. The general classification is an explicit cited theorem.*

By the end of this lecture you will be able to:

- Compute a coadjoint action and its stabilisers.
- Construct the orbit symplectic form and check its signs, using the rules for 1-forms, the exterior derivative and pullback.
- State the nilpotent orbit theorem with its global hypotheses.
- Choose a polarisation for a functional on a two-dimensional Heisenberg orbit and recover the Schrödinger representation by induction.

19.1 The dual action

Throughout, G is a finite-dimensional real Lie group with Lie algebra $\mathfrak{g}$. Its real dual is $\mathfrak{g}^* = \mathrm{Hom}_{\mathbb{R}}(\mathfrak{g}, \mathbb{R})$. We pair a functional ℓ and a vector X by $\ell(X)$. This dual is different from the complex dual of a representation space.

We first say why this dual appears. For $G = \mathbb{R}^n$, a unitary character has the form $x \mapsto e^{i\ell(x)}$ with $\ell \in \mathfrak{g}^*$. So $\mathfrak{g}^*$ labels the irreducible unitary representations of $\mathbb{R}^n$. A noncommutative group acts on $\mathfrak{g}^*$. The guiding idea of this lecture is that two functionals on one orbit should label one representation. In Lecture 17, the scaling subgroup of the $ax + b$ group moved the frequency ξ to ξ/a, and the class of Π_ξ depended only on the orbit.

Definition 19.1. The *coadjoint action* of G on $\mathfrak{g}^*$ is

$$(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X).$$

The *coadjoint orbit* through ℓ is $O_\ell = \{g \cdot \ell : g \in G\}$. Its stabiliser is $G_\ell = \{g : g \cdot \ell = \ell\}$.

The inverse is needed for the action law. Indeed,

$$(g \cdot (h \cdot \ell))(X) = \ell(\mathrm{Ad}(h^{-1})\,\mathrm{Ad}(g^{-1})X) = \ell(\mathrm{Ad}((gh)^{-1})X) = ((gh) \cdot \ell)(X).$$

The stabiliser is closed by continuity. It is therefore an embedded Lie subgroup by the closed-subgroup theorem of Lecture 3. Differentiating the action gives

$$(\mathrm{ad}^*(X)\ell)(Y) = -\ell([X,Y]), \qquad \mathfrak{g}_\ell = \{X : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}. \tag{19.1}$$

The identity $\exp(tX)\cdot\ell = e^{t\,\mathrm{ad}^*(X)}\ell$ shows that $\exp(tX) \in G_\ell$ for every t exactly when $\mathrm{ad}^*(X)\ell = 0$. Thus $\mathrm{Lie}(G_\ell) = \mathfrak{g}_\ell$. For example, a central X lies in every $\mathfrak{g}_\ell$. Central directions therefore do not generate motion on an orbit.

We give $\mathcal{O}_\ell$ the smooth structure of G/G_ℓ. This uses the homogeneous-space theorem from Lecture 3. The natural map

$$G/G_\ell \longrightarrow \mathfrak{g}^*, \qquad gG_\ell \longmapsto g\cdot\ell$$

is injective and an immersion. At the identity coset, its derivative is induced by $X \mapsto \mathrm{ad}^*(X)\ell$, whose kernel is precisely $\mathfrak{g}_\ell$. At other points, equivariance gives the same rank. The orbit dimension is consequently $\dim\mathfrak{g} - \dim\mathfrak{g}_\ell$.

For a general Lie group, an orbit need not be an embedded submanifold of $\mathfrak{g}^*$. All tangent-space statements below use the quotient smooth structure. The Heisenberg orbits will be ordinary affine planes and points, so this distinction disappears there.

Write

$$\xi_X(\eta) = \left.\frac{d}{dt}\right|_0 \exp(tX)\cdot\eta = \mathrm{ad}^*(X)\eta, \qquad \eta \in \mathcal{O}_\ell.$$

The vectors $\xi_X(\eta)$ span the tangent space of $\mathcal{O}_\ell$ at η. Two choices of X give the same vector exactly when their difference belongs to $\mathfrak{g}_\eta$.

A first example: the Heisenberg group

Let H be the Heisenberg group of Lecture 18, with Lie algebra $\mathfrak{h}$ spanned by X, Y and Z. We use exactly its matrix coordinates:

$$(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab'), \qquad [X,Y] = Z.$$

In particular, c is a matrix entry. It is not the central coefficient in the logarithm, which is $c - ab/2$.

Write $\ell = (p,q,\lambda)$ when $\ell(X) = p$, $\ell(Y) = q$ and $\ell(Z) = \lambda$. The adjoint formulas of Lecture 18 are

$$\mathrm{Ad}(a,b,c)X = X - bZ, \qquad \mathrm{Ad}(a,b,c)Y = Y + aZ, \qquad \mathrm{Ad}(a,b,c)Z = Z.$$

The inverse of $g = (a,b,c)$ has first coordinates $(-a,-b)$. Hence $\mathrm{Ad}(g^{-1})X = X + bZ$ and $\mathrm{Ad}(g^{-1})Y = Y - aZ$. Applying ℓ to these vectors gives

$$(a,b,c)\cdot(p,q,\lambda) = (p+\lambda b, q-\lambda a, \lambda). \tag{19.2}$$

For $\lambda \neq 0$, this reaches every point of the plane with fixed last coordinate. Its stabiliser is the centre. For $\lambda = 0$, every point is fixed. Thus the orbit list consists of planes for nonzero λ and individual points $(p,q,0)$.

On a plane, p and q are coordinates. Differentiating the action formula (19.2) gives

$$\xi_X = -\lambda\partial_q, \qquad \xi_Y = \lambda\partial_p, \qquad \xi_Z = 0.$$

Figure 19.1 draws the orbits and these fields.

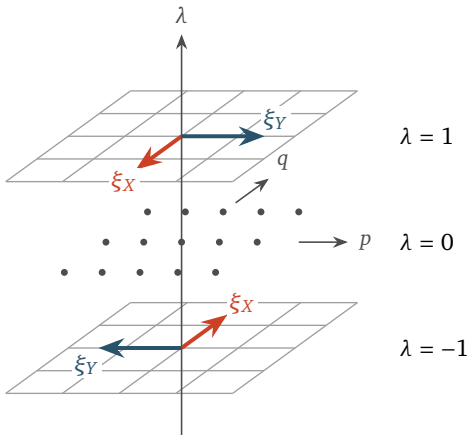


Figure 19.1: Coadjoint orbits of the Heisenberg group in the coordinates (p, q, λ), with λ upward. Each plane $\lambda = \pm 1$ is one orbit, and each dot at $\lambda = 0$ is one orbit. The arrows are the fields ξ_X and ξ_Y at one point of each plane.

19.2 A symplectic form on each orbit

We use the form of this section as follows. First, it shows that orbits have even dimension. Next, at ℓ it is the form B_ℓ on $\mathfrak{g}/\mathfrak{g}_\ell$ below, which decides what a polarisation is. Closedness is not used in this lecture. Lecture 27 needs it for a cited theorem.

We need a small amount of differential-form language. A smooth 2-form ω assigns an alternating bilinear form ω_x to every tangent space, smoothly in local coordinates. On $\mathbb{R}^2$, for example,

$$(dp \wedge dq)((u_1, u_2), (v_1, v_2)) = u_1 v_2 - u_2 v_1.$$

A form ω is *nondegenerate* if, at every point x, the only tangent vector u at x with $\omega_x(u, v) = 0$ for every tangent vector v at x is $u = 0$.

In coordinates $x_1, \dots, x_d$, put $\omega_{ij} = \omega(\partial_i, \partial_j) = -\omega_{ji}$. Then $\omega = \frac{1}{2} \sum_{i,j} \omega_{ij}\, dx_i \wedge dx_j$. The form ω is *closed* when

$$\partial_i \omega_{jk} + \partial_j \omega_{ki} + \partial_k \omega_{ij} = 0 \quad \text{for all } i, j, k.$$

Equivalently, closedness means $d\omega = 0$. A closed nondegenerate 2-form is called a *symplectic form*. These are coordinate-independent conditions. We import this elementary differential-form calculus, including pullbacks and the exterior derivative d, from Warner, Chapter 2. We use the following rules from it.

A smooth 1-*form* α assigns a linear functional α_x on the tangent space to every point x, smoothly in local coordinates. Here dx_j is the 1-form with $dx_j(\partial_k) = \delta_{jk}$. So in coordinates $\alpha = \sum_j a_j\, dx_j$ with $a_j = \alpha(\partial_j)$. The differential of a smooth function f is the 1-form $df = \sum_j (\partial_j f)\, dx_j$. The *wedge* of two 1-forms is the 2-form given by

$$(\alpha \wedge \beta)(u, v) = \alpha(u)\beta(v) - \alpha(v)\beta(u).$$

In particular $\beta \wedge \alpha = -\alpha \wedge \beta$. The exterior derivative of 1-forms is linear, and it satisfies $d(f\, dg) = df \wedge dg$ and $d(dg) = 0$ for smooth functions f and g. In coordinates,

$$\alpha = \sum_j a_j\, dx_j, \qquad d\alpha = \sum_{i<j} (\partial_i a_j - \partial_j a_i)\, dx_i \wedge dx_j.$$

A smooth map F has a *pullback*, given on 1-forms and 2-forms by

$$(F^*\alpha)_x(u) = \alpha_{F(x)}(dF_x u), \qquad (F^*\omega)_x(u,v) = \omega_{F(x)}(dF_x u, dF_x v).$$

It satisfies $F^*(df) = d(f\circ F)$, $F^*(f\alpha) = (f\circ F)F^*\alpha$, $F^*(\alpha\wedge\beta) = F^*\alpha\wedge F^*\beta$ and $d(F^*\alpha) = F^*(d\alpha)$.

For example, on $\mathbb{R}^2$ we have $d(y\,dx) = dy\wedge dx = -dx\wedge dy$. The map $F(s,t) = (s+t, st)$ into the (p,q)-plane gives

$$F^*(dp\wedge dq) = (ds+dt)\wedge(t\,ds + s\,dt) = (s-t)\,ds\wedge dt.$$

The proof below uses two further identities, which it states.

Theorem 19.2 (Orbit symplectic form)**.** *Every coadjoint orbit has a G-invariant symplectic form ω defined by*

$$\omega_\eta(\xi_X(\eta), \xi_Y(\eta)) = \eta([X,Y]). \tag{19.3}$$

This is the Kirillov–Kostant–Souriau form, with the sign convention of the infinitesimal action (19.1).

Proof. Well-definedness. If X changes by $A\in\mathfrak{g}_\eta$, then $\eta([A,Y]) = 0$. Changing Y has the same effect by antisymmetry. Thus the defining formula (19.3) depends only on the two tangent vectors. It is alternating and bilinear.

Nondegeneracy. If $\omega_\eta(\xi_X,\xi_Y) = 0$ for every Y, then $X\in\mathfrak{g}_\eta$ by the stabiliser formula (19.1). Hence $\xi_X(\eta) = 0$.

Smoothness. The fields ξ_X are smooth and span every tangent space. Choose a local frame from these fields. The defining formula (19.3) gives smooth coefficients in that frame, so ω is smooth.

Invariance. For $g\in G$, let g_* be the differential of the map $\eta\mapsto g\cdot\eta$ on the orbit. Since $g\exp(tX) = \exp(t\,\mathrm{Ad}(g)X)g$, differentiating $g\cdot(\exp(tX)\cdot\eta)$ at $t=0$ gives

$$g_*\xi_X(\eta) = \xi_{\mathrm{Ad}(g)X}(g\cdot\eta).$$

Using that $\mathrm{Ad}(g)$ preserves brackets, we get

$$\begin{aligned}\omega_{g\cdot\eta}(g_*\xi_X, g_*\xi_Y) &= (g\cdot\eta)([\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y])\\ &= \eta([X,Y]).\end{aligned}$$

So the form is invariant.

Closedness. Let $\Phi: G\to \mathcal{O}_\ell$ be the orbit map, given by $\Phi(g) = g\cdot\ell$. Put $\beta = \Phi^*\omega$. Invariance makes β a left-invariant form on G, and $\beta_e(X,Y) = \ell([X,Y])$. Exterior differentiation commutes with pullback. For a left-invariant 2-form, the differential at e is

$$(d\beta)_e(X,Y,Z) = -\beta_e([X,Y],Z) + \beta_e([X,Z],Y) - \beta_e([Y,Z],X).$$

We have now stated the two identities imported from Warner, Chapter 2. They are the commutation of d with pullback and the preceding formula for a left-invariant 2-form. Substitution gives

$$(d\beta)_e(X,Y,Z) = -\ell([[X,Y],Z]) + \ell([[X,Z],Y]) - \ell([[Y,Z],X]) = 0$$

by the Jacobi identity. Left invariance gives $d\beta = 0$ everywhere. Therefore $\Phi^*(d\omega) = 0$. Since Φ is a submersion, every tuple of tangent vectors on the orbit lifts to G. This forces $d\omega = 0$. ∎

A nondegenerate alternating form exists only in even dimension. To see this, let A be the matrix of such a form in a basis. Then $A^T = -A$. If the size d of A were odd, then transposition invariance of the determinant would give $\det A = \det(-A) = -\det A$. Hence $\det A = 0$, and the form would be degenerate. In particular, every coadjoint orbit has even dimension. This is a useful first check on a proposed calculation.

Reading the form through linear functions

There is a useful way to remember the sign in the defining formula (19.3). For $X \in \mathfrak{g}$, restrict the linear function $\mu_X(\eta) = \eta(X)$ to an orbit. Its derivative in a tangent direction is

$$(d\mu_X)_\eta(\xi_Y) = (\mathrm{ad}^*(Y)\eta)(X) = -\eta([Y,X]) = \eta([X,Y]).$$

Thus $d\mu_X = \iota_{\xi_X}\omega$, where contraction means $(\iota_V\omega)(W) = \omega(V,W)$. This identifies the infinitesimal action using only the orbit form and the linear functions.

For a smooth real function f, nondegeneracy determines a unique tangent vector field X_f by $\iota_{X_f}\omega = df$. It is called the Hamiltonian vector field in this convention. The preceding calculation says $X_{\mu_X} = \xi_X$. Consequently the bracket

$$\{f,h\} = \omega(X_f, X_h)$$

satisfies $\{\mu_X, \mu_Y\} = \mu_{[X,Y]}$. We only need this identity for linear functions. It says that the Lie bracket is still visible after passing to an orbit.

A central element $W \in \mathfrak{g}$ has $\xi_W = 0$. Thus $d\mu_W = \iota_{\xi_W}\omega = 0$, so μ_W is locally constant on each orbit. It is constant when G is connected. Such a constant can still be the value of a bracket of linear functions. For the Heisenberg basis X, Y and Z of the first example in § 19.1, the identity above gives $\{\mu_X, \mu_Y\} = \mu_Z$, and μ_Z takes one constant value on each orbit.

A change of convention for Hamiltonian vector fields would change some of these signs. Here the definitions, rather than a remembered physics formula, determine them. This calculation concerns finite-dimensional smooth manifolds. It does not differentiate an unbounded operator or assert a quantisation rule.

19.3 Heisenberg planes and points

On a plane, the fields $\xi_X = -\lambda\partial_q$ and $\xi_Y = \lambda\partial_p$ of the first example in § 19.1 span the tangent spaces, and $\xi_Z = 0$. The defining value of the form is $\omega(\xi_X, \xi_Y) = \ell(Z) = \lambda$. Since $(dp \wedge dq)(\xi_X, \xi_Y) = \lambda^2$, we obtain

$$\omega_\lambda = \frac{1}{\lambda}\, dp \wedge dq. \tag{19.4}$$

The coefficient is $1/\lambda$, not λ, because the vectors ξ_X and ξ_Y already carry a factor λ.

By Exercise 19.2, the orbit map pulls ω_λ back to $\lambda\, da \wedge db$ on H. On the vectors (a,b) and (a',b') its value is $\lambda(ab' - a'b)$. This is the exponent of the commutator phase in Lecture 18, where $ab' - a'b$ is drawn as a signed area.

Point orbits have zero-dimensional tangent spaces. The zero form on such a tangent space is nondegenerate, since its kernel is the zero space. The orbit method will associate $(p, q, 0)$ with the one-dimensional character

$$\chi_{p,q}(a, b, c) = e^{i(pa+qb)}.$$

These are different representations for different (p, q). The trivial central character, with $\lambda = 0$, does not mean that the representation is trivial. Only $(0, 0, 0)$ gives the trivial character.

19.4 Polarisations and the orbit theorem

A Lie algebra is *nilpotent* if its lower central series terminates:

$$\mathfrak{g}_1 = \mathfrak{g}, \qquad \mathfrak{g}_{r+1} = [\mathfrak{g}, \mathfrak{g}_r], \qquad \mathfrak{g}_s = 0 \text{ for some } s.$$

For the Heisenberg algebra, $\mathfrak{g}_2 = \mathbb{R}Z$ and $\mathfrak{g}_3 = 0$. Nilpotence is stronger than solvability.

We import a global fact: for a connected simply connected nilpotent group, $\exp : \mathfrak{g} \to G$ is a diffeomorphism, and BCH is a finite polynomial. Every Lie subalgebra $\mathfrak{m}$ gives a closed subgroup $M = \exp \mathfrak{m}$. Indeed, finite BCH keeps products in $\mathfrak{m}$, and the exponential diffeomorphism takes the closed vector subspace $\mathfrak{m}$ to a closed subset. See Knapp, *Lie Groups Beyond an Introduction*, Section I.16.

Definition 19.3. A *real polarisation* for $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{m} \subset \mathfrak{g}$ such that

$$\ell([\mathfrak{m}, \mathfrak{m}]) = 0, \qquad \dim \mathfrak{m} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

A subspace $A \subseteq \mathfrak{g}$ is called *isotropic* for ℓ when

$$\ell([A, A]) = 0.$$

A real polarisation is thus an isotropic Lie subalgebra of the displayed dimension. This dimension says that the image of $\mathfrak{m}$ in $\mathfrak{g}/\mathfrak{g}_\ell$ is as large as the image of an isotropic subspace can be. In fact, the two displayed conditions force $\mathfrak{g}_\ell \subset \mathfrak{m}$, because an isotropic subspace of maximal dimension contains $\mathfrak{g}_\ell$. The subsection on the dimension of a polarisation below proves both statements.

For such a subalgebra, put

$$\chi_\ell(\exp W) = e^{i\ell(W)}, \qquad W \in \mathfrak{m}.$$

This is well-defined because exp is bijective. It is a character of M. Every BCH correction is a bracket of elements of $\mathfrak{m}$, so ℓ kills it and $\chi_\ell(\exp U \exp V) = \chi_\ell(\exp U)\chi_\ell(\exp V)$.

For example, take the Heisenberg algebra and $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Then $\mathfrak{g}_\ell = \mathbb{R}Z$, so a polarisation has dimension two. Both $\operatorname{span}\{Y, Z\}$ and $\operatorname{span}\{X, Z\}$ are polarisations. All of $\mathfrak{h}$ is too large: $\ell([X, Y]) = \lambda \neq 0$, so the formula above does not define a character of H. The line $\mathbb{R}Z$ is isotropic but too small. The representation $\operatorname{Ind}_{Z(H)}^{H} \chi_\lambda$ of Lecture 17 acts on functions of two variables. Right translations of H preserve its space, since $Z(H)$ is central,

and they are unitary and commute with the action. So it is not irreducible. By the Stone–von Neumann theorem of Lecture 18, it is a direct sum of copies of Π_λ.

Theorem 19.4 (Nilpotent orbit theorem; cited)**.** *Let G be connected, simply connected and nilpotent. Every $\ell \in \mathfrak{g}^*$ has a real polarisation. The unitary induced representation*

$$\Pi_\ell = \operatorname{Ind}_{\exp \mathfrak{m}}^{G} \chi_\ell$$

is irreducible. Its equivalence class does not depend on the polarisation. Two such representations are equivalent exactly when their functionals lie on the same coadjoint orbit. Every irreducible strongly continuous unitary representation of G is unitarily equivalent to some Π_ℓ.

This is Kirillov's theorem, imported from *Lectures on the Orbit Method*, Chapter 3. Induction means unitary induction with the quotient measure convention of Lecture 17. The statement concerns complex Hilbert representations, not just finite-dimensional modules.

The conclusion can fail without these hypotheses. A central quotient can restrict which representations descend (Exercise 19.3). For SU(2) the coadjoint orbits are the origin and the spheres of every radius, as Lecture 27 computes. These form a continuum, while the irreducible representations V_m are labelled by integers. Apart from the trivial group, no compact group and no semisimple group satisfies this theorem's hypotheses. We will meet related orbit constructions in Lecture 27, with a different integrality condition and a different representation model.

The hypotheses are sufficient, not necessary. A solvable group need not be nilpotent. The $ax + b$ group is not nilpotent, yet its orbit list in Exercise 19.1 matches Π_+, Π_- and the characters of Lecture 17 (Kirillov, *Lectures on the Orbit Method*, Chapter 4).

The dimension of a polarisation

We now isolate the linear algebra behind a polarisation. Define the alternating form

$$B_\ell(X, Y) = \ell([X, Y])$$

on $\mathfrak{g}$. Its kernel is $\mathfrak{g}_\ell$. Suppose $\dim \mathcal{O}_\ell = 2d$ and $\dim \mathfrak{g}_\ell = k$. Then $\dim \mathfrak{g} = 2d + k$. The form B_ℓ induces a nondegenerate alternating form on the quotient $\mathfrak{g}/\mathfrak{g}_\ell$, which has dimension $2d$. Under the isomorphism $\mathfrak{g}/\mathfrak{g}_\ell \to T_\ell \mathcal{O}_\ell$ given by $X \mapsto \xi_X(\ell)$, this induced form is ω_ℓ. A vector space with a nondegenerate alternating form is called a *symplectic vector space*.

Let L be a subspace of this quotient. Its *orthogonal* $L^\perp$ consists of the vectors whose induced pairing with every vector of L is zero. The subspace L is *isotropic* when the induced form vanishes on pairs of vectors of L, that is, when $L \subseteq L^\perp$. Nondegeneracy gives $\dim L^\perp = 2d - \dim L$, so an isotropic L has $\dim L \leq d$. Let $A \subset \mathfrak{g}$ be an isotropic subspace for ℓ. Its image in the quotient is isotropic, so this image has dimension at most d. Therefore

$$\dim A = \dim(A \cap \mathfrak{g}_\ell) + \dim\big((A + \mathfrak{g}_\ell)/\mathfrak{g}_\ell\big) \leq k + d.$$

Equality forces $\mathfrak{g}_\ell \subset A$. It also forces the image of A to be a d-dimensional isotropic subspace. The dimension in the definition of a polarisation is exactly $k + d$.

There is one extra demand: A must be a Lie subalgebra. A maximal isotropic vector subspace need not have this property. The existence of a suitable subalgebra is a substantial part of the nilpotent orbit theorem. The definition alone does not prove existence.

This count also predicts the number of variables in an induced model. If $\mathfrak{m}$ is a polarisation, then

$$\dim(G/M) = \dim \mathfrak{g} - \dim \mathfrak{m} = d = \tfrac{1}{2} \dim O_\ell .$$

The Heisenberg plane is two-dimensional, but its representation acts on functions of one real variable. The number of these variables is half the orbit dimension. The orbit does not single out which variables to use. A polarisation makes that choice.

19.5 Induction recovers the Schrödinger formula

Take $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Its stabiliser algebra is $\mathbb{R}Z$, so a polarisation must have dimension two. Choose

$$\mathfrak{m} = \operatorname{span}\{Y, Z\}, \qquad M = \{(0, b, c)\}, \qquad \chi_\ell(0, b, c) = e^{i\lambda c}.$$

The subalgebra $\mathfrak{m}$ is abelian and two-dimensional, so it satisfies both polarisation conditions. A section of $H/M \cong \mathbb{R}$ is $s(t) = (t, 0, 0)$. The quotient measure is dt, invariant under the induced translation action.

An induced function satisfies $f(gm) = \chi_\ell(m)^{-1} f(g)$. Write $\varphi(t) = f(s(t))$. Since

$$(a, b, c) = s(a)(0, b, c - ab),$$

its values are

$$f(a, b, c) = e^{-i\lambda(c-ab)} \varphi(a).$$

Left translation acts by $(\Psi_\lambda(g) f)(h) = f(g^{-1}h)$. For $g = (a, b, c)$,

$$g^{-1} s(t) = (t - a, -b, -c + ab).$$

Insert this into the preceding covariance formula. Its central expression becomes $-c + ab - (t - a)(-b) = -c + bt$. Therefore

$$(\Psi_\lambda(a, b, c)\varphi)(t) = e^{i\lambda(c-bt)} \varphi(t - a). \tag{19.5}$$

The operator is unitary: the phase has absolute value one and translation preserves dt. The general unitary induction construction of Lecture 17 supplies the representation law and strong continuity. Both can also be checked from this formula.

Exercise 18.5 carries out the same computation. The sign of the translation differs from the Schrödinger formula of Lecture 18, but it gives an equivalent model. Let $R\varphi(x) = \varphi(-x)$. Then $R^{-1} = R$, and

$$(R\Psi_\lambda(a, b, c)R^{-1}u)(x) = e^{i\lambda(c+bx)} u(x + a) = \Pi_\lambda(a, b, c)u(x).$$

Thus the orbit theorem reproduces the Schrödinger representation with the precise convention already used. The plane's parameter λ determines its nonzero central character.

For a point $(p, q, 0)$, the polarisation is all of $\mathfrak{h}$. Induction from H to itself gives $\chi_{p,q}$. The plane and point calculations therefore recover the complete Heisenberg classification from Lecture 18. The general orbit theorem extends that pattern to every connected simply connected nilpotent group.

The polarisation chooses the variable of the model. On a plane, the group M moves ℓ along lines in the direction of ξ_Y, that is, along ∂_p. By the action formula (19.2), the orbit of $\ell = (0, 0, \lambda)$ has $q = -\lambda a$. So the variable $t = a$ of the model is the transverse coordinate q, up to a factor. The second polarisation $\mathfrak{m}' = \operatorname{span}\{X, Z\}$ moves ℓ along ξ_X, that is, along ∂_q. Its section $s(u) = (0, u, 0)$ gives a model on functions of the transverse coordinate $p = \lambda u$. Figure 19.2 shows both choices. The orbit theorem says that the two models are equivalent, and here the equivalence is a Fourier transform.

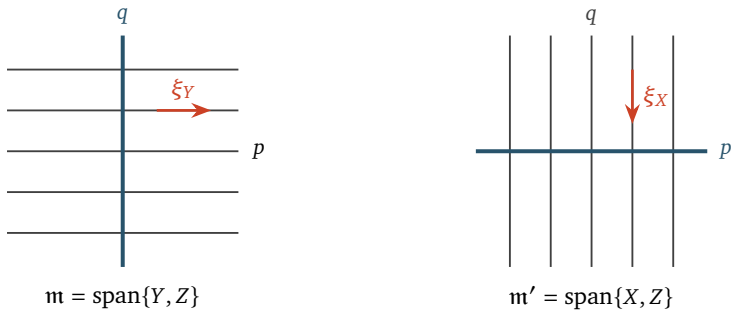


Figure 19.2: Two polarisations on the plane $\lambda = 1$. The parallel lines are the orbits of the subgroup $\exp \mathfrak{m}$ or $\exp \mathfrak{m}'$ on the plane, and the arrow gives their direction. The heavy line is the transverse coordinate, which is the variable of the corresponding induced model up to a factor.

What the Heisenberg orbit list explains

The orbit calculation separates two phenomena. For $\lambda \neq 0$, changing p or q stays within one plane. It cannot change the equivalence class of the representation. For $\lambda = 0$, each point is an orbit of its own, so p and q both survive as representation labels. The discontinuous change in orbit dimension matches the change from infinite-dimensional representations to characters.

One can check the zero-charge conclusion without the general orbit theorem. In an irreducible unitary representation, every central group element acts by a scalar, by the unitary Schur lemma used in Lecture 18. If that scalar character is trivial, the representation factors through $H/Z(H) \cong \mathbb{R}^2$. An irreducible unitary representation of this abelian group is one-dimensional. Its continuous characters are $e^{i(pa+qb)}$, obtained by applying the one-parameter character classification to each coordinate.

For nonzero central character, the Stone–von Neumann theorem gives uniqueness directly. The orbit theorem recovers that result and explains its geometric label. It also shows what to try for a larger nilpotent group: calculate ad^*, find orbit invariants, determine the stabiliser dimension, choose a real polarisation for ℓ, and induce its character. The last exercise carries out the geometric steps in one four-dimensional example.

Remark 19.5. This remark gives a physics dictionary. Compare Lecture 18 with $\lambda = 1/\hbar$. Since $\chi_\ell(\exp W) = e^{i\ell(W)}$, the value $\ell(W)$ corresponds to the operator $-i\,d\Pi(W) = -A_W$. Thus p corresponds to $P/\hbar$ and q to $Q/\hbar$, and $\omega_\lambda = \hbar^{-1} dP \wedge dQ$ is phase-space area in units of $\hbar$. This is a correspondence of labels, not an equality of operators.

Self-checks

1. Why must the stabiliser algebra contain the centre?

Answer. If X is central, then $[X, Y] = 0$ for every Y. The stabiliser formula (19.1) puts X in $\mathfrak{g}_\ell$ for every ℓ.

2. Does the formula $\omega_\lambda = \lambda\, dp \wedge dq$ have the right normalisation?

Answer. No. Its value on $(-\lambda\partial_q, \lambda\partial_p)$ would be λ^3. The required value is λ, so the coefficient is $1/\lambda$.

3. Can one apply the orbit theorem directly to a Heisenberg group with circle centre?

Answer. No. That group is not simply connected. Apply the theorem to its simply connected cover. Then use the Descent theorem of Lecture 4. Only representations trivial on the covering kernel descend.

Summary

The coadjoint action turns $\mathfrak{g}^*$ into a union of even-dimensional symplectic orbits. For connected simply connected nilpotent groups, a polarisation converts an orbit into an irreducible induced representation. For the Heisenberg group, planes give Schrödinger representations and points give characters. The cited classification is exact in this setting. Its hypotheses are part of the result.

Reading

- Kirillov, *Lectures on the Orbit Method*, Chapters 2–3: Heisenberg representations, polarisations and the nilpotent correspondence.
- Kirillov, *Lectures on the Orbit Method*, Chapter 4: the correspondence for exponential groups, which include the $ax + b$ group.
- Knapp, *Lie Groups Beyond an Introduction*, Section I.16: simply connected nilpotent groups.
- Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Chapter 2: differential forms and exterior differentiation. The lecture uses only the rules stated before the orbit form theorem and the two identities stated in its proof.

Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. The orbit theorem of Lecture 19 concerns nilpotent groups and does not apply to such a group, so we start from its structure. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k \exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \ \operatorname{ad} Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad} X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad} H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

Every complex semisimple Lie algebra has both types. The Lie Algebras course does not treat real forms, so we cite this fact from Knapp Beyond, Section VI.1 (Theorem 6.6, Corollary 6.10 and Theorem 6.11).

The construction starts from a *Chevalley basis*. This basis consists of the coroots H_α and of root vectors E_α with $[E_\alpha, E_{-\alpha}] = H_\alpha$. Its structure constants are real, and the linear map given by $E_\alpha \mapsto -E_{-\alpha}$ and $H_\alpha \mapsto -H_\alpha$ is an automorphism. Knapp scales these vectors by real factors, which does not change the real spans below. The real span of the Chevalley basis is a split form, with $\mathfrak{h}_0$ the real span of the H_α. The elements iH_α, $E_\alpha - E_{-\alpha}$ and $i(E_\alpha + E_{-\alpha})$ span a compact form. For $\mathfrak{sl}(2,\mathbb{C})$, with $E_\alpha = E$, $E_{-\alpha} = F$ and $H_\alpha = H$, these elements are the matrices A, B and C of the next example.

Example 20.3 proves both claims for $\mathfrak{sl}(2,\mathbb{C})$, and Proposition 20.5 proves them for $\mathfrak{sl}(n,\mathbb{C})$. The general case stays cited.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X, Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X, \theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X, Y) = \kappa_\theta(Y, X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X, Y] = [-X, -Y] = [X, Y]$. We cite from Knapp Beyond, Section VI.2, that every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism.

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{R})$ and $\theta(X) = -X^T$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n, \mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n, \mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^T Y^T - Y^T X^T = (YX - XY)^T = \theta[X, Y].$$

On $\mathfrak{sl}(n, \mathbb{C})$ we import $\kappa(X, Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X, X) = 2n \operatorname{tr}(XX^T) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n, \mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^* = X^T - iY^T$. So $Z^* = -Z$ exactly when $X^T = -X$ and $Y^T = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n, \mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z, Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^* Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n, \mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form. This general fact is cited from Knapp Beyond, Sections VI.1–VI.2. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X, Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The name refers to the roots of the complexification $\mathfrak{g}_{\mathbb{C}}$. For a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$ containing $\mathfrak{a}$, each restricted root is the restriction to $\mathfrak{a}$ of a root of $\mathfrak{g}_{\mathbb{C}}$. For example, § 20.3 takes $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{a} = \mathbb{R}H$, and finds the restricted roots $\pm\lambda_1$ with $\lambda_1(H) = 2$.

Fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker\lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^{T}$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

The real rank in the table does not depend on the choices of θ and $\mathfrak{a}$. We cite this from Knapp Beyond, Chapter VI. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Theorem 20.8 (Cartan and Iwasawa decompositions; cited)**.** *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) A and N are closed subgroups, and $K \times A \times N \to G$, $(k, a, n) \mapsto kan$, is a diffeomorphism.

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

For $\mathrm{SL}(n, \mathbb{R})$ these decompositions are matrix factorisations. The form $G = KAN$ is the QR decomposition $g = k(an)$, where an is upper triangular with positive diagonal. Gram–Schmidt produces it (Exercise 20.2). The form $G = K \exp \mathfrak{p}$ writes $g = kP$ with $P = \exp X$ symmetric and positive definite. This is the polar decomposition of g, and $g^{T}g = P^{2} = \exp 2X$ determines P. In § 20.3 we prove both forms for $\mathrm{SL}(2, \mathbb{R})$, and we also obtain the singular value decomposition $G = KAK$.

A real form keeps the finite-dimensional complex modules of $\mathfrak{g}_{\mathbb{C}}$, by restriction and complex-linear extension. It does not keep their unitarity. For example, $\mathrm{SL}(2, \mathbb{R})$ acts on $V_m = \operatorname{Sym}^{m}\mathbb{C}^{2}$. For $m \geq 1$ and $t \neq 0$ the matrix $\operatorname{diag}(e^{t}, e^{-t})$ has the eigenvalue e^{mt} on V_m, and $|e^{mt}| \neq 1$. So no inner product makes V_m unitary for $\mathrm{SL}(2, \mathbb{R})$, whereas $\mathrm{SU}(2)$ has one by Lecture 5. To build unitary representations of $\mathrm{SL}(2, \mathbb{R})$ we use infinite-dimensional spaces, with K in Lecture 21 and KAN in Lecture 22.

In physics language, the adjoint action identifies $\mathfrak{sl}(2, \mathbb{R})$ with $\mathfrak{so}(2, 1)$. Then $\mathfrak{k}$ is the rotation generator and $\mathfrak{p}$ consists of the boosts, and $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$ says that the commutator of two boosts is a rotation (Lecture 24).

20.3 The rank-one model $\mathrm{SL}(2, \mathbb{R})$

Let $G = \mathrm{SL}(2, \mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n, \mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E - F)$ and $\mathfrak{p} = \operatorname{span}(H, E + F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$). *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^t = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^2}.$$

Proof. The first column of g is nonzero, so $r > 0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta - \beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^t & e^t x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^t = r$ and $x = (\alpha\beta+\gamma\delta)/r^2$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^T$ with $p>0$ and unit length, so $p=1$. The second column is a unit vector orthogonal to $(1,0)^T$ with positive second entry, so it is $(0,1)^T$. Hence $k = k'$ and $an = a'n'$. Comparing diagonals gives $a = a'$, and then $n = n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r > 0$. ∎

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u) = \sum_{k\ge 0}\frac{u^k}{(2k)!}, \qquad S(u) = \sum_{k\ge 0}\frac{u^k}{(2k+1)!}.$$

For $u = s^2 \ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s=0$. We use $u = s^2$ rather than $s = \sqrt{-\det Y}$, because s is not smooth at $Y = 0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y \in \mathfrak{p}$, set $u = -\det Y \ge 0$. Then $Y^2 = uI$ and*

$$\exp Y = C(u)I + S(u)Y.$$

The map $\exp\colon \mathfrak{p} \to \mathcal{P}$ *is a bijection. Its inverse is*

$$L(Q) = \frac{Q - cI}{\mathcal{S}(C^{-1}(c))}, \qquad c = \tfrac{1}{2}\operatorname{tr} Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ *among symmetric matrices.*

Proof. Write $Y = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$. Then $Y^2 = (p^2+q^2)I = uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y = (\exp \frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det \exp Y = e^{\operatorname{tr} Y} = 1$.

Every coefficient of the series for C' is positive, so $C' > 0$ on $(0,\infty)$. Also, $C'(0) = \frac{1}{2}$. So $C' > 0$ on some interval $(-\varepsilon, \infty)$. Thus C is a strictly increasing bijection $[0,\infty) \to [1,\infty)$, with smooth inverse near $[1,\infty)$. Also $\mathcal{S} > 0$ near $[0,\infty)$.

For injectivity, let $Q = \exp Y$. Then $\operatorname{tr} Q = 2C(u)$, which determines u because C is strictly increasing on $[0,\infty)$. The formula for $\exp Y$ then gives $Y = (Q - C(u)I)/\mathcal{S}(u) = L(Q)$.

For surjectivity, let $Q \in \mathcal{P}$ have eigenvalues $\mu, \mu^{-1} > 0$. Then $c = \frac{1}{2}(\mu + \mu^{-1}) \geq 1$. Put $u = C^{-1}(c)$ and $Y = L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2 = 2cQ - I$, so $(Q - cI)^2 = (c^2 - 1)I$. Since $u\,\mathcal{S}(u)^2 = C(u)^2 - 1 = c^2 - 1$, we get $Y^2 = uI$. Therefore

$$\exp Y = C(u)I + \mathcal{S}(u)Y = cI + (Q - cI) = Q. \qquad \blacksquare$$

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2,\mathbb{R})$)**.** *The map* $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, *is a diffeomorphism. Its inverse is*

$$g \mapsto \bigl(g \exp(-X),\, X\bigr), \qquad X = \tfrac{1}{2} L(g^T g).$$

Proof. Suppose first that $g = k \exp X$ with $k \in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^T g = \exp(X) k^T k \exp(X) = \exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X = L(g^T g)$, so X and then k are forced. Conversely, given g, the matrix $g^T g$ lies in $\mathcal{P}$. Define X and $k = g \exp(-X)$ as stated. Then

$$k^T k = \exp(-X)\exp(2X)\exp(-X) = I, \qquad \det k = 1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. $\blacksquare$

Corollary 20.12 ($G = KAK$)**.** *Every* $g \in G$ *can be written as* $g = k_1 a_t k_2$ *with* $k_1, k_2 \in K$ *and* $t \geq 0$*, and* t *is unique. The number* e^{2t} *is the larger eigenvalue of* $g^T g$*. So* e^t *is the larger singular value of* g*.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk') a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^T g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. $\blacksquare$

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^t = \sqrt{2}$ and $x = \frac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^{T} g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \frac{3}{2}$. The equation $C(u) = \cosh \sqrt{u} = \frac{3}{2}$ gives $\sqrt{u} = 2 \log \tau$, since $\tau^2 + \tau^{-2} = 3$. Also $\sinh \sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2} L(Q) = \frac{2 \log \tau}{\sqrt{5}} \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log \tau}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log \tau)^2$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g \exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm \log \tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log \tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{ z \in \mathbb{C} : \operatorname{Im} z > 0 \}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma)\operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

These numbers x and t are not those of the Iwasawa decomposition, Proposition 20.9, of g. From $g = n_x a_t k$ we get $g^{-1} = k^{-1}a_{-t}n_{-x}$. So they are the negatives of the parameters of g^{-1} in $G = KAN$.

Proof. We have $a_t \cdot z = e^{2t}z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \frac{1}{2}\log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

The action of G on $\mathbb{H}$ is smooth and transitive. So, by the homogeneous-space theorem of Lecture 3, the bijection $G/K \to \mathbb{H}$ is a diffeomorphism.

So Iwasawa coordinates are Cartesian coordinates (x, y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t}z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16. Under this identification the left Haar measure $dy\,db/y^2$ of Lecture 16 becomes the measure $dx\,dy/y^2$ on $\mathbb{H}$.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

Figure 20.1 draws both descriptions. It marks the point $g \cdot i = \frac{1}{2}(1 + i)$ for the matrix g of Example 20.13. This point has Iwasawa coordinates $x = \frac{1}{2}$ and $e^{2t} = \frac{1}{2}$. Its KAK parameter

is $t = \log \tau$, so it lies on the K-orbit of $i\tau^2$. The derivative of $z \mapsto k_\varphi \cdot z$ at i is $e^{-2i\varphi}$, so k_φ with $\varphi > 0$ turns clockwise around i.

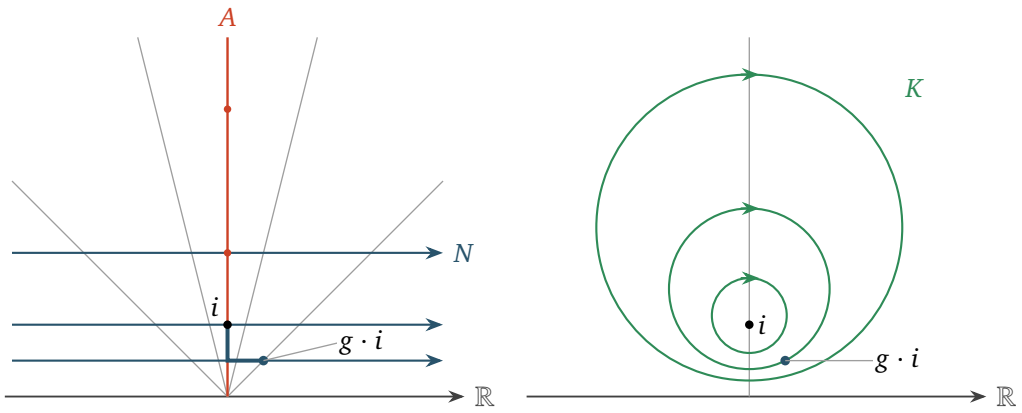


Figure 20.1: Coordinates on $\mathbb{H}$, with the point $g \cdot i$ of the lower triangular example. The first panel shows N-orbits (horizontal lines), A-orbits (rays from 0, with the imaginary axis through i) and a heavy path from i along A and then along N. The second panel shows K-orbits (circles, and the point i itself), with arrows for k_φ, $\varphi > 0$.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n, \mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n, \mathbb{R})$ is $n - 1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p} = 0$, so $A = N = \{I\}$ and the Iwasawa decomposition reads $G = K$.

2. Find the Iwasawa decomposition of $g = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$.
   Answer. Here $r = 1$, $\cos\varphi = 0$ and $\sin\varphi = -1$, so $k = k_{-\pi/2}$ and $a = I$. Also $x = (0 \cdot 1 + (-1) \cdot 2)/1 = -2$. Indeed
   $$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

3. Where does $-I$ sit in $G = KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G = KAN$, the element $-I$ has factors $k = k_\pi$ and $a = n = I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. The compact form $\mathfrak{k} \oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n, \mathbb{R})$ has real rank $n - 1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G = K \exp \mathfrak{p}$ and $G = KAN$. For $\mathrm{SL}(2, \mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t \geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G = KAN$ to construct the principal series.

Lecture 21

Smooth Vectors, K-Types and $(\mathfrak{g}, K)$-Modules

Objectives. *A representation of a noncompact group usually acts on an infinite-dimensional Hilbert space. We use its derivatives on smooth vectors. These operators are in general unbounded, and the space of smooth vectors has no convenient basis. Let K be a compact subgroup. If $\Pi|_K$ is unitary, then Lecture 10 splits the Hilbert space into a Hilbert sum of blocks, one for each K-type. So we pass to the smooth vectors that are finite under K. On these vectors $\mathfrak{g}$ and K act together, subject to compatibility conditions. For $\mathrm{SL}(2,\mathbb{R})$ with $K = \mathrm{SO}(2)$, the study of a representation becomes a raising and lowering calculation.*

By the end of this lecture you will be able to:

- Define smooth vectors, K-finite vectors, K-types and admissibility.
- Prove that the K-finite smooth vectors of a strongly continuous representation on a Hilbert space form a $(\mathfrak{g}, K)$-module.
- Use the compatibility conditions to show that E and F shift the circle K-types of $\mathrm{SL}(2,\mathbb{R})$ by ± 2.
- State which global questions the $(\mathfrak{g}, K)$-module leaves open, and which cited theorems address them.

21.1 Smooth vectors and K-finite vectors

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$. The subgroup $K \subseteq G$ is compact, with Lie algebra $\mathfrak{k} \subseteq \mathfrak{g}$. The representation Π of G on a Hilbert space $\mathcal{H}$ is strongly continuous. Here each operator $\Pi(g)$ is bounded, linear and invertible. The map $g \mapsto \Pi(g)v$ from G to $\mathcal{H}$ is continuous for every $v \in \mathcal{H}$. Unitarity is assumed only where stated.

Definition 21.1. A vector $v \in \mathcal{H}$ is *smooth* if the orbit map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is smooth. This means that in every chart all iterated partial derivatives exist as norm limits and are continuous. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ we set

$$d\Pi(X)v = \left.\frac{d}{dt}\right|_{t=0} \Pi(\exp tX)v.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, because a linear combination of smooth orbit maps is smooth.

For unitary representations, the following facts were imported in Lecture 15. We now also import the density and invariance facts (F1)–(F2) for strongly continuous representations by bounded operators. A source for this more general smooth-vector theorem is Knapp RT, Sections III.3–4. Here Knapp RT is A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (2001).

(F1) *Density* (Gårding). The space $\mathcal{H}^\infty$ is dense in $\mathcal{H}$.

(F2) *Invariance.* Each operator $d\Pi(X)$ maps $\mathcal{H}^\infty$ into itself, and $X \mapsto d\Pi(X)$ is a Lie algebra representation of $\mathfrak{g}$ on $\mathcal{H}^\infty$.

(F3) *Skew-symmetry.* If Π is unitary, then $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for all $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$.

We extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$ by $d\Pi(X + iY) = d\Pi(X) + i\, d\Pi(Y)$ for $X, Y \in \mathfrak{g}$. By the invariance property (F2), this is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{H}^\infty$.

One elementary fact is proved here. If $v \in \mathcal{H}^\infty$ and $h \in G$, then $\Pi(h)v \in \mathcal{H}^\infty$. Indeed, the orbit map of $\Pi(h)v$ is $g \mapsto \Pi(gh)v$. This is the orbit map of v composed with the smooth right translation $g \mapsto gh$.

Definition 21.2. A vector $v \in \mathcal{H}$ is *K-finite* if $\operatorname{span}\{\Pi(k)v : k \in K\}$ is finite-dimensional. We write $\mathcal{H}_K$ for the set of K-finite vectors.

The set $\mathcal{H}_K$ is a subspace, because for $a, b \in \mathbb{C}$ the span of the vectors $\Pi(k)(av + bw)$, $k \in K$, lies in $\operatorname{span}\Pi(K)v + \operatorname{span}\Pi(K)w$.

Let $\widehat{K}$ be the set of equivalence classes of irreducible continuous representations of K. Each class δ has a finite-dimensional representative space V_δ (Lecture 10). Let $\operatorname{Hom}_K(V_\delta, \mathcal{H})$ be the space of linear maps $V_\delta \to \mathcal{H}$ that commute with K.

Definition 21.3. The *multiplicity* of $\delta \in \widehat{K}$ in Π is $m_\delta(\Pi) = \dim \operatorname{Hom}_K(V_\delta, \mathcal{H})$. The class δ is a *K-type* of Π if $m_\delta(\Pi) \geq 1$. The representation Π is *admissible* if $m_\delta(\Pi) < \infty$ for every $\delta \in \widehat{K}$.

Let $v \in \mathcal{H}_K$ and $W_v = \operatorname{span}\Pi(K)v$. Then W_v is finite-dimensional and K-invariant. All norms on W_v are equivalent, so strong continuity makes K act continuously on W_v. By complete reducibility (Lecture 5), W_v is a direct sum of irreducible subspaces. Hence every K-finite vector is a finite sum of vectors from K-types.

For the circle we write k_θ, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, with $k_\theta k_\phi = k_{\theta+\phi}$. Its irreducible continuous representations are the characters $\chi_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 7). We label the K-type χ_n by n.

Example 21.4. Let $G = K = \mathrm{SO}(2)$ and $\mathcal{H} = L^2(\mathbb{R}/2\pi\mathbb{Z})$, with

$$\langle f, h\rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{f(x)}\, h(x)\, dx, \qquad (\Pi(k_\theta)f)(x) = f(x + \theta).$$

This representation is unitary and strongly continuous. The functions $e_n(x) = e^{inx}$ form an orthonormal basis (Lecture 10). Put $\mathcal{H}(n) = \{v : \Pi(k_\theta)v = e^{in\theta}v \text{ for all } \theta\}$. Then $e_n \in \mathcal{H}(n)$.

We show that $\mathcal{H}(n) = \mathbb{C}e_n$. Let $a_m(v) = \langle e_m, v\rangle$. For continuous v, the substitution $y = x + \theta$ and periodicity give

$$a_m(\Pi(k_\theta)v) = \frac{1}{2\pi}\int_0^{2\pi} e^{-im(y-\theta)}v(y)\,dy = e^{im\theta}a_m(v).$$

Both sides are continuous linear functionals of v, so density extends the identity to all L^2 vectors. If $v \in \mathcal{H}(n)$, the left side is also $e^{in\theta}a_m(v)$. For $m \neq n$ take $\theta = \pi/(m-n)$. Then $e^{im\theta} = -e^{in\theta}$, so $a_m(v) = 0$. Hence $v = a_n(v)e_n$.

So $\mathcal{H}_K$ is the space of trigonometric polynomials $f = \sum_{|n|\le N} c_n e_n$. Each such f is smooth, because its orbit map

$$\theta \mapsto \Pi(k_\theta)f = \sum_{|n|\le N} c_n\, e^{in\theta}\, e_n$$

is a finite sum of smooth scalar functions times fixed vectors. Thus every K-finite vector is smooth. Every multiplicity is 1, so Π is admissible. The space $\mathcal{H}_K$ is dense but proper. For example, $\sum_n 2^{-|n|}e_n$ is not K-finite.

A K-finite vector need not be smooth. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by translation, and let $K = \{0\}$. Every vector is K-finite. The indicator function of $[0,1]$ is not smooth (Exercise 21.5). The domain example of Lecture 15 computes the same difference quotients. So we impose both conditions. We write $\mathcal{H}_K^\infty$ for the space of smooth K-finite vectors:

$$\mathcal{H}_K^\infty = \mathcal{H}^\infty \cap \mathcal{H}_K.$$

21.2 The compatibility conditions

Definition 21.5. A $(\mathfrak{g}, K)$-*module* is a complex vector space V with a Lie algebra representation of $\mathfrak{g}$ on V, written $v \mapsto X \cdot v$, and a group homomorphism $K \to \mathrm{GL}(V)$, written $v \mapsto k \cdot v$, such that:

(a) (*finiteness condition*) every $v \in V$ lies in a finite-dimensional K-invariant subspace W on which K acts continuously;

(b) (*derivative condition*) for $X \in \mathfrak{k}$ and $v \in V$, with W as in the finiteness condition (a),

$$X \cdot v = \left.\frac{d}{dt}\right|_{t=0} \exp(tX)\cdot v \qquad \text{(derivative computed in } W\text{)};$$

(c) (*covariance condition*) for $k \in K$, $X \in \mathfrak{g}$ and $v \in V$,

$$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v).$$

A *morphism* of $(\mathfrak{g}, K)$-modules is a linear map that commutes with both actions. A $(\mathfrak{g}, K)$-module V is *admissible* if $\dim \mathrm{Hom}_K(V_\delta, V) < \infty$ for every $\delta \in \widehat{K}$.

The derivative in the derivative condition (b) does not depend on the choice of W. Indeed, two choices W and W' both lie in the finite-dimensional space $W + W'$, and all norms on $W + W'$ are equivalent.

The $\mathfrak{g}$-action extends complex-linearly to $\mathfrak{g}_\mathbb{C}$. Both sides of the covariance condition (c) are complex-linear in X, so the covariance condition also holds for $X \in \mathfrak{g}_\mathbb{C}$.

The derivative condition (b) involves only elements $\exp(tX)$, which lie in the identity component of K. Hence, if K is disconnected, the K-action carries information that $\mathfrak{k}$ cannot detect (Exercise 21.7).

The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. In § 21.3 it gives the shifts of the circle K-types for $\mathrm{SL}(2,\mathbb{R})$.

Proposition 21.6. *Let Π be a strongly continuous representation of G on $\mathcal{H}$, and let $K \subseteq G$ be compact. Then $\mathcal{H}_K^\infty$ is invariant under $\Pi(k)$ for $k \in K$ and under $d\Pi(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. With these actions $\mathcal{H}_K^\infty$ is a $(\mathfrak{g}, K)$-module. Let Π' be a strongly continuous representation of G on a Hilbert space $\mathcal{H}'$. Let $T : \mathcal{H} \to \mathcal{H}'$ be bounded and linear, with $T\Pi(g) = \Pi'(g)T$ for all $g \in G$. Then T maps $\mathcal{H}_K^\infty$ into $\mathcal{H}_K'^\infty$, and its restriction is a morphism.*

Proof. Step 1: the K-action. Let $v \in \mathcal{H}_K^\infty$ and $k \in K$. Then $\Pi(k)v$ is smooth by the right-translation argument in § 21.1. Since $Kk = K$,

$$\operatorname{span} \Pi(K)\Pi(k)v = \operatorname{span} \Pi(Kk)v = \operatorname{span} \Pi(K)v,$$

which is finite-dimensional. So $\Pi(k)v \in \mathcal{H}_K^\infty$. In particular, $W_v = \operatorname{span} \Pi(K)v$ is contained in $\mathcal{H}_K^\infty$.

Step 2: the covariance identity. Let $k \in K$, $X \in \mathfrak{g}$ and $w \in \mathcal{H}^\infty$. We claim

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w. \tag{21.1}$$

By the conjugation formula for exponentials (Lecture 2), $k \exp(tX)k^{-1} = \exp(t\,\mathrm{Ad}(k)X)$. Hence, for $t \neq 0$,

$$\Pi(k)\Big[\frac{\Pi(\exp tX)w - w}{t}\Big] = \frac{\Pi(\exp(t\,\mathrm{Ad}(k)X))\,\Pi(k)w - \Pi(k)w}{t}.$$

Let $t \to 0$. On the left, the bracket tends to $d\Pi(X)w$, and $\Pi(k)$ is bounded. So the left side tends to $\Pi(k)\, d\Pi(X)w$. On the right, $\Pi(k)w$ is smooth, so the quotient tends to $d\Pi(\mathrm{Ad}(k)X)\Pi(k)w$.

Step 3: $\mathfrak{g}$ preserves K-finiteness. This is the main step. The difficulty is that $d\Pi(X)$ does not commute with $\Pi(k)$. So the span of $\Pi(K)\, d\Pi(X)v$ is not obviously finite-dimensional. The covariance identity (21.1) replaces commutation. It moves X to $\mathrm{Ad}(k)X$, which still lies in the finite-dimensional space $\mathfrak{g}$.

Let $v \in \mathcal{H}_K^\infty$ and $W = W_v$. By Step 1, $W \subseteq \mathcal{H}^\infty$, so $d\Pi(X)w$ is defined for $w \in W$. Choose bases $X_1, \dots, X_d$ of $\mathfrak{g}$ and $w_1, \dots, w_r$ of W. Put

$$U = \operatorname{span}\{d\Pi(X_i)w_j : 1 \le i \le d,\ 1 \le j \le r\}.$$

Then $\dim U \le dr$. Let $X \in \mathfrak{g}$, $w \in W$ and $k \in K$. Expand

$$\mathrm{Ad}(k)X = \sum_{i=1}^{d} a_i(k)X_i, \qquad \Pi(k)w = \sum_{j=1}^{r} b_j(k)w_j,$$

with scalars $a_i(k)$ and $b_j(k)$. The second expansion uses $\Pi(k)w \in W$. By the covariance identity (21.1) and linearity of $d\Pi$ in both arguments,

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w = \sum_{i=1}^{d}\sum_{j=1}^{r} a_i(k)\, b_j(k)\, d\Pi(X_i)w_j \in U.$$

In particular $\Pi(K)\, d\Pi(X)v \subseteq U$. So $d\Pi(X)v$ is K-finite. It is smooth by the invariance property (F2). Complex combinations stay in the subspace $\mathcal{H}_K^\infty$, so the same holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

Step 4: the axioms. By the invariance property (F2) and Step 3, $d\Pi$ is a representation of $\mathfrak{g}$ on $\mathcal{H}_K^\infty$. The finiteness condition (a) holds with $W = W_v$, by Step 1 and the continuity of the K-action on W_v shown in § 21.1. For the derivative condition (b), let $X \in \mathfrak{k}$. The curve $t \mapsto \Pi(\exp tX)v$ lies in W_v. Its derivative in $\mathcal{H}$ is $d\Pi(X)v$. A finite-dimensional subspace is closed, and W_v carries the restricted norm. So the difference quotients converge in W_v to $d\Pi(X)v$. The covariance condition (c) is the covariance identity (21.1).

Step 5: intertwiners. Let $v \in \mathcal{H}_K^\infty$. The orbit map $g \mapsto \Pi'(g)Tv = T\Pi(g)v$ is smooth, because T is bounded and linear. Also $\operatorname{span} \Pi'(K)Tv = T(\operatorname{span} \Pi(K)v)$ is finite-dimensional. Differentiating $T\Pi(\exp tX)v = \Pi'(\exp tX)Tv$ at $t = 0$ gives $T\, d\Pi(X)v = d\Pi'(X)Tv$. Commutation with K is the hypothesis. ∎

Definition 21.7. The $(\mathfrak{g}, K)$-*module of* Π is $\mathcal{H}_K^\infty$ with the actions given in the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6. Two representations are *infinitesimally equivalent* if their $(\mathfrak{g}, K)$-modules are isomorphic.

21.3 Circle K-types for $\mathrm{SL}(2, \mathbb{R})$

Now $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. Put

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

The matrix k_θ here is $k_{-\theta}$ in the notation of Lecture 20. Then $\mathfrak{k} = \mathbb{R}W$, and $\theta \mapsto k_\theta$ identifies $\mathbb{R}/2\pi\mathbb{Z}$ with K. So K is connected.

Let V be a $(\mathfrak{g}, K)$-module. For $n \in \mathbb{Z}$ put

$$V(n) = \{v \in V : k_\theta \cdot v = e^{in\theta} v \text{ for all } \theta\}.$$

By the finiteness condition (a) and complete reducibility (Lecture 5), every $v \in V$ is a finite sum of vectors from the spaces $V(n)$. The sum is direct, because the functions $e^{in\theta}$ are linearly independent. So $V = \bigoplus_{n\in\mathbb{Z}} V(n)$, and $\dim V(n)$ is the multiplicity of the K-type n.

Let $H = \operatorname{diag}(1, -1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then H, S, W is a basis of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{R})$. Multiplying matrices gives

$$[H, S] = 2W, \qquad [W, H] = -2S, \qquad [W, S] = 2H.$$

In $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$ we set

$$H' = -iW = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}(H + iS) = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}(H - iS) = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}.$$

The real brackets give

$$\begin{aligned} [H', E] &= -\tfrac{i}{2}\big([W, H] + i[W, S]\big) = -\tfrac{i}{2}(-2S + 2iH) = H + iS = 2E, \\ [H', F] &= -\tfrac{i}{2}\big([W, H] - i[W, S]\big) = -\tfrac{i}{2}(-2S - 2iH) = -(H - iS) = -2F, \\ [E, F] &= \tfrac{1}{4}\big(-i[H, S] + i[S, H]\big) = -\tfrac{i}{2}[H, S] = -iW = H'. \end{aligned}$$

Thus (H', E, F) is an $\mathfrak{sl}_2$-triple. The $\mathfrak{sl}_2$ theory of the prerequisite Lie Algebras course applies to it. The element H' spans $\mathfrak{k}_{\mathbb{C}}$, whereas $H \notin \mathfrak{k}$. Also $F = \overline{E}$, where $\overline{X + iY} = X - iY$ for $X, Y \in \mathfrak{g}$.

These E and F are not the real matrices of Lecture 20. Let $u_\pm = (1, \pm i)^T$, and let Q be the matrix with columns u_+ and u_-. Then

$$H' = Q \operatorname{diag}(1, -1) Q^{-1}, \qquad E = Q \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} Q^{-1}, \qquad F = Q \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} Q^{-1}.$$

The vectors $u_\pm$ are eigenvectors of K, as Example 21.10 shows. In the K-eigenbasis (u_+, u_-), the matrices of (H', E, F) are the standard triple of Lecture 20. Up to a scalar, Q^{-1} is the matrix c of Exercise 20.1.

Lemma 21.8. *For every $\theta \in \mathbb{R}$, the complex-linear extension of $\operatorname{Ad}(k_\theta)$ to $\mathfrak{g}_{\mathbb{C}}$ satisfies $\operatorname{Ad}(k_\theta)H' = H'$, $\operatorname{Ad}(k_\theta)E = e^{2i\theta}E$ and $\operatorname{Ad}(k_\theta)F = e^{-2i\theta}F$.*

Proof. Since $W = iH'$, we get

$$\operatorname{ad}(W)E = i[H', E] = 2iE, \qquad \operatorname{ad}(W)F = i[H', F] = -2iF, \qquad \operatorname{ad}(W)H' = 0.$$

By the relation between Ad and ad in Lecture 2, $\operatorname{Ad}(\exp \theta W) = e^{\theta \operatorname{ad} W}$, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. On an eigenvector of $\operatorname{ad} W$ with eigenvalue μ this operator is multiplication by $e^{\theta\mu}$. The eigenvalues 0, $2i$ and $-2i$ of $\operatorname{ad} W$ on H', E and F give the three formulas. ∎

Exercise 21.1 checks the adjoint lemma by matrix multiplication.

Proposition 21.9. *Let V be a $(\mathfrak{g}, K)$-module for $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. If $v \in V(n)$, then*

$$H' \cdot v = nv, \qquad E \cdot v \in V(n+2), \qquad F \cdot v \in V(n-2).$$

Proof. Let $v \in V(n)$. Since $\exp(\theta W) \cdot v = e^{in\theta} v$, the derivative condition (b) with $X = W$ gives

$$W \cdot v = \frac{d}{d\theta}\Big|_{\theta=0} e^{in\theta} v = in\, v, \qquad \text{so} \qquad H' \cdot v = -i\, W \cdot v = nv.$$

By the covariance condition (c), extended to $X \in \mathfrak{g}_{\mathbb{C}}$, and the adjoint lemma, Lemma 21.8,

$$k_\theta \cdot (E \cdot v) = (\operatorname{Ad}(k_\theta)E) \cdot (k_\theta \cdot v) = e^{2i\theta} E \cdot (e^{in\theta} v) = e^{i(n+2)\theta}\, E \cdot v.$$

So $E \cdot v \in V(n+2)$. The argument for F is the same with $e^{-2i\theta}$, and gives $F \cdot v \in V(n-2)$. ∎

In this proof the shift by 2 comes from the covariance condition (c). For this connected group K, the shift also follows from the derivative condition (b) and the bracket $[H', E] = 2E$. Indeed, this bracket and $H' \cdot v = nv$ show that H' multiplies $E \cdot v$ by $n + 2$. The operator H' acts on each summand $V(m)$ of $V = \bigoplus_{m \in \mathbb{Z}} V(m)$ by m, so $E \cdot v \in V(n+2)$. The bracket $[H', F] = -2F$ gives $F \cdot v \in V(n-2)$ in the same way.

Let Ω be the element

$$\Omega = H'^2 + 2EF + 2FE$$

of the universal enveloping algebra $U(\mathfrak{g}_{\mathbb{C}})$. It is the Casimir element of the triple. By the prerequisite Lie Algebras course, it is central in $U(\mathfrak{g}_{\mathbb{C}})$. Substituting $EF = FE + H'$ gives

$$\Omega = H'^2 + 2H' + 4FE, \qquad \Omega \cdot v = n(n+2)v + 4F \cdot E \cdot v \quad (v \in V(n)). \tag{21.2}$$

By the shift proposition, Proposition 21.9, $F \cdot E \cdot v \in V(n)$. So Ω preserves each $V(n)$.

Example 21.10. The group $\mathrm{SL}(2,\mathbb{R})$ acts on $\mathbb{C}^2$ by matrix multiplication, and hence on $V_m = \mathrm{Sym}^m \mathbb{C}^2$, of dimension $m+1$. The orbit maps are polynomial in the matrix entries, so every vector is smooth. Since V_m is finite-dimensional, every vector is also K-finite. Hence V_m is its own $(\mathfrak{g}, K)$-module. The Lie algebra acts on products by the Leibniz rule. For $u_\pm = (1, \pm i)^T$ we have

$$k_\theta u_+ = \begin{pmatrix} \cos\theta + i\sin\theta \\ -\sin\theta + i\cos\theta \end{pmatrix} = e^{i\theta} u_+, \qquad k_\theta u_- = e^{-i\theta} u_-.$$

Direct multiplication (Exercise 21.2) gives $Eu_- = u_+$, $Eu_+ = 0$, $Fu_+ = u_-$ and $Fu_- = 0$. Let $w_j = u_+^j u_-^{m-j}$ for $0 \le j \le m$. The group acts on each factor, so

$$k_\theta w_j = (e^{i\theta} u_+)^j (e^{-i\theta} u_-)^{m-j} = e^{i(2j-m)\theta} w_j.$$

Hence the K-types of V_m are $-m, -m+2, \dots, m$, each with multiplicity one. The Leibniz rule gives

$$E \cdot w_j = (m-j) w_{j+1} \quad (0 \le j < m), \qquad F \cdot w_j = j w_{j-1} \quad (0 < j \le m).$$

The endpoint actions are $Ew_m = 0$ and $Fw_0 = 0$, in agreement with the shift proposition, Proposition 21.9. Since $E \cdot w_m = 0$ and $w_m \in V(m)$, the Casimir formula (21.2) gives $\Omega \cdot w_m = m(m+2) w_m$. The K-types are the familiar $\mathfrak{sl}_2$-weights, read with respect to the compact element H' rather than the diagonal element H.

For $m \ge 1$ no inner product makes this module unitary. Indeed, $\mathrm{diag}(e^t, e^{-t})$ has the eigenvalue e^{mt} on V_m, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. For this reason we allow nonunitary Π in this lecture.

If Π is unitary and Ω acts on $\mathcal{H}_K^\infty$ by a scalar λ, the skew-symmetry (F3) turns the Casimir formula (21.2) into formulas for $\|d\Pi(E)v\|^2$ and $\|d\Pi(F)v\|^2$. Exercise 21.4 derives them. Here $\langle d\Pi(E)u, w\rangle = -\langle u, d\Pi(F)w\rangle$ for smooth vectors u and w, because $F = \overline{E}$. In a unitary representation of $\mathrm{SU}(2)$ the same triple gives the sign +, because $F = E^*$ as matrices. In the physics notation $n = 2j$, the value $n(n+2)$ in (21.2) is $4j(j+1)$.

Exercise 21.3 builds infinite-dimensional $(\mathfrak{g}, K)$-modules from the same raising and lowering pattern. Its module V_λ has a basis v_n, $n \in 2\mathbb{Z}$, with $E \cdot v_n = v_{n+2}$ and $F \cdot v_n = \frac{1}{4}\big(\lambda - (n-2)n\big) v_{n-2}$. Figure 21.1 compares it with the module V_4 of Example 21.10. The shift proposition, Proposition 21.9, says that every arrow joins neighbouring dots of one parity. A missing arrow means that the operator gives zero. In the figure, generic means that $\lambda + 1$ is not the square of an odd integer. Exercise 21.3 shows that V_λ is then irreducible.

21.4 What the $(\mathfrak{g}, K)$-module does not record

A $(\mathfrak{g}, K)$-module has no topology. It does not record the norm or the action of elements of G outside K. We cite theorems that relate it back to representations. None of them is proved or used later in this lecture.

Admissibility. A *linear connected reductive group* is a closed connected group of real or complex matrices that is stable under conjugate transpose (Knapp RT, Section I.1). Examples are

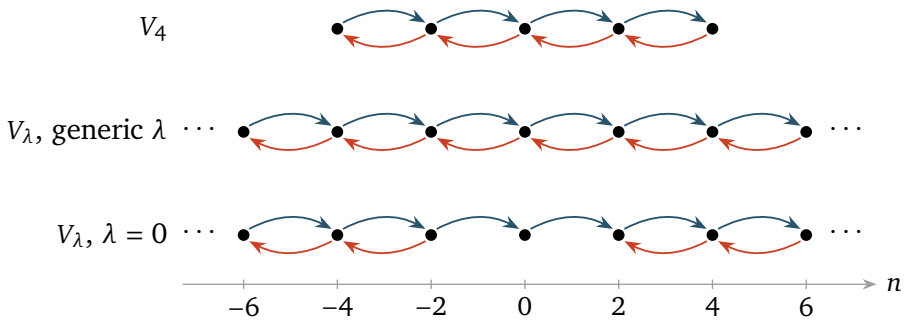


Figure 21.1: Ladders of K-types. Each dot is a one-dimensional space $V(n)$. Upper arrows show E and lower arrows show F. The rows show V_4, the module V_λ of Exercise 21.3 for generic λ, and the same family at $\lambda = 0$, where F vanishes on v_0 and v_2.

$\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(p,q)$. This class is not the class of Lecture 20, as the note at the end of this lecture explains.

A *maximal compact subgroup* of a group is a compact subgroup contained in no strictly larger compact subgroup. For a linear connected reductive group $G \subseteq \mathrm{GL}(n,\mathbb{C})$, the subgroup $K = G \cap \mathrm{U}(n)$ is a maximal compact subgroup (Knapp RT, Section I.1, Proposition 1.2). For $G = \mathrm{SL}(2,\mathbb{R})$ it is $\mathrm{SO}(2)$. If G is also semisimple, then $\theta(X) = -X^*$ is a Cartan involution of $\mathfrak{g}$, and this K is the group K of the Cartan and Iwasawa theorem of Lecture 20.

Let G be a linear connected reductive group and $K = G \cap \mathrm{U}(n)$. The following two statements are in Knapp RT, Chapter VIII, and the note at the end of this lecture gives their numbers.

(i) If a strongly continuous Hilbert representation Π is admissible and $\Pi|_K$ is unitary, then every K-finite vector is smooth, so $\mathcal{H}_K^\infty = \mathcal{H}_K$.

(ii) (Harish-Chandra) Every irreducible unitary representation of G is admissible.

The circle example, Example 21.4, shows the conclusion of the smoothness statement (i) directly in the case $G = K$.

Globalization. Passing from a $(\mathfrak{g}, K)$-module to a continuous group representation is called globalization. A Harish-Chandra module is an admissible $(\mathfrak{g}, K)$-module finitely generated over $U(\mathfrak{g}_\mathbb{C})$. A positive definite Hermitian form on a Harish-Chandra module is invariant if the K-action is unitary and the $\mathfrak{g}$-action is skew-symmetric for real $X \in \mathfrak{g}$.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, an irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization. It is unique up to unitary equivalence. This theorem is cited from P. Etingof's MIT 18.757 notes, Theorem 7.5 and Proposition 7.1. The theorem supplies an action of G on a Hilbert space completing the module, together with its strong continuity. A completion of the vector space alone supplies neither.

One $(\mathfrak{g}, K)$-module can arise from completions of different kinds. Take $G = K = \mathrm{SO}(2)$ as in the circle example, Example 21.4. Translation also acts continuously on the Banach space $C(\mathbb{R}/2\pi\mathbb{Z})$ with the supremum norm, which is not a Hilbert space. With the same definitions, the smooth K-finite vectors of $C(\mathbb{R}/2\pi\mathbb{Z})$ are again the trigonometric polynomials.

So $L^2(\mathbb{R}/2\pi\mathbb{Z})$ and $C(\mathbb{R}/2\pi\mathbb{Z})$ have the same $(\mathfrak{g}, K)$-module. So the module does not record the norm.

This common module has infinitely many distinct circle types and is not finitely generated over $U(\mathfrak{k}_{\mathbb{C}})$. Thus this example is outside the irreducible Harish-Chandra theorem. The more general Casselman–Wallach globalization theorem belongs to further study.

Integration. The identities among $d\Pi(E)$, $d\Pi(F)$ and $d\Pi(H')$ were valid because Π was given. Formal commutation relations among unbounded operators do not by themselves produce a group representation. For the modules of Exercise 21.3, the question of which ones come from unitary representations is taken up in Lecture 22 and Lecture 23.

21.5 Self-checks

Check 1. In Step 3 of the proof of the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6, why must all of W_v consist of smooth vectors?

Answer. The space U is spanned by vectors $d\Pi(X_i)w_j$ with $w_j \in W_v$. These are defined only for smooth w_j. Step 1 shows that each $\Pi(k)v$ is smooth.

Check 2. What are the K-types of Ad on $\mathfrak{g}_{\mathbb{C}}$? Which V_m has the same K-types?

Answer. By the adjoint lemma, Lemma 21.8, the elements E, H' and F have K-types 2, 0, -2, each with multiplicity one. The module V_2 has the same K-types.

Check 3. Could the eigenvalues of $H = \operatorname{diag}(1, -1)$ serve as K-type labels?

Answer. No. The derivative condition (b) relates the K-action only to elements of $\mathfrak{k}_{\mathbb{C}} = \mathbb{C}H'$. Since $H \notin \mathfrak{k}_{\mathbb{C}}$, the action of H is not determined by the K-action. In the V_m example, Example 21.10, the element $H = E + F$ moves $w_j \in V(n)$, where $n = 2j - m$, into $V(n+2) \oplus V(n-2)$.

Summary

Let K be a compact subgroup of G. For a strongly continuous representation of G on a Hilbert space, the smooth K-finite vectors carry an action of $\mathfrak{g}$ and of K. The derivative condition (b) relates the two actions on $\mathfrak{k}$. The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. The proof that $\mathfrak{g}$ preserves K-finiteness uses the covariance condition.

For the circle, the K-finite vectors are the trigonometric polynomials. For $\mathrm{SL}(2, \mathbb{R})$ with $K = \mathrm{SO}(2)$, the covariance condition (c) shows that E and F shift the circle K-types by 2 and -2, respectively. The derivative condition (b) shows that H' acts on the K-type n by multiplication by n. For this connected K, the brackets $[H', E] = 2E$ and $[H', F] = -2F$ then give the same shifts.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, three results are cited. Irreducible unitary representations are admissible. In an admissible strongly continuous Hilbert representation Π with $\Pi|_K$ unitary, K-finite vectors are smooth. An irreducible Harish-Chandra module

with an invariant positive definite Hermitian form has an irreducible unitary globalization, unique up to unitary equivalence.

Sources and hypotheses

The class of groups. Knapp RT, Section I.1, defines linear connected reductive groups, and calls such a group with finite centre linear connected semisimple. Its Proposition 1.2 states that K is compact, connected and maximal compact, and that $K \times \mathfrak{p} \to G$ is a diffeomorphism. This class differs from the class of Lecture 20, which consists of connected semisimple Lie groups with finite centre. For example, the identity component $\mathrm{GL}(n, \mathbb{R})_0$ lies in the present class and is not semisimple, whereas a nonlinear finite cover of $\mathrm{SL}(2, \mathbb{R})$ lies only in the class of Lecture 20. The group $\mathrm{SL}(2, \mathbb{R})$ lies in both.

Admissibility. The smoothness statement (i) is Knapp RT, Proposition 8.5. The definition immediately after the proof of Theorem 8.1 includes the unitarity of $\Pi|_K$ and finite multiplicities, which are the hypotheses of the smoothness statement. The admissibility statement (ii) is Knapp RT, Theorem 8.1.

Globalization. In the notes of Etingof, the unitary globalization is Theorem 7.5, printed p. 38, and its uniqueness is Proposition 7.1, printed p. 37.

Reading

- Knapp RT, Section I.1: linear connected reductive groups.
- Knapp RT, Sections III.3–4: smooth vectors and the facts (F1)–(F3).
- Knapp RT, Chapter VIII: admissible representations and Harish-Chandra's admissibility theorem.
- P. Etingof, MIT 18.757 *Representations of Lie Groups*, lecture notes, Lectures 2 and 5–7: a second account of this material.
- Etingof, Lecture 7, Theorem 7.5 (printed p. 38) and Proposition 7.1 (printed p. 37): the unitary globalization result stated here and its uniqueness. For further study, Wallach, *Real Reductive Groups II*, develops Casselman–Wallach globalization.

Lecture 22

The principal series of $\mathrm{SL}(2,\mathbb{R})$

Objectives. *This lecture builds the principal series of $G = \mathrm{SL}(2,\mathbb{R})$. The principal series is induced from characters of the upper triangular subgroup. We realise it on functions on a circle. We prove unitarity when the parameter is purely imaginary. Then we compute how the Lie algebra moves vectors between K-types. Finally we use this ladder to decide which of these representations are irreducible.*

By the end of this lecture you will be able to:

- State the inducing data (ε, ν) and the role of the modular function.
- Pass between induced functions, homogeneous functions on the plane and the circle model.
- Prove that $\Pi_{\varepsilon,\nu}$ is unitary for $\nu \in i\mathbb{R}$.
- Compute the ladder of K-types, and prove that $\Pi_{\varepsilon,\nu}$ has a proper closed invariant subspace exactly at the points where the ladder breaks.

22.1 Inducing data

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$. We use the matrices

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Let $K = \{k_\theta\}$, $A = \{a_t\}$, $N = \{n_x\}$ and $M = \{\pm I\}$. The group M is the centre of G.

Definition 22.1. The *minimal parabolic subgroup* of G is the group $P = MAN$ of upper triangular matrices in G.

The subgroup P is the stabiliser of the line $\mathbb{R}(1,0)^{\mathsf{T}}$. So G/P is the set of lines through the origin of $\mathbb{R}^2$. This set is a circle, and it is compact. We induce from P for this reason: the functions below live on a circle, and their integrals over it are finite.

Each $p \in P$ has a unique form $p = ma_tn_x$. Its diagonal determines m and t; its upper-right entry then determines x. The subgroup N is normal in P, so $ma_tn_x \mapsto ma_t$ is a homomorphism. Note that $K \cap P = M$.

The tangent algebra of P has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Direct multiplication gives

$$\mathrm{Ad}(a_t)H = H, \qquad \mathrm{Ad}(a_t)E = e^{2t}E, \qquad \mathrm{Ad}(n_x)H = H - 2xE, \qquad \mathrm{Ad}(n_x)E = E,$$

and $\mathrm{Ad}(-I) = \mathrm{id}$. Since Ad and det are multiplicative, these formulas give $|\det \mathrm{Ad}(ma_tn_x)| = e^{2t}$. We use the modular convention of Lecture 16, namely

$$\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx.$$

With this convention $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. Therefore

$$\Delta_P(ma_tn_x) = e^{-2t}. \tag{22.1}$$

Up to the centre, P is the $ax + b$ group of Lecture 16 and Lecture 17 with $a = e^{2t}$, by Exercise 22.2. So this is its modular function $\Delta = 1/a$.

The group G itself is unimodular because it is connected with semisimple Lie algebra, by Lecture 16.

Definition 22.2. Let $\varepsilon \in \{0, 1\}$ and $\nu \in \mathbb{C}$. The *inducing character* $\chi_{\varepsilon,\nu} \colon P \to \mathbb{C}^\times$ is

$$\chi_{\varepsilon,\nu}(ma_tn_x) = \sigma_\varepsilon(m)\, e^{\nu t}, \qquad \sigma_\varepsilon(\pm I) = (\pm 1)^\varepsilon.$$

It is a continuous character, since it factors through $P \to MA$. It is unitary exactly when $\nu \in i\mathbb{R}$.

Definition 22.3. The space $I_{\varepsilon,\nu}$ consists of the continuous $f \colon G \to \mathbb{C}$ with

$$f(gp) = \Delta_P(p)^{1/2} \chi_{\varepsilon,\nu}(p)^{-1} f(g) \qquad (g \in G,\ p \in P).$$

For $g \in G$, the operator $\Pi_{\varepsilon,\nu}(g)$ on $I_{\varepsilon,\nu}$ is given by $(\Pi_{\varepsilon,\nu}(g)f)(h) = f(g^{-1}h)$ for $h \in G$.

By the modular formula (22.1), the defining condition reads

$$f(g\, ma_tn_x) = \sigma_\varepsilon(m)\, e^{-(\nu+1)t} f(g). \tag{22.2}$$

Left and right translations commute, so $\Pi_{\varepsilon,\nu}(g)$ preserves $I_{\varepsilon,\nu}$. Also $\Pi(g_1g_2) = \Pi(g_1)\Pi(g_2)$. We drop subscripts when (ε, ν) is fixed.

Remark 22.4. The factor $\Delta_P^{1/2}$ gives another form of the normalisation in Lecture 17. In Lecture 17 the square root appears in the action.

Let q be the rho-function of Lecture 17. It is a positive continuous function on G with $q(gp) = \Delta_P(p)q(g)$ for $g \in G$ and $p \in P$. Let f satisfy the unitary covariance condition of Lecture 17, and set $F(g) = \sqrt{q(g)}f(g)$. By the transformation law of q, the square root moves into the covariance condition. The action becomes plain left translation. For $\nu \in i\mathbb{R}$ the resulting condition is

$$|F(gp)|^2 = \Delta_P(p)|F(g)|^2.$$

This condition allows integration over G/P. Below we give the circle integral and prove its invariance directly. Nothing else from Lecture 17 is used.

The shift $e^{-t} = \Delta_P(a_t)^{1/2}$ is often written $e^{-\rho(tH)}$. Here ρ is the linear functional on $\mathbb{R}H$ with $\rho(H) = 1$, half the eigenvalue 2 of $\mathrm{ad}\, H$ on E. It is the analogue of the ρ of Lecture 9.

22.2 The plane model and the circle model

Let $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$ and $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}} = k_\theta \mathbf{e}_1$. Multiplying g on the right by n_x leaves the first column of g unchanged and adds a multiple of it to the second column. So a function with $f(gn_x) = f(g)$ depends only on the first column $g\mathbf{e}_1$, which can be any nonzero vector.

Lemma 22.5. *The map $g \mapsto g\mathbf{e}_1$ induces a homeomorphism $G/N \to \mathbb{R}^2 \setminus \{0\}$.*

Proof. If $g\mathbf{e}_1 = \mathbf{e}_1$, the first column of g is $\mathbf{e}_1$, and $\det g = 1$ forces $g \in N$. So the fibres of $g \mapsto g\mathbf{e}_1$ are the cosets gN. The map is onto, with continuous section

$$s(v) = \begin{pmatrix} v_1 & -v_2/|v|^2 \\ v_2 & v_1/|v|^2 \end{pmatrix}, \qquad s(v)\mathbf{e}_1 = v, \qquad \det s(v) = 1.$$

Thus $v \mapsto s(v)N$ is a continuous inverse. ∎

Proposition 22.6. *The rule $F(g\mathbf{e}_1) = f(g)$ is a bijection from $I_{\varepsilon,\nu}$ onto the continuous functions F on $\mathbb{R}^2 \setminus \{0\}$ satisfying*

$$F(cv) = \operatorname{sgn}(c)^{\varepsilon}\,|c|^{-(\nu+1)} F(v) \qquad (c \in \mathbb{R}^\times). \tag{22.3}$$

Under it, $\Pi_{\varepsilon,\nu}(g)$ becomes $F \mapsto F(g^{-1}\cdot)$.

Proof. Taking $p = n_x$ in the covariance law (22.2) shows $f(gn_x) = f(g)$. By the plane identification (Lemma 22.5), F is well defined, and $F = f \circ s$ is continuous. For $c = \pm e^t$ we have $g(\pm a_t)\mathbf{e}_1 = c\,g\mathbf{e}_1$. So the covariance law with $p = \pm a_t$ gives the homogeneity rule (22.3).

Conversely, given F, put $f(g) = F(g\mathbf{e}_1)$. Since $g\,m a_t n_x \mathbf{e}_1 = \pm e^t g\mathbf{e}_1$ for $m = \pm I$, the homogeneity rule gives the covariance law. Finally, $(\Pi(g)f)(h) = f(g^{-1}h) = F(g^{-1}h\mathbf{e}_1)$ for $h \in G$. So $\Pi(g)$ becomes $F \mapsto F(g^{-1}\cdot)$. ∎

A homogeneous function F is determined by its values on the unit circle. Put $\varphi(\theta) = F(u_\theta) = f(k_\theta)$. Then

$$F(ru_\theta) = r^{-(\nu+1)}\varphi(\theta) \quad (r > 0), \qquad \varphi(\theta+\pi) = (-1)^{\varepsilon}\varphi(\theta).$$

The second rule is the homogeneity rule (22.3) with $c = -1$. Let C_ε be the space of continuous 2π-periodic functions with this parity. For every ν, restriction to K identifies $I_{\varepsilon,\nu}$ with C_ε. This identification is the *compact picture*. In the compact picture the space C_ε does not depend on ν.

Restriction to K loses no data because of the Iwasawa decomposition of Lecture 20. It gives $G = KAN$, so $G = KP$. Every $g \in G$ has the form kp, and the defining condition gives $f(kp) = \Delta_P(p)^{1/2}\chi_{\varepsilon,\nu}(p)^{-1} f(k)$. So f is determined by its values on K.

Definition 22.7. For $h \in G$, the map τ_h of $\mathbb{R}/2\pi\mathbb{Z}$ is defined by $hu_\theta = |hu_\theta|\,u_{\tau_h(\theta)}$.

Apply the plane model (Proposition 22.6) with $g^{-1}u_\theta = |g^{-1}u_\theta|\,u_{\tau_{g^{-1}}(\theta)}$. This gives the circle model:

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\,\varphi\big(\tau_{g^{-1}}(\theta)\big). \tag{22.4}$$

The parameter ν enters only through the scalar factor.

Figure 22.1 draws τ_h for $h = a_{1/2}$. The matrix h turns the unit circle into an ellipse, and each point of the ellipse is projected back to the circle along its ray. Directions crowd towards the horizontal axis.

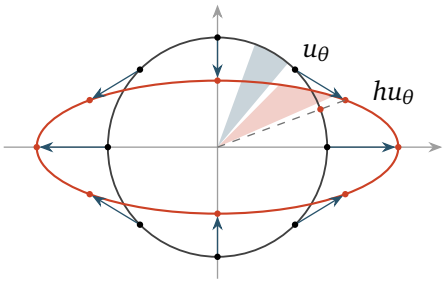


Figure 22.1: The map τ_h for $h = a_{1/2}$. Black dots are the points u_θ at multiples of $\pi/4$, and arrows lead to their images on the ellipse. The dashed ray projects $hu_{\pi/4}$ back to the circle at the angle $\tau_h(\pi/4)$. The thin sector of the disk and its image in the ellipse have equal area.

Example 22.8. For rotations $k_\alpha^{-1} u_\theta = u_{\theta-\alpha}$, so

$$(\Pi(k_\alpha)\varphi)(\theta) = \varphi(\theta - \alpha), \qquad \Pi(-I)\varphi = (-1)^\varepsilon \varphi.$$

The second formula uses $-I = k_\pi$. Thus the centre M acts by σ_ε for every ν. We call ε the central parity.

For the diagonal subgroup, $a_s^{-1} u_\theta = (e^{-s}\cos\theta,\ e^{s}\sin\theta)^{\mathsf{T}}$. Hence

$$r_s(\theta)^2 := |a_s^{-1} u_\theta|^2 = e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta, \qquad \tan \tau_{a_s^{-1}}(\theta) = e^{2s}\tan\theta.$$

When $\varepsilon = 0$ the constant function $e_0 = 1$ lies in C_0, and $\Pi_{0,\nu}(a_s)e_0 = r_s^{-(\nu+1)}$.

22.3 Unitarity on the imaginary axis

Let $\mathcal{H}_\varepsilon$ be the closure of C_ε in $L^2(\mathbb{R}/2\pi\mathbb{Z})$, with inner product

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \overline{\varphi(\theta)}\, \psi(\theta)\, d\theta.$$

By Fourier analysis on the circle (Lecture 10), $\mathcal{H}_\varepsilon$ is the closed span of the functions $e_n(\theta) = e^{in\theta}$ with $n \equiv \varepsilon \pmod 2$. The key step towards unitarity is a change of variables on the circle.

Lemma 22.9. *Each τ_h is an orientation-preserving diffeomorphism. Moreover $\tau_{h_1 h_2} = \tau_{h_1} \circ \tau_{h_2}$ and*

$$\tau_h'(\theta) = |hu_\theta|^{-2}.$$

Proof. Linear maps commute with positive scaling, so $h_1 h_2 u_\theta$ is a positive multiple of $h_1 u_{\tau_{h_2}(\theta)}$. This gives the composition law, and $\tau_{h^{-1}}$ inverts τ_h. Locally $\tau_h(\theta)$ is a smooth branch of the argument of the nonvanishing smooth curve hu_θ. So τ_h is smooth.

For the derivative, write $r = |hu_\theta|$ and $\tau = \tau_h(\theta)$. Differentiate $hu_\theta = r\,u_\tau$ in θ:

$$hu'_\theta = r'u_\tau + r\tau' u'_\tau, \qquad u'_\theta = (-\sin\theta, \cos\theta)^{\mathsf{T}}.$$

Now compute the determinant of the pair (hu_θ, hu'_θ) in two ways. By multiplicativity of the determinant it equals $\det h \cdot \det(u_\theta, u'_\theta) = 1$. By the first display it also equals

$$\det(ru_\tau,\ r'u_\tau + r\tau' u'_\tau) = r^2\tau' \det(u_\tau, u'_\tau) = r^2\tau'.$$

Hence $\tau' = r^{-2} > 0$. The inverse $\tau_{h^{-1}}$ is smooth by the same argument. ∎

Figure 22.1 shows the reason. A thin sector of angle $d\theta$ in the unit disk has area $\frac{1}{2}d\theta$. Its image under h is, to first order, a sector of radius r and angle $d\tau$, of area $\frac{1}{2}r^2 d\tau$. Since $\det h = 1$, the matrix h preserves area, so $d\theta = r^2 d\tau$.

As a check, take $h = a_s^{-1}$ and differentiate $\tan\tau = e^{2s}\tan\theta$ from the diagonal example (Example 22.8):

$$\tau' = \frac{e^{2s}\sec^2\theta}{1 + e^{4s}\tan^2\theta} = \frac{e^{2s}}{\cos^2\theta + e^{4s}\sin^2\theta} = \frac{1}{e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta} = \frac{1}{r_s(\theta)^2}.$$

Proposition 22.10. *For $\nu \in \mathbb{C}$, $\varphi, \psi \in C_\varepsilon$ and $g \in G$,*

$$\langle \Pi_{\varepsilon,-\bar\nu}(g)\varphi,\ \Pi_{\varepsilon,\nu}(g)\psi\rangle = \langle \varphi, \psi\rangle.$$

Proof. Put $h = g^{-1}$, $r(\theta) = |hu_\theta|$ and $\theta' = \tau_h(\theta)$. Since $r > 0$, we have $\overline{r^{-(-\bar\nu+1)}} = r^{\nu-1}$. By the circle-model formula (22.4), the integrand is

$$\overline{(\Pi_{\varepsilon,-\bar\nu}(g)\varphi)(\theta)}\,(\Pi_{\varepsilon,\nu}(g)\psi)(\theta) = r^{\nu-1}r^{-\nu-1}\,\overline{\varphi(\theta')}\psi(\theta') = \overline{\varphi(\theta')}\psi(\theta')\,r(\theta)^{-2}.$$

By the Jacobian lemma (Lemma 22.9), $r(\theta)^{-2}\,d\theta = d\theta'$. As θ runs once around the circle, so does θ', in the same direction. So the integral equals $\int_0^{2\pi} \overline{\varphi(\theta')}\psi(\theta')\,d\theta'$. After division by 2π, this is $\langle\varphi,\psi\rangle$. ∎

The Jacobian lemma rewrites the circle formula. With $\tau = \tau_{g^{-1}}$ we have $|g^{-1}u_\theta|^{-2} = \tau'(\theta)$, so

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = \tau'(\theta)^{(\nu+1)/2}\,\varphi(\tau(\theta)).$$

For $\nu = 0$ the factor is the square root of the Jacobian, and it keeps $|\varphi|^2 d\theta$ fixed when the circle is stretched. This is the square root of Lecture 17. For $\nu \in i\mathbb{R}$ the factor gains a phase of modulus one. For $\nu = -1$ the factor is 1, so φ transforms as a function. For $\nu = 1$ the factor is the Jacobian, so φ transforms as a density and its integral is invariant. Without $\Delta_P^{1/2}$, the two powers of r in the proof above would multiply to 1, and the Jacobian would be left over.

Theorem 22.11. *Let $\nu \in i\mathbb{R}$. Each $\Pi_{\varepsilon,\nu}(g)$ extends uniquely from C_ε to a unitary operator on $\mathcal{H}_\varepsilon$. The map $g \mapsto \Pi_{\varepsilon,\nu}(g)$ is a strongly continuous unitary representation of G.*

Proof. For $\nu \in i\mathbb{R}$ we have $-\bar{\nu} = \nu$. So the pairing identity (Proposition 22.10) says that $\Pi(g)$ preserves norms on C_ε. As C_ε is dense, $\Pi(g)$ extends uniquely to an isometry of $\mathcal{H}_\varepsilon$.

The relation $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$ holds on C_ε by the plane model (Proposition 22.6). By continuity it holds on $\mathcal{H}_\varepsilon$. So $\Pi(g^{-1})$ inverts $\Pi(g)$. An invertible isometry is unitary.

For strong continuity, fix $\varphi \in C_\varepsilon$ and $g_0 \in G$. By the circle-model formula (22.4), the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is continuous. It is uniformly continuous on $U \times [0, 2\pi]$ for a compact neighbourhood U of g_0. Hence $\sup_\theta |(\Pi(g)\varphi - \Pi(g_0)\varphi)(\theta)| \to 0$ as $g \to g_0$. The L^2 norm is at most this supremum.

For general $\psi \in \mathcal{H}_\varepsilon$ and $\delta > 0$, choose $\varphi \in C_\varepsilon$ with $\|\psi - \varphi\| < \delta$. Since every $\Pi(g)$ is unitary,

$$\|\Pi(g)\psi - \Pi(g_0)\psi\| \le 2\delta + \|\Pi(g)\varphi - \Pi(g_0)\varphi\|,$$

and the right side is below 3δ for g near g_0. ∎

Example 22.12. Unitarity can fail off the imaginary axis. For example, take $\varepsilon = 0$ and $\nu \in \mathbb{R}$. By the diagonal example (Example 22.8),

$$\|\Pi_{0,\nu}(a_s)e_0\|^2 = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^{\nu+1}}, \qquad A = e^{-2s},\ B = e^{2s}.$$

For $A, B > 0$, the substitution $u = \tan\theta$ on each quarter period gives

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = 4\int_0^{\infty} \frac{du}{A + Bu^2} = \frac{2\pi}{\sqrt{AB}}.$$

Applying $-\partial_A - \partial_B$ to this identity gives

$$\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^2} = \frac{\pi(A+B)}{(AB)^{3/2}}.$$

Here $AB = 1$. For $\nu = 0$ the norm is 1, as the unitarity theorem (Theorem 22.11) predicts. For $\nu = 1$ the squared norm is $\frac{1}{2}(e^{2s} + e^{-2s}) = \cosh 2s$, which exceeds 1 for $s \neq 0$. Exercise 22.6 gives the general skew-symmetry test.

This shows only that the L^2 norm is not preserved. For $\varepsilon = 0$ and $0 < |\nu| < 1$, Lecture 23 uses a different inner product, which is preserved.

For general $\nu \in \mathbb{C}$ the substitution in the proof of the pairing identity (Proposition 22.10) still gives

$$\|\Pi_{\varepsilon,\nu}(g)\varphi\|^2 \le \max_\theta |g^{-1}u_\theta|^{-2\operatorname{Re}\nu}\,\|\varphi\|^2.$$

So each $\Pi_{\varepsilon,\nu}(g)$ is bounded, locally uniformly in g. The proof of the unitarity theorem (Theorem 22.11) then gives a strongly continuous representation on $\mathcal{H}_\varepsilon$. In its density step, 2δ is replaced by $2C\delta$, where C bounds $\|\Pi_{\varepsilon,\nu}(g)\|$ on a compact neighbourhood of g_0. By the pairing identity (Proposition 22.10), the adjoint is $\Pi_{\varepsilon,\nu}(g)^* = \Pi_{\varepsilon,-\bar{\nu}}(g^{-1})$.

22.4 K-types and the Lie algebra action

By the rotation example (Example 22.8), $\Pi(k_\alpha)e_n = e^{-in\alpha}e_n$. So the character $k_\alpha \mapsto e^{-in\alpha}$ of K occurs exactly once when $n \equiv \varepsilon \pmod 2$, and not at all otherwise. This matches the centre: $k_\pi = -I$ acts on e_n by $(-1)^n = (-1)^\varepsilon$. Let

$$V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\} \qquad \text{(finite sums)}.$$

For $\nu \in i\mathbb{R}$ this is the space of K-finite vectors of Lecture 21.

For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ put

$$d\Pi(X)\varphi = \frac{d}{ds}\Big|_{s=0} \Pi_{\varepsilon,\nu}(\exp sX)\varphi.$$

By the circle-model formula (22.4), the function $(s,\theta) \mapsto (\Pi(\exp sX)\varphi)(\theta)$ is smooth. So the difference quotients converge uniformly in θ. Hence they converge in $\mathcal{H}_\varepsilon$.

More generally, the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is jointly smooth. On compact sets in a group chart, the difference quotients for its derivatives converge uniformly in θ. Hence φ is a smooth vector in the Hilbert representation, in the sense used in Lecture 21.

For $\nu \in i\mathbb{R}$ this derivative agrees with the derivative on smooth vectors in Lecture 15. We extend $d\Pi$ complex linearly to $\mathfrak{g}_{\mathbb{C}}$.

Lemma 22.13. *For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g}$, using the dot product of $\mathbb{R}^2$,*

$$(d\Pi(X)\varphi)(\theta) = (\nu+1)\,(u_\theta \cdot Xu_\theta)\,\varphi(\theta) - \det(u_\theta, Xu_\theta)\,\varphi'(\theta).$$

Proof. Let $v(s) = \exp(-sX)u_\theta = r(s)u_{\tau(s)}$. Then $r(0)=1$, $\tau(0)=\theta$ and $v'(0) = -Xu_\theta$. As in the proof of the Jacobian lemma (Lemma 22.9), $\det(v,v') = r^2\tau'$, so $\tau'(0) = -\det(u_\theta, Xu_\theta)$. Also $(r^2)' = 2\,v\cdot v'$, so $r'(0) = -u_\theta \cdot Xu_\theta$. Now differentiate $r(s)^{-(\nu+1)}\varphi(\tau(s))$ at $s=0$. The product rule with these values of $r'(0)$ and $\tau'(0)$ gives the stated formula. ∎

We use the basis

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with $\exp(\alpha W) = k_\alpha$. We have $Wu_\theta = u_\theta'$, $Hu_\theta = (\cos\theta, -\sin\theta)^{\mathsf T}$ and $Qu_\theta = (\sin\theta, \cos\theta)^{\mathsf T}$. The derivative formula (Lemma 22.13) gives the table:

X	$u_\theta \cdot Xu_\theta$	$\det(u_\theta, Xu_\theta)$	$d\Pi(X)\varphi$
W	0	1	$-\varphi'$
H	$\cos 2\theta$	$-\sin 2\theta$	$(\nu+1)\cos 2\theta\,\varphi + \sin 2\theta\,\varphi'$
Q	$\sin 2\theta$	$\cos 2\theta$	$(\nu+1)\sin 2\theta\,\varphi - \cos 2\theta\,\varphi'$

Set $X_\pm = H \pm iQ \in \mathfrak{g}_{\mathbb{C}}$. Since $\sin 2\theta \mp i\cos 2\theta = \mp i e^{\pm 2i\theta}$,

$$d\Pi(X_\pm)\varphi = e^{\pm 2i\theta}\big((\nu+1)\varphi \mp i\varphi'\big).$$

The element W used here is the negative of the W in Lecture 21, so k_α here is $k_{-\alpha}$ there. In the notation of Lecture 21, $E = X_+/2$, $F = X_-/2$ and $H' = iW$. Thus the Fourier vector e_n has type n in that lecture's parametrisation, even though its character here is $e^{-in\alpha}$. These E and F are not the real matrices of § 22.1, where E is upper triangular.

Proposition 22.14. *For $n \equiv \varepsilon \pmod 2$,*

$$d\Pi(W)e_n = -in\,e_n, \qquad d\Pi(X_+)e_n = (\nu+1+n)\,e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu+1-n)\,e_{n-2}.$$

In the letters of Lecture 21, $d\Pi(E)e_n = \tfrac{1}{2}(\nu+1+n)e_{n+2}$ and $d\Pi(F)e_n = \tfrac{1}{2}(\nu+1-n)e_{n-2}$.

Proof. Insert $e'_n = in\,e_n$ into the table row for W and into the display for $d\Pi(X_\pm)$. For example, $(\nu+1) - i(in) = \nu+1+n$. ∎

The ladder formulas also show that $d\Pi$ respects brackets, for every $\nu \in \mathbb{C}$. On e_n the eigenvalues of $d\Pi(W)$ give

$$[d\Pi(W), d\Pi(X_\pm)]e_n = \big(-i(n\pm 2) + in\big)(\nu+1\pm n)\,e_{n\pm2} = \mp 2i\,d\Pi(X_\pm)e_n.$$

The products of the two ladder operators are

$$d\Pi(X_\mp)d\Pi(X_\pm)e_n = (\nu+1\pm n)(\nu-1\mp n)\,e_n = \big(\nu^2 - (1\pm n)^2\big)e_n.$$

Subtracting the two cases gives

$$[d\Pi(X_+), d\Pi(X_-)]e_n = \big((1+n)^2 - (1-n)^2\big)e_n = 4n\,e_n = 4i\,d\Pi(W)e_n.$$

In $\mathfrak{g}_{\mathbb{C}}$ the brackets $[W,H] = 2Q$, $[W,Q] = -2H$ and $[H,Q] = -2W$ give $[W, X_\pm] = \mp 2iX_\pm$ and $[X_+, X_-] = 4iW$ (Exercise 22.4). The elements W, X_+ and X_- span $\mathfrak{g}_{\mathbb{C}}$, and $d\Pi$ is complex linear. Hence $d\Pi([Y,Z]) = [d\Pi(Y), d\Pi(Z)]$ for all $Y, Z \in \mathfrak{g}_{\mathbb{C}}$. So $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $V_{\varepsilon,\nu}$ for every $\nu \in \mathbb{C}$.

Proposition 22.15. *A subspace $U \subseteq V_{\varepsilon,\nu}$ is stable under $\Pi(K)$ and $d\Pi(\mathfrak{g})$ exactly when $U = \operatorname{span}\{e_n : n \in S\}$ for a set S such that the following two conditions hold. If $n \in S$ and $\nu+1+n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu+1-n \neq 0$, then $n-2 \in S$. Hence the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is irreducible if and only if ν is not an integer with $\nu \equiv \varepsilon + 1 \pmod 2$.*

Proof. Let U be K-stable and let $u = \sum c_n e_n \in U$ be a finite sum. The e_n with $c_n \neq 0$ span a finite-dimensional K-stable space F. The continuous curve $\alpha \mapsto e^{in\alpha}\Pi(k_\alpha)u$ lies in the closed subspace $U \cap F$ of F. Hence

$$\frac{1}{2\pi}\int_0^{2\pi} e^{in\alpha}\,\Pi(k_\alpha)u\,d\alpha = c_n e_n$$

lies in U. Thus U is spanned by the e_n it contains. Since H and Q are linear combinations of $X_\pm$, and $d\Pi(W)$ preserves every span of the e_n, stability under $d\Pi(\mathfrak{g})$ means stability under $d\Pi(X_\pm)$. By the ladder formulas (Proposition 22.14), this is the stated property of S.

If no coefficient $\nu+1\pm n$ vanishes, every nonempty admissible S contains all $n \equiv \varepsilon$. If $\nu+1+n_0 = 0$, then $S = \{n \le n_0\}$ is admissible and proper. If $\nu+1-n_0 = 0$, use $S = \{n \ge n_0\}$. Finally, $\nu = -(n_0+1)$ or $\nu = n_0 - 1$ with $n_0 \equiv \varepsilon$ means exactly that $\nu \in \mathbb{Z}$ and $\nu \equiv \varepsilon + 1 \pmod 2$. ∎

Figure 22.2 shows the criterion at four parameters.

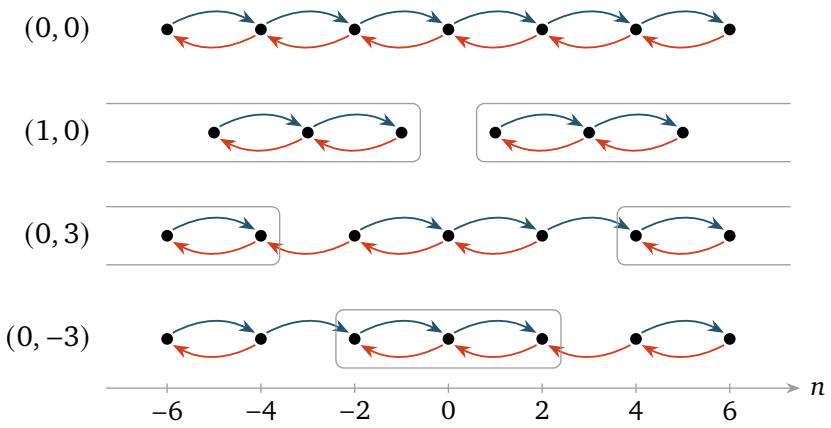


Figure 22.2: The ladder of $V_{\varepsilon,\nu}$, with rows labelled (ε, ν). Each dot is the one-dimensional space $\mathbb{C}e_n$; the rows continue in both directions. Upper arrows show $d\Pi(X_+)$ and lower arrows show $d\Pi(X_-)$. An arrow is missing where its coefficient vanishes. An invariant set of dots is one that no arrow leaves, and the boxes mark such sets.

22.5 Closed invariant subspaces

We now pass from the module $V_{\varepsilon,\nu}$ to the Hilbert representation on $\mathcal{H}_\varepsilon$. A $(\mathfrak{g}, K)$-submodule is an algebraic object and a closed invariant subspace is an analytic one, so the submodule criterion does not settle the question alone. In general the passage between them is Harish-Chandra's correspondence for admissible representations. For this family we prove what we need directly. First, a closed invariant subspace contains Fourier vectors. Next, a half-ladder has an invariant closed span.

Lemma 22.16. *Let $Y \neq 0$ be a closed subspace of $\mathcal{H}_\varepsilon$ invariant under $\Pi_{\varepsilon,\nu}(G)$. If $v \in Y$ and $\langle e_k, v\rangle \neq 0$, then $e_k \in Y$. If $e_k \in Y$, then $d\Pi(X_\pm)e_k \in Y$.*

Proof. By the rotation example (Example 22.8), the Fourier coefficient of $\Pi(k_\alpha)v$ at m is $e^{-im\alpha}\langle e_m, v\rangle$. So the integral

$$\frac{1}{2\pi}\int_0^{2\pi} e^{ik\alpha}\,\Pi(k_\alpha)v\,d\alpha = \langle e_k, v\rangle e_k$$

has Fourier coefficient $\langle e_k, v\rangle$ at k and 0 elsewhere. It is a norm limit of Riemann sums of vectors in Y, because the integrand is continuous. Since Y is closed, $e_k \in Y$.

The vector e_k is smooth, as shown before the derivative formula (Lemma 22.13). For $X \in \mathfrak{g}$, the quotient $t^{-1}(\Pi(\exp tX)e_k - e_k)$ lies in Y and converges in norm to $d\Pi(X)e_k$. So $d\Pi(X)e_k \in Y$. Complex combinations give $d\Pi(X_\pm)e_k \in Y$. ∎

Let $n \geq 1$ be an integer and let $\varepsilon \equiv n \pmod 2$. At $\nu = n - 1$ the ladder shows the half-ladder $k \geq n$. We define the closed span

$$H_n^+ = \overline{\operatorname{span}}\{e_{n+2m} : m \geq 0\} \subset \mathcal{H}_\varepsilon,$$

and likewise $H_n^- = \overline{\mathrm{span}}\{e_{-n-2m} : m \geq 0\}$. The ladder alone does not show that H_n^+ is invariant under the group, since $\Pi(g)$ is not a polynomial in $d\Pi$. We use power series in $z = e^{2i\theta}$. For a polynomial f put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = e_{n+2m}.$$

We use polynomials because a general vector of H_n^+ need not extend to a function holomorphic near the closed unit disk.

We also fix the complex matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$. For $g \in G$ with $g^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiplying out gives

$$\gamma g^{-1} \gamma^{-1} = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{22.5}$$

The determinant gives $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. So $|\beta| < |\alpha|$.

Proposition 22.17. *Let $n \geq 1$, $\varepsilon \equiv n \pmod 2$, $g \in G$, and let f be a polynomial. With $z = e^{2i\theta}$ and α, β as in (22.5),*

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = e^{in\theta}\,(\bar\beta z + \bar\alpha)^{-n}\, f\Big(\frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}\Big). \tag{22.6}$$

Proof. For a real vector v we have $\gamma v = (v_1 + iv_2,\ v_1 - iv_2)^{\mathsf T}$. Hence $\gamma u_\theta = e^{-i\theta}(z, 1)^{\mathsf T}$. Write $g^{-1} u_\theta = r u_\tau$, with $r = |g^{-1} u_\theta|$ and $\tau = \tau_{g^{-1}}(\theta)$. Apply γ to both sides. Since $\gamma g^{-1} = (\gamma g^{-1} \gamma^{-1})\gamma$, this gives

$$r e^{-i\tau} \begin{pmatrix} e^{2i\tau} \\ 1 \end{pmatrix} = e^{-i\theta} \begin{pmatrix} \alpha z + \beta \\ \bar\beta z + \bar\alpha \end{pmatrix}.$$

The second entries give $\bar\beta z + \bar\alpha = r e^{i(\theta - \tau)}$. Dividing the first entries by the second gives $e^{2i\tau} = (\alpha z + \beta)/(\bar\beta z + \bar\alpha)$. Now the circle-model formula (22.4) with $\nu + 1 = n$ gives

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = r^{-n} e^{in\tau} f(e^{2i\tau}) = e^{in\theta}\big(r e^{i(\theta-\tau)}\big)^{-n} f(e^{2i\tau}),$$

which is the stated formula. ∎

Proposition 22.18. *Let $n \geq 1$ and $\varepsilon \equiv n \pmod 2$. The subspaces H_n^+ and H_n^- are invariant under $\Pi_{\varepsilon,n-1}$. Their orthogonal complements in $\mathcal{H}_\varepsilon$ are invariant under $\Pi_{\varepsilon,-(n-1)}$.*

Proof. The half H_n^+. Let $g \in G$, and let f be a polynomial. Since $|\beta/\alpha| < 1$, the geometric series

$$(\bar\beta w + \bar\alpha)^{-1} = \bar\alpha^{-1} \sum_{l \geq 0} (-\bar\beta w/\bar\alpha)^l$$

converges absolutely and uniformly on $|w| \leq 1$, and so do its powers. Therefore the function of w obtained from the right side of (22.6) has a power series $\sum_{l \geq 0} b_l w^l$ that converges uniformly on $|w| = 1$. Setting $w = e^{2i\theta}$, we see that $\Pi_{\varepsilon,n-1}(g) U_n f$ is a uniform limit, hence an L^2 limit, of finite combinations of the e_{n+2l}. So it lies in H_n^+. The vectors $U_n f$ span a dense subspace of H_n^+, and $\Pi_{\varepsilon,n-1}(g)$ is bounded. Hence $\Pi_{\varepsilon,n-1}(g) H_n^+ \subset H_n^+$ for every g.

The half H_n^-. For real ν the factor in (22.4) is real, so complex conjugation commutes with $\Pi_{\varepsilon,\nu}(g)$. Conjugation carries H_n^+ onto H_n^-. So H_n^- is invariant at $\nu = n - 1$.

The complements. By the adjoint formula after the pairing identity (Proposition 22.10), $\Pi_{\varepsilon,-(n-1)}(g)^* = \Pi_{\varepsilon,n-1}(g^{-1})$. Let $v \perp H_n^{\pm}$ and $u \in H_n^{\pm}$. Then

$$\langle u, \Pi_{\varepsilon,-(n-1)}(g)v\rangle = \langle \Pi_{\varepsilon,n-1}(g^{-1})u, v\rangle = 0.$$

■

Theorem 22.19 (Irreducibility; the two equivalences are cited). *Let $\nu \in \mathbb{C}$. The representation $\Pi_{\varepsilon,\nu}$ on $\mathcal{H}_\varepsilon$ has no closed invariant subspace other than 0 and $\mathcal{H}_\varepsilon$ exactly when $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$. For $\nu \in i\mathbb{R}$, all $\Pi_{\varepsilon,\nu}$ are irreducible except $\Pi_{1,0}$. The space $\mathcal{H}_1$ of $\Pi_{1,0}$ is the orthogonal sum of the closed spans of $\{e_n : n > 0\}$ and $\{e_n : n < 0\}$, and each summand is irreducible.*
Moreover, for $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$ we have $V_{\varepsilon,\nu} \cong V_{\varepsilon,-\nu}$ as $(\mathfrak{g}, K)$-modules. For $\nu \in i\mathbb{R}$, the representations $\Pi_{\varepsilon,\nu}$ and $\Pi_{\varepsilon,-\nu}$ are unitarily equivalent. These two equivalences are cited from Knapp, Representation Theory of Semisimple Groups*, §II.5, and Etingof's 18.757 notes, Lecture 19.*

Proof. We prove the first paragraph.

Irreducibility. Let $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$, and let $W \neq 0$ be a closed invariant subspace. A nonzero vector of W has a nonzero Fourier coefficient, so $e_k \in W$ for some k, by Lemma 22.16. No coefficient $\nu + 1 \pm n$ with $n \equiv \varepsilon$ vanishes. So the ladder formulas (Proposition 22.14) and Lemma 22.16 put every e_n with $n \equiv \varepsilon$ in W. Hence $W = \mathcal{H}_\varepsilon$.

Reducibility. Let $\nu \in \varepsilon + 1 + 2\mathbb{Z}$. Then $\nu = n - 1$ or $\nu = -(n-1)$ for an integer $n \geq 1$ with $n \equiv \varepsilon \pmod 2$. By Proposition 22.18, H_n^+ or its orthogonal complement is a closed invariant subspace in the two cases. Each is nonzero and proper.

The imaginary axis. A number $\nu \in i\mathbb{R}$ lies in $\varepsilon + 1 + 2\mathbb{Z}$ only for $\nu = 0$ and $\varepsilon = 1$. So the exception is $\Pi_{1,0}$.

The point $(1,0)$. Take $n = 1$. Then H_1^+ and H_1^- are invariant, mutually orthogonal, and together they span $\mathcal{H}_1$. Let $W \neq 0$ be a closed invariant subspace of H_1^+. A nonzero vector of W has a nonzero coefficient at some odd $k \geq 1$, so $e_k \in W$. For odd $n \geq 1$ we have $1 + n \neq 0$, and for odd $n \geq 3$ we have $1 - n \neq 0$. So the ladder puts every e_n with odd $n \geq 1$ in W. Hence $W = H_1^+$. Complex conjugation gives the same result for H_1^-. ■

The pieces at the reducibility points return in Lecture 23.

The reducible module at $(\varepsilon, \nu) = (1, 0)$

The simplest reducibility point is $(\varepsilon, \nu) = (1, 0)$. The Fourier labels are the odd integers, and the ladder formulas become

$$d\Pi(X_+)e_n = (1+n)e_{n+2}, \qquad d\Pi(X_-)e_n = (1-n)e_{n-2}.$$

At $n = -1$, the upward coefficient vanishes. At $n = 1$, the downward coefficient vanishes. Thus the space $V_{1,0}$ is the direct sum

$$V_{1,0} = \operatorname{span}\{e_1, e_3, e_5, \ldots\} \;\oplus\; \operatorname{span}\{e_{-1}, e_{-3}, e_{-5}, \ldots\}.$$

Each summand is preserved by both ladder operators and by K. Within either chain, the remaining connecting coefficients are nonzero. The weight-projection argument in the

proof of the submodule criterion (Proposition 22.15) shows that each chain is algebraically irreducible.

The irreducibility theorem (Theorem 22.19) gives the two closed irreducible summands. Their group invariance is Proposition 22.18, which goes beyond the algebraic calculation above. Lecture 23 identifies them as limits of discrete series and explains their norms. The full odd ladder breaks at the gap between -1 and 1.

For comparison, $(\varepsilon,\nu)=(0,0)$ has only even labels. Neither equation $1+n=0$ nor $1-n=0$ has an even solution. Its algebraic module is therefore irreducible.

We compare the two modules through the Casimir element $\Omega = H^2+Q^2-W^2$ of $U(\mathfrak{g}_{\mathbb{C}})$. Exercise 22.4 shows $\Omega = X_+X_- - 2iW - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Since $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$, it extends to $U(\mathfrak{g}_{\mathbb{C}})$, and this identity holds for the operators. By the ladder formulas (Proposition 22.14), the three terms act on e_n by $\nu^2-(1-n)^2$, $-2n$ and n^2. Hence Ω acts on $V_{\varepsilon,\nu}$ by the scalar

$$\nu^2-(1-n)^2-2n+n^2=\nu^2-1.$$

By the dictionary of Lecture 21 above, $2EF+2FE=H^2+Q^2$ and $H'^2=-W^2$. So this Ω is the Casimir element $H'^2+2EF+2FE$ of Lecture 21.

Lecture 21 found $\Omega = m(m+2)$ on V_m. This equals ν^2-1 at $\nu=\pm(m+1)$. Indeed, take $\nu=-(m+1)$ with ε of the parity of m. For a function F on the plane, the homogeneity rule (22.3) then reads $F(cv)=c^mF(v)$. The homogeneous polynomials of degree m on $\mathbb{R}^2$ form a finite-dimensional invariant space. On the circle they span the e_n with $|n|\le m$ and $n\equiv m \pmod 2$.

At $(\varepsilon,\nu)=(1,0)$ and at $(0,0)$ this scalar is -1. The two modules nevertheless have different central parity and different reducibility. A Casimir value alone is not a classification label.

22.6 Self-checks

1. What is $\Pi_{\varepsilon,\nu}(-I)$?
   Answer. $(-I)^{-1}u_\theta = u_{\theta+\pi}$ has length 1. So $(\Pi(-I)\varphi)(\theta)=\varphi(\theta+\pi)=(-1)^\varepsilon\varphi(\theta)$. The centre acts by σ_ε for every ν.
2. For which (ε,ν) is e_0 fixed by G?
   Answer. We need $e_0\in C_\varepsilon$, so $\varepsilon=0$. We also need $d\Pi(X_+)e_0=(\nu+1)e_2=0$, so $\nu=-1$. Conversely, for $\nu=-1$ the factor in the circle-model formula (22.4) is 1, so $\Pi(g)e_0=e_0$.
3. Why is $V_{1,0}$ reducible?
   Answer. Here $\nu+1+n=0$ at $n=-1$ and $\nu+1-n=0$ at $n=1$. So $d\Pi(X_+)e_{-1}=0$ and $d\Pi(X_-)e_1=0$. The sets $\{n\ge 1\}$ and $\{n\le -1\}$ are admissible, and $V_{1,0}$ is the direct sum of their spans.

22.7 Summary

The principal series $\Pi_{\varepsilon,\nu}$ is induced from the character $\sigma_\varepsilon\otimes e^{\nu t}$ of $P=MAN$, shifted by $\Delta_P^{1/2}$. Restriction to K identifies every $I_{\varepsilon,\nu}$ with the continuous parity-ε functions on the circle. The group acts on these functions by the circle-model formula (22.4). The identity $\tau'_h=|hu_\theta|^{-2}$

makes the pairing of ν with $-\bar{\nu}$ invariant. Hence, for $\nu \in i\mathbb{R}$, the operators $\Pi_{\varepsilon,\nu}(g)$ extend to a strongly continuous unitary representation on $\mathcal{H}_\varepsilon$.

Each K-type occurring in $V_{\varepsilon,\nu}$ occurs once and is spanned by one e_n with $n \equiv \varepsilon \pmod 2$. The elements $X_\pm$ move n by ± 2 with coefficients $\nu + 1 \pm n$. We proved that the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is reducible exactly at integers ν of parity opposite to ε. We also proved the same criterion for closed invariant subspaces of $\mathcal{H}_\varepsilon$. Fourier projection and the ladder give irreducibility away from these points. At $\nu = n - 1$ the power series of (22.6) show that the closed span of the half-ladder $k \geq n$ is invariant, and at $\nu = -(n-1)$ its orthogonal complement is. At $(\varepsilon, \nu) = (1, 0)$ the two half-ladders split $\mathcal{H}_1$ into irreducible summands. The module equivalence of $\pm\nu$ and its unitary version stay cited.

22.8 Further reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (Princeton, 2001), §II.5 for $\mathrm{SL}(2,\mathbb{R})$, and §§VII.1–2 for induction from parabolic subgroups in general.
- P. Etingof, lecture notes for MIT 18.757, *Representations of Lie Groups*, Lectures 9 and 19.

Lecture 23

Discrete series and the rank one unitary picture

Objectives. *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules, and a map* U_n *identifies its polynomials with a half-ladder of the principal series. Finally we state the full list of irreducible unitary representations, and we prove that a finite-dimensional unitary representation is trivial.*

By the end of this lecture you will be able to:

- Prove that Π_n is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its K-types $n, n{+}2, \dots$ and the action of H', E, F, and identify its polynomials with a half-ladder of the principal series by the map U_n.
- Decide which lowest weights λ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification, derive the positivity bounds behind it, and prove that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial.

23.1 The disk model

Lecture 22 found half-ladders at $\nu = n-1$. For $n \geq 2$ the L^2 norm of Lecture 22 is not invariant there, so a half-ladder needs a new norm. A half-ladder has one sign of K-type. Power series in z have the same shape, because rotation multiplies z^m by a power of $e^{2i\theta}$. So we look for holomorphic functions on a disk.

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation Π, a vector v has *K-type k* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all θ. If such a v is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for a, b, c, d shows that φ is an isomorphism of Lie groups from G onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by γ is complex linear on all 2×2 matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

The matrix $\bar{\gamma}$, which is the matrix c of Exercise 20.1, gives the triple in the order of Lecture 21. Our γ reverses it, so that Π_n below has positive K-types.

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}, \qquad j(h, z) = \bar{\beta} z + \bar{\alpha}.$$

Lemma 23.1. *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

(a) $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

(b) h *maps* D *into* D *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

(c) $d(h \cdot z)/dz = j(h,z)^{-2}$.

Proof. (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so h maps D into D. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So h maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar{\beta}_1(\alpha_2 z + \beta_2) + \bar{\alpha}_1(\bar{\beta}_2 z + \bar{\alpha}_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z, 1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z, 1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar{\beta} z + \bar{\alpha}) - \bar{\beta}(\alpha z + \beta) = 1$. ∎

Before we choose a norm, we look for an invariant area. By the identity (23.4) and Lemma 23.1(c), the area form $dA/(1 - |z|^2)^2$ on D is invariant under $\mathrm{SU}(1,1)$. This is the *hyperbolic area*. The matrix $\bar{\gamma}$ moves the upper half-plane $\mathbb{H}$ of Lecture 20 to D, with i going to 0, and this form becomes $\frac{1}{4} y^{-2} dx\, dy$. So it is a constant multiple of the invariant measure of Lecture 20, which is the left Haar measure of Lecture 16. Figure 23.1 shows this chart. Our γ uses the lower half-plane instead, with $-i$ going to 0, so that the K-types of Π_n are positive.

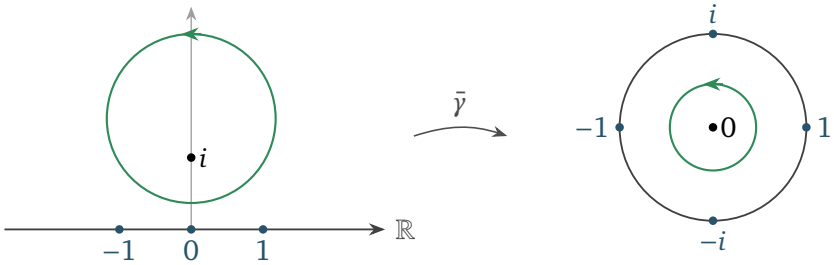


Figure 23.1: The chart $z \mapsto \bar{\gamma} \cdot z$. The points -1, 0 and 1 of the real line go to i, -1 and $-i$ on the boundary circle, and the point at infinity goes to 1. The point i goes to 0, and the K-orbit of ie goes to a centred circle. Arrows show k_θ for $\theta > 0$ in the parametrisation of this lecture.

Now take f and $\Pi_n(h) f$ as in (23.5) below. At $z = h \cdot w$ the value $|\Pi_n(h) f|^2$ gains the factor $|j(h,w)|^{2n}$, and $(1 - |z|^2)^n$ gains the factor $|j(h,w)|^{-2n}$ by (23.4). These cancel. So the natural norm integrates $|f|^2 (1 - |z|^2)^n$ against the hyperbolic area. This is why the power below is $n - 2$.

Definition 23.2. Let $n \geq 2$ be an integer and $dA = dx\, dy$. For f holomorphic on D put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1 - |z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic f on D with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in D with smooth boundary.

Lemma 23.3. *If $f(z)=\sum_{m\ge 0} a_m z^m$ is holomorphic on D, then*

$$\|f\|_n^2=\sum_{m\ge 0} c_m|a_m|^2,\qquad c_m=\|z^m\|_n^2=\frac{m!\,(n-1)!}{(n+m-1)!}=\binom{n+m-1}{m}^{-1}.$$

So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m=c_m^{-1/2}z^m$. Moreover $\|g\|_n\le \sup_D|g|$ for bounded holomorphic g.

Proof. On $|z|=s<1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi}|f(se^{i\theta})|^2\,d\theta=2\pi\sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi}\int_{|z|\le r}|f|^2(1-|z|^2)^{n-2}\,dA=\sum_m |a_m|^2\,(n-1)\int_0^r 2s^{2m+1}(1-s^2)^{n-2}\,ds.$$

The terms are nonnegative and increase with r, so we may let $r\to 1$ termwise. With $t=s^2$ and repeated integration by parts,

$$(n-1)\int_0^1 t^m(1-t)^{n-2}\,dt=(n-1)\frac{m!\,(n-2)!}{(n+m-1)!}=c_m.$$

Conversely, let $(a_m)_{m\ge 0}$ be complex numbers with $C=\sum_m c_m|a_m|^2<\infty$. Then $|a_m|^2\le C\binom{n+m-1}{m}\le C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on D. By the power-series import, its sum f is holomorphic on D. The formula above then gives $\|f\|_n^2=C$. Hence $f\mapsto (c_m^{1/2}a_m)_m$ is an isometry of $\mathcal{A}_n$ onto ℓ^2. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2\le \sup|g|^2\,\|1\|_n^2=\sup|g|^2$, since $c_0=1$. ∎

Theorem 23.4. *For $h\in \mathrm{SU}(1,1)$ and $f\in\mathcal{A}_n$ define*

$$(\Pi_n(h)f)(z)=j(h^{-1},z)^{-n}\,f(h^{-1}\cdot z). \tag{23.5}$$

Then $g\mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of G on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.

Proof. Homomorphism. Apply the cocycle identity of Lemma 23.1(b) to h_2^{-1} and h_1^{-1}:

$$(\Pi_n(h_1)\Pi_n(h_2)f)(z)=j(h_1^{-1},z)^{-n}\,j(h_2^{-1},h_1^{-1}\cdot z)^{-n}f(h_2^{-1}h_1^{-1}\cdot z)=(\Pi_n(h_1h_2)f)(z).$$

Isometry. Substitute $z=h\cdot w$. The cocycle identity with $h^{-1}h=I$ gives $j(h^{-1},h\cdot w)=j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\&\quad=|j(h,w)|^{2n}|f(w)|^2\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R \subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over R.

The map $w \mapsto h\cdot w$ is a bijection of D with inverse $w \mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n = \|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

Continuity. Let f, g be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n \geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood U of I. So $h \mapsto \langle g, \Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\| = 1$, so an $\varepsilon/3$ argument extends this to all f, g. Then

$$\|\Pi_n(h)f - f\|_n^2 = 2\|f\|_n^2 - 2\,\mathrm{Re}\langle f, \Pi_n(h)f\rangle_n \to 0 \qquad (h \to I).$$

Unitarity moves this to every other point. ∎

Remark 23.5. Put $\delta = \mathrm{diag}(1,-1)$. Since $\bar\gamma = \gamma\delta$, the matrix $\bar\gamma$ in place of γ gives $\Pi_n^-(g) = \Pi_n(\delta g \delta^{-1})$. Conjugation by δ preserves G, although $\delta \notin G$. Since $\delta k_\theta \delta^{-1} = k_{-\theta}$, the K-types of Π_n^- are the negatives of those of Π_n. Which one is called holomorphic depends on the Cayley matrix and the sign of the K-angle. We label them by K-types.

23.2 K-types and Lie algebra operators

Proposition 23.6. *For all θ and $m \geq 0$,*

$$\Pi_n(k_\theta)z^m = e^{i(n+2m)\theta}z^m.$$

Hence the K-types of Π_n are $n, n+2, n+4, \dots$, each with multiplicity one, and the K-finite vectors are the polynomials.

Proof. By the formula (23.2), the element $h = \varphi(k_\theta)$ has $h^{-1} = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. So $h^{-1}\cdot z = e^{2i\theta}z$ and $j(h^{-1}, z) = e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer k define

$$P_k f = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m = e_m$ if $k = n+2m$, and $P_k e_m = 0$ otherwise. Being bounded, P_k is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of K-type k.

If v is K-finite, its orbit spans a finite-dimensional space U. Every $P_k v$ lies in U, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of v is therefore a polynomial. Conversely, every polynomial is a finite sum of K-types. ∎

Lemma 23.7. *Polynomials are smooth vectors for Π_n. For $Y = \begin{pmatrix} p & q \\ r & s\end{pmatrix} \in \mathfrak{su}(1,1)$ and a*

polynomial f,

$$d\Pi_n(Y)f = n(rz+s)f + \big(rz^2 + (s-p)z - q\big)f'. \tag{23.6}$$

Proof. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), $(h,z) \mapsto (\Pi_n(h)f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in h of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So f is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y)f$ is the pointwise t-derivative at 0 of $(\Pi_n(h_t)f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz+s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1-tp)z - tq}{1 - t(rz+s)} = z + t\big(rz^2 + (s-p)z - q\big) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz+s) + O(t^2)$, and the product and chain rules give (23.6). ∎

Proposition 23.8. *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H')z^m &= (n+2m)z^m, \\ d\Pi_n(E)z^m &= (n+m)z^{m+1}, \\ d\Pi_n(F)z^m &= -m\,z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F)1 = 0. \end{aligned}$$

The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n-2)$.

Proof. The right side of (23.6) is complex linear in (p,q,r,s). So it also computes $d\Pi_n$ on the matrices in (23.3). For H' we have $p=-1$, $s=1$, giving $nf + 2zf'$. For E we have $r=1$, giving $nzf + z^2f'$. For F we have $q=1$, giving $-f'$. Apply these to z^m. Since $[E,F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On z^m its eigenvalue is

$$(n+2m)^2 + 2(n+2m) - 4(n+m)(m+1) = n^2 - 2n.$$ ∎

For a unitary representation Π and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices H_0, S and W are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators H_0 and S. It also turns the coefficient i of S into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors u and v satisfy

$$\langle d\Pi(E)u, v\rangle = -\langle u, d\Pi(F)v\rangle, \qquad \langle d\Pi(H')u, v\rangle = \langle u, d\Pi(H')v\rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for c_m in Lemma 23.3. Conversely, distinct K-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

Theorem 23.9. *Π_n is irreducible for every integer $n \geq 2$.*

Proof. Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each P_k maps V into V, since its Riemann sums do and V is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some e_m lies in V. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in V. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in V. By the operator formulas of Proposition 23.8, E sends z^m to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, F sends z^m to $-mz^{m-1}$ with $m \neq 0$. So every z^k lies in V. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on G. The family Π_n, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of e_0. The end of § 23.4 explains the word discrete.

23.3 Descent conditions

Proposition 23.10. *Let Π be a strongly continuous representation of G on a Hilbert space. Let $v \neq 0$ have K-type k.*

(a) $k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such Π has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its K-finite part.

(b) $\Pi(-I)v = (-1)^k v$.

(c) $\Pi_n(-I) = (-1)^n \mathrm{id}$. Hence Π_n factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if n is even.

Proof. (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight K-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of K-type λ, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of K-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I,z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So Π_n factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every K-type $n+2m$ of Π_n has the parity of n. The integrality argument of Proposition 23.10(a) uses the period 2π of K in G. On the universal covering group the lift of K is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

Remark 23.11. The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure μ on D gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on D. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

23.4 Comparison and the unitary picture

Let V be a $(\mathfrak{g}, K)$-module of G (Lecture 21) on which Ω acts by a scalar c. Write v_k for a vector of K-type k. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

Example 23.12. Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In Π_n the K-types are $n, n+2, \dots$. F kills z^0, and E is injective.

(ii) Put $d = n-2$ and replace n by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends f to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most d. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills z^d. The K-types are $-d, \dots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a K-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n-1$, and with ε chosen so that its K-types have the parity of n. By Lecture 22, Exercise 22.4, Ω acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$.
The ladder of Lecture 22 says which arrow vanishes. At $\nu = n-1$ the arrows E from $-n$ and F from n vanish. So the half-ladders $k \geq n$ and $k \leq -n$ are submodules, and the finite segment $2-n \leq k \leq n-2$ is what remains when both are removed. At $\nu = -(n-1)$ the arrows E from $n-2$ and F from $2-n$ vanish, so the finite segment is a submodule. In both cases the three pieces have the K-types and Casimir value of Π_n^-, of the finite-dimensional module Π_{-d} with $d = n-2$, and of Π_n.

We now identify the piece $k \geq n$ with Π_n. In this lecture e_m is the basis of $\mathcal{A}_n$, so we write $\mathsf{e}_k(\theta) = e^{ik\theta}$ for the Fourier vectors of Lecture 22. For $n \geq 1$ and f holomorphic on a neighbourhood of $\overline{D}$, put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = \mathsf{e}_{n+2m}.$$

This is the map U_n of Lecture 22, with the same formula. Formula (23.5) also makes sense for $n = 1$ on such functions.

Proposition 23.13. *Let $n \geq 1$ and let $\varepsilon \equiv n \pmod 2$. For a polynomial f and $g \in G$, the function $\Pi_n(g)f$ is holomorphic on a neighbourhood of $\overline{D}$, and*

$$U_n \Pi_n(g) f = \Pi_{\varepsilon,n-1}(g) U_n f.$$

Proof. Put $h = \varphi(g)^{-1} = \gamma g^{-1} \gamma^{-1}$. By (23.5), $(\Pi_n(g)f)(z) = j(h,z)^{-n} f(h \cdot z)$. By Lemma 23.1(a), $j(h, \cdot)$ has no zero on $\overline{D}$, and its only possible zero lies outside. So $\Pi_n(g)f$ is holomorphic near $\overline{D}$.

Let $z = e^{2i\theta}$ and $g^{-1}u_\theta = r u_\tau$, with $u_\theta = (\cos\theta, \sin\theta)^T$ as in Lecture 22. Then $\gamma u_\theta = e^{-i\theta}(z, 1)^T$. Applying $\gamma g^{-1} = h\gamma$ to u_θ gives $h \cdot z = e^{2i\tau}$ and $j(h,z) = re^{i(\theta-\tau)}$, as in the half-ladder formula of Lecture 22. Hence

$$e^{in\theta} j(h,z)^{-n} f(h \cdot z) = r^{-n} e^{in\tau} f(e^{2i\tau}) = (\Pi_{\varepsilon,n-1}(g) U_n f)(\theta). \qquad \blacksquare$$

This identity uses only matrices, so the opposite sign of k_θ in this lecture does not enter. It says that the polynomials of Π_n are the piece $k \geq n$ of $V_{\varepsilon,n-1}$. The ladders agree too: for $\nu = n-1$ and $k = n+2m$ we have $\frac{1}{2}(\nu+1+k) = n+m$ and $\frac{1}{2}(\nu+1-k) = -m$, which are the coefficients of Proposition 23.8. The boundary point $z = e^{2i\theta}$ is the line $\mathbb{R}u_\theta$ of Lecture 22.

Only the norm differs. The squared disk norm of z^m is c_m, and the squared circle norm of e_{n+2m} is 1. So for $n \geq 2$ the map U_n is not an isometry. Also a general vector of $\mathcal{A}_n$ need not have boundary values, so we do not define U_n on all of $\mathcal{A}_n$. At $n = 1$ we have $c_m = 1$, and the two norms agree.

Corollary 23.14. *Let $\mathcal{A}_1$ be the completion of the polynomials in the norm $\|\sum a_m z^m\|^2 = \sum |a_m|^2$. Then U_1 extends to a unitary map from $\mathcal{A}_1$ onto the closed span H_1^+ of the vectors $\mathtt{e}_1, \mathtt{e}_3, \dots$. The operators $\Pi_1(g) = U_1^{-1}\Pi_{1,0}(g)U_1$ form an irreducible strongly continuous unitary representation of G on $\mathcal{A}_1$, and on polynomials they are given by (23.5) with $n = 1$. Put $\Pi_1^-(g) = \Pi_1(\delta g \delta^{-1})$. Then $\Pi_{1,0}$ is the orthogonal sum of a copy of Π_1 on H_1^+ and a copy of Π_1^- on the closed span of $\mathtt{e}_{-1}, \mathtt{e}_{-3}, \dots$.*

Proof. By Parseval, $\|U_1 f\|^2 = \sum |a_m|^2$, and the vectors $\mathtt{e}_{1+2m}$ are an orthonormal basis of H_1^+. So U_1 extends to a unitary map onto H_1^+. The representation $\Pi_{1,0}$ is unitary and strongly continuous, since $\nu = 0$ is imaginary. By the irreducibility theorem of Lecture 22, H_1^+ is invariant and irreducible. These properties pass to Π_1 through U_1. A function holomorphic near $\overline{D}$ has a Taylor series that converges in $\mathcal{A}_1$, so the extended U_1 agrees on it with the formula above. Hence, on polynomials, Proposition 23.13 gives the stated formula for $\Pi_1(g)$.

For the second half, put $(R\varphi)(\theta) = \varphi(-\theta)$. This is unitary and carries H_1^+ onto the negative half. Since $\delta u_\theta = u_{-\theta}$, the circle formula of Lecture 22 gives $\Pi_{1,0}(\delta g \delta^{-1}) = R\,\Pi_{1,0}(g)R$. Hence $RU_1\Pi_1^-(g) = \Pi_{1,0}(g)RU_1$. ∎

We now compare the sign in the two real forms of $\mathfrak{sl}(2,\mathbb{C})$. In a unitary representation of SU(2) the same triple has, in physics notation, $H' = 2J_3$, $E = J_+$ and $F = J_-$, and now $\langle Eu, v\rangle = \langle u, Fv\rangle$. Then this sign and (23.8) give $\|Ev_k\|^2 = \frac{1}{4}\big(c - k(k+2)\big)\|v_k\|^2$. For $c = 4j(j+1)$ and $k = 2m$, this is $j(j+1) - m(m+1)$ times $\|v_k\|^2$. For large k the right side is negative, so the ladder must stop. For $\mathrm{SL}(2,\mathbb{R})$ the relation (23.7) has a minus sign, and the sign of the norm formula is reversed. So a ladder need not stop.

Proposition 23.15. *Let V be an irreducible $(\mathfrak{g}, K)$-module of G. Assume each K-type has multiplicity at most one and Ω acts by c. Suppose V has a positive definite inner product satisfying (23.7) for the module action. If k and $k+2$ are K-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

Consequently:

(i) if the K-types are $\lambda, \lambda+2, \lambda+4, \dots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;

(ii) if the K-types are $2\mathbb{Z}$, then $c < 0$;

(iii) if the K-types are $2\mathbb{Z}+1$, then $c < -1$.

Proof. By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that c is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the K-type spaces of that parity is a submodule. By irreducibility, all K-types have the parity of k. Since E raises and F lowers K-types by 2, the span of the v_j with $j \leq k$ is invariant. It is nonzero and misses v_{k+2}, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda-2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda}v_\lambda$, which makes λ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even k and -1 over odd k. ∎

Figure 23.2 shows formula (23.9). At $c = 8$ the dots on the line are $k = 2$ and $k = -4$, where the step vanishes. Between them the even dots $k = -2$ and $k = 0$ are the steps of the finite-dimensional module of K-types $-2, 0, 2$. For $\mathrm{SL}(2,\mathbb{R})$ these steps have negative squared norm. For $\mathrm{SU}(2)$ they are exactly the steps with positive squared norm.

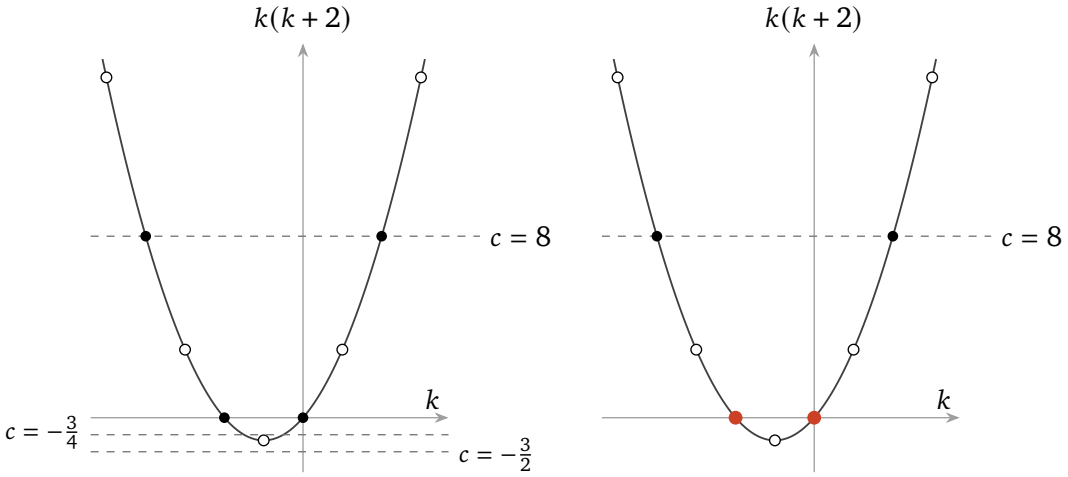


Figure 23.2: Formula (23.9) for $\mathrm{SL}(2,\mathbb{R})$ in the first panel and its $\mathrm{SU}(2)$ analogue in the second. Filled dots are even k and hollow dots are odd k. In the first panel a nonzero step from k to $k+2$ needs its dot strictly above the line $y = c$; a dot on the line is a zero step. In the second panel the step needs its dot strictly below the line, and the coloured dots are the steps of the module with K-types $-2, 0, 2$.

Twisting by δ from Remark 23.5 interchanges E, F, sends H' to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \geq 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \ldots$.

Theorem 23.16 (Bargmann; cited). *Every irreducible unitary representation of G is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its K-types and its Casimir value c. Here Π_1 and Π_1^- are the representations of Corollary 23.14, whose unitarity and irreducibility are proved there.*

Family	*K-types*	c	ν
trivial	$\{0\}$	0	± 1
even principal series	$2\mathbb{Z}$	$c \leq -1$	$i\mathbb{R}$
odd principal series	$2\mathbb{Z}+1$	$c < -1$	$i\mathbb{R} \setminus \{0\}$
complementary series	$2\mathbb{Z}$	$-1 < c < 0$	$(-1,1) \setminus \{0\}$
Π_n, $n \geq 1$	$n, n+2, \ldots$	$n(n-2)$	$\pm(n-1)$
Π_n^-, $n \geq 1$	$-n, -n-2, \ldots$	$n(n-2)$	$\pm(n-1)$

For $n \geq 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.

The last column gives the parameters of Lecture 22 with $c = \nu^2 - 1$. In the principal and complementary rows the entry has the module $V_{\varepsilon,\nu}$, with $\varepsilon = 0$ in the even rows and $\varepsilon = 1$ in the odd row, and ν and $-\nu$ give the same module. In the other rows the entry is only one

piece of a reducible module $V_{\varepsilon,\nu}$. So a value of ν can carry several entries of the list.

Proposition 23.15 proves the necessity of the c-ranges for these K-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its K-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z}+1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the L^2 norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

Figure 23.3 sets the list against c. We use c and not ν because of the last remark on the table. At $c = -1$ the odd principal series splits into Π_1 and Π_1^-. At $c = 0$ the representations Π_2, Π_2^- and the trivial representation are distinct.

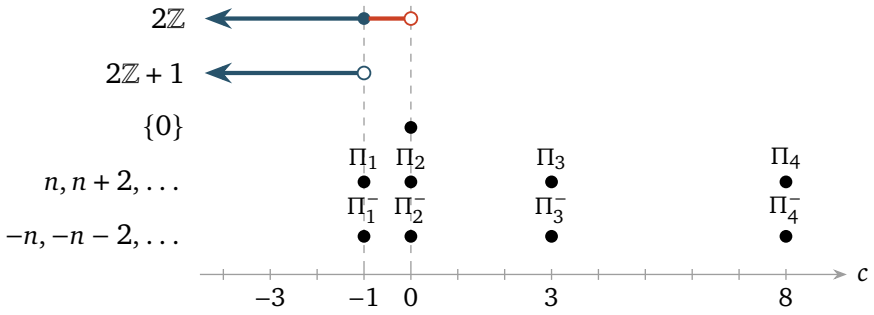


Figure 23.3: Bargmann's list against the Casimir value c, with one row for each pattern of K-types. In the top row the long segment $c \le -1$ is the even principal series, and the short segment $-1 < c < 0$ is the complementary series. The second row is the odd principal series, the third is the trivial representation, and the last two rows show the first few Π_n and Π_n^-.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its K-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one K-types on which Ω is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of G. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

We can now explain the word discrete. For a compact group, the Plancherel theorem of Lecture 11 writes the squared norm of a function as a sum over the irreducible representations. For G the corresponding formula has a sum and integrals. Let h be smooth with compact support on G, and put $\Pi(h) = \int_G h(g)\Pi(g)\,dg$. For a suitable normalisation of Haar measure,

$$h(I) = \int_{\mathbb{R}} \operatorname{tr} \Pi_{0,i\nu}(h)\, \nu \tanh \frac{\pi\nu}{2}\, d\nu + \int_{\mathbb{R}} \operatorname{tr} \Pi_{1,i\nu}(h)\, \nu \coth \frac{\pi\nu}{2}\, d\nu + \sum_{n \ge 2} 4(n-1) \operatorname{tr}\bigl(\Pi_n(h) + \Pi_n^-(h)\bigr).$$

We cite this inversion formula from Knapp, §II.7, formula (2.25), and we do not use it. Knapp proves it as Theorem 11.6 in §XI.3 and credits it to Harish-Chandra (1952). The Plancherel formula for G follows from it. The discrete series enters through a discrete sum, and the

unitary principal series enters through integrals over $\nu = i\nu$. Neither the complementary series nor $\Pi_1^{\pm}$ appears. The number $n-1$ is the one in Exercise 23.6, where the coefficient of e_0 has squared norm $C/(n-1)$.

Finally we return to finite dimensions. The next statement does not use Bargmann's theorem, and it needs no irreducibility hypothesis.

Corollary 23.17. *Every finite-dimensional continuous unitary representation Π of G is trivial.*

Proof. By the theorem of Lecture 3 on continuous homomorphisms, Π is smooth. The complex-linear extension of $d\Pi$ makes the space V a finite-dimensional module of $\mathfrak{sl}(2,\mathbb{C})$. The real element $H_0 = \operatorname{diag}(1,-1)$ is the diagonal element of the triple of Lecture 20. So complete reducibility and the highest-weight calculation of the prerequisite Lie Algebras course make $d\Pi(H_0)$ diagonalisable with integer eigenvalues. Unitarity makes the same operator skew-Hermitian, with imaginary eigenvalues. Therefore $d\Pi(H_0) = 0$.

The kernel of the complexified differential is an ideal of $\mathfrak{sl}(2,\mathbb{C})$ containing the nonzero element H_0. Since $\mathfrak{sl}(2,\mathbb{C})$ is simple, this kernel is everything, so $d\Pi = 0$. Since G is connected, it is generated by exponentials (Lecture 2), and $\Pi(\exp X) = \exp d\Pi(X) = I$. Hence Π is trivial. ∎

So a nontrivial unitary representation of G is infinite-dimensional. Exercise 23.3 checks this directly for the finite-dimensional modules Π_{-d} of Example 23.12(ii).

23.5 Self-checks

1. For $n = 3$, compute c_1 and c_2. Check (23.7) for $u = z$ and $v = z^2$.

   Answer. $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does Π_5 factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   Answer. No, since $\Pi_5(-I) = -\mathrm{id}$. By the K-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with K-types $1, 3, 5, \ldots$ and $c = -\frac{1}{2}$?

   Answer. No. By the cited Harish-Chandra facts after Theorem 23.16, the K-finite vectors of such a representation satisfy the hypotheses of Proposition 23.15. Its lowest-weight case (i) shows that the lowest K-type 1 forces $c = 1\cdot(1-2) = -1$.

23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation Π_n on $\mathcal{A}_n$. Its K-types are $n, n+2, \ldots$, each once. The operator E raises the K-type by 2 and F lowers it by 2. The Casimir element Ω acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral K-types. The representation Π_n descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even n. In an irreducible unitary module with K-types $\lambda, \lambda+2, \ldots$, the lowest weight is an integer $\lambda \geq 1$, and Ω acts

by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

The map U_n carries the polynomials of Π_n onto the half-ladder $k \geq n$ of the principal series at $\nu = n-1$, and only the norm differs. For $n=1$ the norms agree, and this constructs Π_1 as a summand of the odd principal series at $\nu = 0$. Every finite-dimensional continuous unitary representation of G is trivial. We also cited the inversion formula for G, whose discrete sum runs over the discrete series.

23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$. §II.7 states the inversion formula cited above, and §XI.3 proves it.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- Harish-Chandra, Plancherel formula for the 2×2 real unimodular group, *Proc. Nat. Acad. Sci. USA* 38 (1952), 337–342.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.

Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$, and identify $\mathrm{SL}(2,\mathbb{R})$ as the double cover of $\mathrm{SO}^+(1,2)$ inside it.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w \rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x \rangle = \langle w, x \rangle = w_0 x_0 - \vec{w} \cdot \vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x \rangle = \langle x, x \rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det \Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det \Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T \eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim \mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1, 2, 3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(t K_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

The quantity $x_0^2 - x_3^2 = (x_0 + x_3)(x_0 - x_3)$ does not change, so a point moves along a hyperbola. Figure 24.1 shows this motion in the plane $x_1 = x_2 = 0$.

Parity $\mathrm{diag}(1, -1, -1, -1)$ and time reversal $\mathrm{diag}(-1, 1, 1, 1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Lecture 4 built the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ as the adjoint action on $\mathfrak{su}(2)$. It also identified this with the action $\xi \mapsto U \xi U^*$ on traceless Hermitian matrices. We now allow all Hermitian matrices and all of $\mathrm{SL}(2,\mathbb{C})$, with the action $X \mapsto A X A^*$. The same construction gives the Lorentz group.

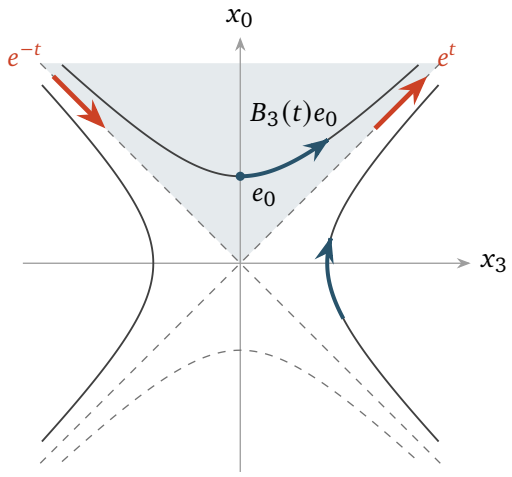


Figure 24.1: The boost $B_3(t)$ in the plane $x_1 = x_2 = 0$. The dashed diagonals are the null lines $x_0 = \pm x_3$, and the shaded wedge is the slice of T^+. The solid curves are the hyperbolas $x_0^2 - x_3^2 = \pm 1$. The coloured arrows show that $B_3(t)$ stretches the diagonal $x_0 = x_3$ by e^t and shrinks the diagonal $x_0 = -x_3$ by e^{-t}. The heavy arcs show the motion of two points for one fixed $t > 0$: the point e_0, which is the lowest point of the upper branch, and a point with $x_0^2 - x_3^2 = -1$. The dashed lower branch is the slice of the sheet on which Λe_0 lies when $\Lambda_{00} \le -1$.

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x \rangle, \qquad x_0 = \tfrac{1}{2} \operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2, \mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4, \mathbb{R})$ by

$$X(\varphi(A)x) = A\, X(x)\, A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements). Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2 (x_0 + x_3) & (a/\bar{a})(x_1 - i x_2) \\ (\bar{a}/a)(x_1 + i x_2) & |a|^{-2}(x_0 - x_3) \end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + i x_2 \mapsto e^{i\theta}(x_1 + i x_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp t i X_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j} |A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W\operatorname{diag}(\lambda, \lambda^{-1})W^*$ with W unitary and $\lambda > 0$. Put $P = W\operatorname{diag}(\lambda^{1/2}, \lambda^{-1/2})W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^{\perp} = \operatorname{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \operatorname{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k\sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W'\sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W'\operatorname{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\operatorname{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW' \exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\operatorname{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk}X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk}J_k, \qquad [J_i, K_j] = \epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\epsilon_{ijk}J_k.$$

Steps 2 and 4 of the proof are the Cartan decomposition of Lecture 20, for the real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ with $\mathfrak{k} = \mathfrak{su}(2)$ and $\mathfrak{p} = i\,\mathfrak{su}(2)$. Step 2 writes $A = UP$, with $U \in \mathrm{SU}(2)$ and $P = \exp Y_2$ for a traceless Hermitian Y_2. So $\varphi(A) = \varphi(U)\varphi(P)$, which is a boost followed by a rotation. Step 4 writes the same kind of element in the other order, as a rotation followed by a boost.

The same construction one dimension down

The construction also works for the real subgroup $\mathrm{SL}(2,\mathbb{R})$ of $\mathrm{SL}(2,\mathbb{C})$. It then relates $\mathrm{SL}(2,\mathbb{R})$ to a Lorentz group with two space coordinates. Let $\mathbb{R}^{1,2}$ be the subspace $\{x \in \mathbb{R}^4 : x_2 = 0\}$, with the form $x_0^2 - x_1^2 - x_3^2$. Let $\mathrm{SO}^+(1,2)$ be the group of linear maps of $\mathbb{R}^{1,2}$ that preserve this form, have determinant one and satisfy $\Lambda_{00} \geq 1$.

> **Proposition 24.6.** *Each $\varphi(A)$ with $A \in \mathrm{SL}(2,\mathbb{R})$ fixes e_2 and preserves $\mathbb{R}^{1,2}$. Restriction to $\mathbb{R}^{1,2}$ gives a surjective homomorphism $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$ with kernel $\{\pm I\}$.*

Proof. Let A be real, so $A^* = A^T$. If $x_2 = 0$, then $X(x)$ is real symmetric, and so is $AX(x)A^T$. Hence $\varphi(A)$ preserves $\mathbb{R}^{1,2}$. Next, $X(e_2) = \sigma_2$, and $A\sigma_2 A^T = (\det A)\sigma_2 = \sigma_2$, so $\varphi(A)$ fixes e_2. Thus $\varphi(A)$ is the identity on $\mathbb{R}e_2$ and its restriction on $\mathbb{R}^{1,2}$. By Theorem 24.5, this restriction preserves the form, has determinant one and has $\Lambda_{00} \geq 1$. If the restriction is the identity, then $\varphi(A) = I$, so $A = \pm I$.

For surjectivity we repeat Step 4 with real matrices. Let $\Lambda \in \mathrm{SO}^+(1,2)$ and $v = \Lambda e_0$. The matrix $V = X(v)$ is real symmetric with positive eigenvalues and determinant one. A real orthogonal diagonalisation gives a real positive definite P with $\det P = 1$ and $P^2 = V$, so $\varphi(P)e_0 = v$. Then $\varphi(P)^{-1}\Lambda$ fixes e_0, so it acts on the (x_3, x_1)-plane by an element of $\mathrm{SO}(2)$. Direct multiplication shows that the rotation matrix with first column $(\cos\theta, \sin\theta)^T$ fixes e_0 and rotates the (x_3, x_1)-plane through the angle 2θ. So this element is $\varphi(k)$ for a rotation matrix k, and $\Lambda = \varphi(Pk)$ on $\mathbb{R}^{1,2}$. ∎

So Lecture 21, Lecture 22 and Lecture 23 studied the double cover of the Lorentz group of two space coordinates and one time. This is why Bargmann's paper, cited in Lecture 23, has the title *Irreducible unitary representations of the Lorentz group*.

The earlier models of that block appear here as well. The real matrices $\frac{1}{2}\sigma_3$ and $\frac{1}{2}\sigma_1$ map to the boost generators K_3 and K_1. The stabiliser of e_0 in $\mathrm{SL}(2,\mathbb{R})$ is $\{A : AA^T = I\}$, which is the subgroup K of rotation matrices. The proof above shows that the group acts transitively on the upper sheet $x_0^2 - x_1^2 - x_3^2 = 1$ with $x_0 > 0$. So this sheet is $\mathrm{SL}(2,\mathbb{R})/K$. As a homogeneous space it is the half-plane of Lecture 20 and the disk of Lecture 23. Finally, a real line $\mathbb{R}v$ in $\mathbb{R}^2$ goes to the matrix vv^T. This matrix has determinant zero and trace $|v|^2 > 0$, so it is a future light-like vector, and $A(vv^T)A^T = (Av)(Av)^T$. Thus the circle of real lines in Lecture 22 is the circle of light rays in $\mathbb{R}^{1,2}$.

24.3 Complexifying the Lorentz algebra

This section rests on one change of basis. In the complexified Lorentz algebra $\mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$, the elements $\frac{1}{2}(J_k - iK_k)$ obey the brackets of $\mathfrak{su}(2)$. So do the elements $\frac{1}{2}(J_k + iK_k)$, and the two sets commute. We now derive this.

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

The real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ carries its own operator M, given by $M(Y) = iY$. It satisfies $M^2 = -1$ and $[MY, Z] = M[Y, Z]$. After complexifying, $M \otimes 1$ has the eigenvalues i and $-i$. Its eigenspaces are ideals, since $[(M \otimes 1)u, v] = (M \otimes 1)[u, v]$. They commute: if u and v have eigenvalues i and $-i$, then $i[u, v] = (M \otimes 1)[u, v] = -i[u, v]$. The next proposition identifies these ideals with two copies of $\mathfrak{sl}(2,\mathbb{C})$. Under it, $M \otimes 1$ becomes multiplication by i on the first copy and by $-i$ on the second.

Proposition 24.7. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\ z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\ zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\ \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. In other words, X_k^L and X_k^R are the images of $X_k \otimes 1$ under the projections $\frac{1}{2}(1 - iM \otimes 1)$ and $\frac{1}{2}(1 + iM \otimes 1)$ onto the two ideals. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On $\mathrm{SU}(2)$, V_m is the spin-$m/2$ representation.

Definition 24.8. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L, j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L + 1)(2j_R + 1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.9. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L, j_R)}(-I) = (-1)^{2j_L + 2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L + 2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L, j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Here $(Y_1, Y_2) \in \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ acts on $V_m \boxtimes V_n = V_m \otimes V_n$ by $dS^m(Y_1) \otimes 1 + 1 \otimes dS^n(Y_2)$. We import from the prerequisite course Lie Algebras (Lectures 13 and 26): these modules are irreducible and pairwise non-isomorphic, since their highest weights (m, n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m, n). This is the complexified differential of $D^{(m/2, n/2)}$, so $\Pi \cong D^{(m/2, n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon) S^n(U) S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the prerequisite course Lie Algebras (Lectures 6 and 27) gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

The representations $(j, 0)$ are the holomorphic extensions from $\mathrm{SU}(2)$ given by Lecture 14, since the entries of $S^{2j}(A)$ are polynomials in the entries of A.

Only $(0, 0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0, 0)$. Exercise 24.6 gives the proof. So every nontrivial continuous unitary representation of this group is infinite-dimensional. Such representations exist. They are induced from the upper triangular subgroup, as in Lecture 22, and their models use the sphere of light rays. We do not study them; see Knapp, *Representation Theory of Semisimple Groups*, Chapter II, Section 4.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.10 (Weyl spinors)**.** The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.9, the two representations are inequivalent. A direct check uses traces. For $A = \operatorname{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2, \mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.11 (Four-vectors)**.** Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2, \mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.9. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.9(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0) \oplus (0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on $\mathrm{SU}(2)$	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac{1}{2},0)$, $(0,\frac{1}{2})$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac{1}{2},\frac{1}{2})$	4	$V_0 \oplus V_2$	yes	four-vector
$(1,\frac{1}{2})$	6	$V_1 \oplus V_3$	no	–

In physics language, the two summands of $\Lambda^2\mathbb{C}^4$ are spanned by the complex combinations $\vec{E} + i\vec{B}$ and $\vec{E} - i\vec{B}$ of the electric and magnetic fields. Which one is called $(1,0)$ depends on the convention for hands.

24.6 Self-checks

(1) Compute $\varphi(\mathrm{diag}(2, \frac{1}{2}))$.
(2) Find all $A \in \mathrm{SL}(2,\mathbb{C})$ with $\varphi(A) = R_3(\pi)$.
(3) Does $(\frac{3}{2}, \frac{1}{2})$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to $\mathrm{SU}(2)$.

Answer. (1) By the diagonal formula of Example 24.4 with $a = 2$, we get $x_0 \pm x_3 \mapsto 4^{\pm 1}(x_0 \pm x_3)$. Since $a/\bar{a} = 1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t = 4$, so $\cosh t = \frac{17}{8}$ and $\sinh t = \frac{15}{8}$. Thus

$$\varphi\big(\mathrm{diag}(2, \tfrac{1}{2})\big) = \begin{pmatrix} \frac{17}{8} & 0 & 0 & \frac{15}{8} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{15}{8} & 0 & 0 & \frac{17}{8} \end{pmatrix}.$$

Check: $17^2 - 15^2 = 64$.

(2) The element $\exp(\pi X_3) = \mathrm{diag}(-i, i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A = \pm\,\mathrm{diag}(-i, i)$.

(3) Here $2j_L + 2j_R = 4$ is even, so it descends. Its dimension is $4 \cdot 2 = 8$. The restriction is $V_2 \oplus V_4$, of dimension $3 + 5 = 8$.

24.7 Summary

The action $X \mapsto AXA^*$ on Hermitian matrices preserves $\det X = \langle x, x\rangle$. It gives a two-sheeted covering $\varphi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its restriction to real matrices is the double cover $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L, j_R), given by $S^{2j_L}(A) \otimes S^{2j_R}(\bar{A})$. Exactly those with $2j_L + 2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, and four-vectors are $(\frac{1}{2}, \frac{1}{2})$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.

Lecture 25

Poincaré representations and little groups

Objectives. *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, apply it to the $ax+b$ group, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where x_0 is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the 2×2 Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g)\colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\,\Phi(x)\,g^*.$$

This is the construction of Lecture 24, with Φ in place of X, Λ in place of φ, and $x \cdot y$ in place of $\langle x, y\rangle$.

Lemma 25.1. *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of g. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

Proof. The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on x. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on h. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of g, an entry of $\Phi(x)$ and the complex conjugate of an entry of g. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of g.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, Λ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

Definition 25.2. The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a,g)(a',g') = (a + \Lambda(g)a',\; gg'). \tag{25.2}$$

Proposition 25.3. *G is a Lie group of dimension 10. Its identity is $(0,I)$ and $(a,g)^{-1} = (-\Lambda(g)^{-1}a,\; g^{-1})$. The translations $A = \{(a,I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0,g)(a,I)(0,g)^{-1} = (\Lambda(g)a,\; I). \tag{25.3}$$

The map $(a,g) \mapsto (a,\Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0,\pm I)\}$.

Proof. Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and Λ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a,g)(-\Lambda(g)^{-1}a, g^{-1}) = (a - a,\; I), \qquad (-\Lambda(g)^{-1}a, g^{-1})(a,g) = (-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\; I).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since g^{-1} is the adjugate of g. Hence G is a Lie group. The projection $(a,g) \mapsto g$ is a continuous homomorphism with kernel A, so A is closed and normal. Next, $(0,g)(a,I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0,g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of Λ. ∎

From now on, the Poincaré group means G. The central element $z = (0,-I)$ maps to the identity.

25.2 Momentum characters and mass shells

Let U be a strongly continuous unitary representation of G on a Hilbert space. On A it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

> **Lemma 25.4.** *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation χ_p and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and p_0 is the energy.

To describe all these operators together we use a projection-valued measure. We recall this object from *Linear Algebra for Physics*, Lecture 14. There the spectral theorem attaches to a self-adjoint operator a projection $P(\Omega)$ for each Borel subset Ω of its spectrum. The Borel sets are obtained from the open sets by countable unions, intersections and complements. For multiplication by x on $L^2[0,1]$, the projection $P(\Omega)$ is multiplication by the indicator function of Ω, and each single point has $P(\{\lambda\}) = 0$.

The same definition works on $\mathbb{R}^4$. A *projection-valued measure* E on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that E is *concentrated* on a Borel set S when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure E on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

Here $E(S)$ is the projection onto the part of the space whose momentum lies in S. For a vector v, the map $S \mapsto \langle v, E(S)v\rangle$ is a positive Borel measure of total mass $\|v\|^2$. For a unit vector v, it is the probability distribution of the momentum in the state v, as in the Born rule of Lecture 14 of that course. Unlike the measures of Lecture 17, it may vanish on open sets. The operator integral has the following meaning: for every vector v,

$$\langle v, U(a, I)v\rangle = \int_{\mathbb{R}^4} \chi_p(a)\, d\langle v, E(p)v\rangle.$$

By polarisation, these scalar identities determine the bounded operator $U(a, I)$.

Example 25.5 (A multiplication model)**.** On $L^2(\mathbb{R}^4, d^4p)$ let $(U(a)\psi)(p) = e^{-ip\cdot a}\psi(p)$, and let $E(S)$ multiply ψ by the indicator function of S. Then

$$\langle \psi, E(S)\psi\rangle = \int_S |\psi(p)|^2\, d^4p, \qquad \langle \psi, U(a)\psi\rangle = \int_{\mathbb{R}^4} e^{-ip\cdot a}|\psi(p)|^2\, d^4p.$$

So this E is the measure that the SNAG theorem attaches to U. A single point has Lebesgue measure zero, so $E(\{p\}) = 0$ for every p. This representation therefore has no momentum eigenvector in the Hilbert space, although every vector has a momentum distribution.

The *support* of E is the complement of the union of open sets O with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set F with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0 \, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2 \, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of E on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all a, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. In particular, if g fixes p, then $U(0, g)\psi$ again has momentum p. So the stabiliser of p acts on the formal eigenvectors of momentum p. Moving this action around the orbit of p gives the whole space. This is the idea of induction, and it is why stabilisers appear in the classification below.

The same calculation with E, together with the uniqueness of E in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

Definition 25.6. The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

Theorem 25.7. *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, Y_κ and $\{0\}$.*

Proof. Invariants. The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1\mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g\Phi g^* v = (g^* v)^* \Phi(g^* v)$ and g is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

Transitivity. Write $\Phi(p) = u\ \mathrm{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts u in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their Φ-images are $\mathrm{diag}(m, m)$, $\mathrm{diag}(2, 0)$ and $\mathrm{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \mathrm{diag}(s, s^{-1})$. Then

$$d_s\ \mathrm{diag}(\alpha, \beta)\, d_s^* = \mathrm{diag}(s^2\alpha,\ s^{-2}\beta). \tag{25.5}$$

On X_m^+ we have $\mu_1, \mu_2 > 0$ and $\mu_1\mu_2 = m^2$. Take $s^2 = \mu_1/m$. Then $s^{-2}m = \mu_2$, so $p = \Lambda(ud_s)k_m$. On C^+ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1/2$. On Y_κ we have $\mu_1 > 0 > \mu_2$ and $\mu_1\mu_2 = -\kappa^2$; take $s^2 = \mu_1/\kappa$. Since $\Lambda(g)$ is linear, X_m^- and C^- are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

Figure 25.1 draws these orbits with one spatial coordinate suppressed. The base points k_m, k and ℓ_κ all have $p_1 = p_2 = 0$, and their stabilisers are computed in the next section.

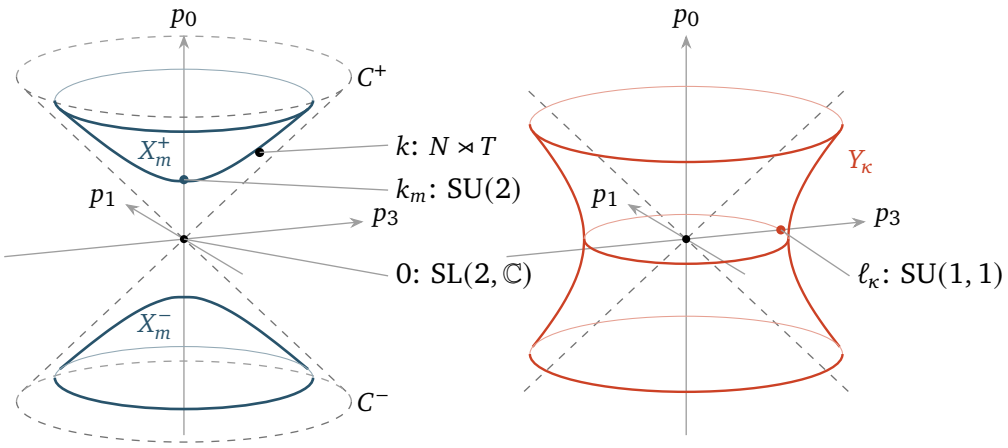


Figure 25.1: The orbits of Theorem 25.7 in the slice $p_2 = 0$. Each drawn surface is a slice of a three-dimensional orbit, since the coordinate p_2 is suppressed. In the panel with the two bowls, the bowls are $X_m^\pm$ and the dashed cone is $C^\pm$ with its vertex 0 removed. In the other panel, the connected waisted surface is Y_κ, around the same cone. Each marked base point carries the name of its stabiliser. Positive energy keeps the upper bowl, the upper cone and the origin. It excludes all of Y_κ, including the points of Y_κ with $p_0 > 0$.

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by s^2 and $p_0 - p_3$ by s^{-2}. It is a boost along the x_3-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, C^+ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with r rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

These orbits lie in the dual of the translation subgroup. They are not the coadjoint orbits of Lecture 19, which lie in the dual of the whole Lie algebra. The two kinds are related. Restricting a functional to the translations maps a coadjoint orbit onto one orbit in the dual of the translations, because translations act trivially on that dual. With $\chi_\ell = e^{i\ell}$ in Lecture 19 and $\chi_p = e^{-ip}$ here, the restriction of ℓ is the functional $-p$. For the $ax + b$ group, this restriction sends the half-planes of Exercise 19.1 onto the frequency half-lines of Lecture 17.

25.3 Little groups and Mackey's classification

Definition 25.8. The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2,\mathbb{C}) : \Lambda(g)p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}.$$

Proposition 25.9. *(a)* $H_{k_m} = \mathrm{SU}(2)$.
(b) $H_k = \{n(b)r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda)n(b)r(\lambda)^{-1} = n(\lambda^2 b)$.
(c) $\Lambda(r(e^{-i\theta/2}))$ *rotates the* (x_1, x_2)*-plane by the angle* $+\theta$ *and fixes* x_0 *and* x_3.
(d) $H_{\ell_\kappa} = \{g : g\sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1,1)$, *where* $\sigma_3 = \mathrm{diag}(1,-1)$. *Also* $H_0 = \mathrm{SL}(2,\mathbb{C})$.

Proof. (a) The condition is $g(mI)g^* = mI$, that is, $gg^* = I$.

(b) Since $\Phi(k) = 2e_1e_1^*$, the condition is $vv^* = e_1e_1^*$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar v_2 \\ v_2\bar v_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar\lambda \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such g fixes k. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \bar\lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar\lambda^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

The group in part (d) is the group $\mathrm{SU}(1,1)$ of Lecture 23. It is also a copy of the real subgroup of Lecture 24. Every $g \in \mathrm{SL}(2,\mathbb{C})$ satisfies $g\sigma_2 g^T = \sigma_2$, so $g\sigma_2 g^* = \sigma_2$ exactly when g is real. Thus $\mathrm{SL}(2,\mathbb{R})$ is the stabiliser of the spacelike vector e_2. The Cayley matrix γ of Lecture 23 satisfies $\gamma\gamma^* = 2I$, so a scalar multiple u of γ lies in $\mathrm{SU}(2)$. It also satisfies $\gamma\sigma_2\gamma^{-1} = -\sigma_3$, so $\Lambda(u)e_2 = -e_3$. Hence $g \mapsto \gamma g \gamma^{-1}$ carries $\mathrm{SL}(2,\mathbb{R})$ onto the stabiliser of e_3, which is H_{ℓ_κ}. This does not make the spacelike orbit relevant to positive energy. It identifies the little-group data of Y_κ with the classification of Lecture 23.

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.9(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call N the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing k, not spacetime translations. For $b = 1$ this map is the null rotation of Exercise 24.2.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.9(c), this generates positive-angle rotations about x_3, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

Theorem 25.10 (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where H is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^*$ put $\chi_p(a) = e^{-ip(a)}$. Let H act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let H_p be the stabiliser of p. For a strongly continuous unitary representation U of G, let E be the projection-valued measure on $(\mathbb{R}^n)^*$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

(a) If U is irreducible, then E is concentrated on one orbit Ω.

(b) Let $p \in \Omega$ and let σ be a strongly continuous irreducible unitary representation of H_p. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\operatorname{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure E is concentrated on Ω.

(c) Every strongly continuous irreducible unitary representation whose measure E is concentrated on Ω is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations σ are unitarily equivalent.

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our G it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^*$ of the theorem with the vectors p used above. The remarks after the orbit theorem, Theorem 25.7, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

Before the ten-dimensional group we run the theorem on two small groups from earlier lectures.

Example 25.11 (The $ax+b$ and Heisenberg groups). The $ax+b$ group of Lecture 17 is $\mathbb{R} \rtimes \mathbb{R}_{>0}$, where $a > 0$ acts on the translations n_c by scaling. We keep the frequency ξ of Lecture 17, with $\chi_\xi(n_c) = e^{i\xi c}$, so the momentum of the theorem is $p = -\xi$. The element a sends ξ to ξ/a. The orbits are $\xi > 0$, $\xi < 0$ and $\{0\}$. These three sets are locally closed and separate the orbits, so the action is regular.

The orbits $\xi > 0$ and $\xi < 0$ have trivial stabiliser. By part (b), each gives one irreducible representation, induced from χ_ξ on N. These are Π_+ and Π_- of Lecture 17, so both are irreducible. The orbit $\{0\}$ has stabiliser $\mathbb{R}_{>0}$. The irreducible unitary representations of $\mathbb{R}_{>0}$ are the characters $a \mapsto a^{is}$, with $s \in \mathbb{R}$. They give the one-dimensional representations $(a, b) \mapsto a^{is}$. By parts (a) and (c), this is the whole list of irreducible unitary representations. In particular, the representation Π_0 of Lecture 17 is not irreducible. Translations act trivially in it, so its measure is concentrated on $\{0\}$, and it is infinite-dimensional.

The Heisenberg group of Lecture 18 is also a semidirect product. The subgroup $A = \{(0, b, c)\}$ is a closed normal vector group, the subgroup $\{(a, 0, 0)\}$ is a copy of $\mathbb{R}$, and

$$(a, 0, 0)(0, b, c)(-a, 0, 0) = (0, b, c + ab).$$

Write a character of A as $e^{i(\mu b + \lambda c)}$. The element $(a, 0, 0)$ sends (μ, λ) to $(\mu - \lambda a, \lambda)$. For $\lambda \neq 0$, the line with this value of λ is one orbit with trivial stabiliser. So exactly one irreducible unitary representation has central character λ. By Exercise 18.5 it is the Schrödinger representation Π_λ. This is the Stone–von Neumann theorem for irreducible representations. For $\lambda = 0$, the orbits are single points $(\mu, 0)$ with stabiliser $\mathbb{R}$. The characters of this stabiliser give the characters $\chi_{\alpha,\beta}$ of Lecture 18. All these orbits are closed. Rational levels of λ, and of μ on the line $\lambda = 0$, separate them. So this action is regular too.

Corollary 25.12. *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of G with E concentrated on X_m^+ are labelled by $n \geq 0$. The label is the $\mathrm{SU}(2)$-representation V_n, of dimension $n+1$ and spin $n/2$.*

(b) Up to unitary equivalence, those with E concentrated on C^+ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.

Proof. Combine Mackey's theorem, Theorem 25.10, with the little groups computed in Proposition 25.9. In the massive case, strongly continuous irreducible unitary representations of the compact group $\mathrm{SU}(2)$ are finite-dimensional by Lecture 10. Hence they are the V_n of Lecture 8. ∎

25.4 Massless sectors: helicity and continuous spin

Proposition 25.13. *Let σ be a finite-dimensional continuous unitary representation of H_k on V. Then $\sigma(n(b)) = I$ for all b. If σ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

Proof. The operators $\sigma(n(b))$ are unitary and commute. So V is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let V_ζ be the joint eigenspace for ζ. Let S be the finite set of ζ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.9(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\, b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some λ^2. If some $\zeta \neq 0$ lay in S, the whole circle $|\zeta'| = |\zeta|$ would lie in S. This contradicts finiteness. So $S = \{0\}$, and N acts trivially.

Then σ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto $\mathrm{U}(1)$, since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If σ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real ν. Periodicity in θ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where σ is a finite-dimensional continuous irreducible unitary representation of H_k. By Proposition 25.13, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

Remark 25.14. Mackey's theorem, Theorem 25.10, also applies to $H_k = N \rtimes T$. The T-orbits on the characters of N are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the σ_ν. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of H_k that are nontrivial on N (Exercise 25.7). Inducing them to G gives the *continuous-spin* representations. Their spectral measure is concentrated on C^+, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has E concentrated on C^+. Its closed support meets C^+ and is Lorentz-invariant by covariance of E. Transitivity then puts all of C^+ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes X_m^- and C^-, where $p_0 < 0$. It also excludes Y_κ, because this orbit contains points with $p_0 < 0$ and the group moves every point of it to such points. In detail, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of S cover Y_κ. By second countability, countably many of these translates already cover Y_κ. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of E makes every translate of S an E-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of E on that orbit in a nonzero representation.

The table collects the sectors. Here $\rho > 0$ is the radius of a T-orbit on the characters of N, and the sign is the value at $r(-1)$ of a character of its stabiliser $\{r(\pm 1)\}$.

Orbit	Base point	Stabiliser	Labels	Energy ≥ 0
X_m^+	k_m	$\mathrm{SU}(2)$	V_n, $n \geq 0$	yes
C^+, N trivial	k	$N \rtimes T$	σ_ν, $\nu \in \mathbb{Z}$	yes
C^+, N nontrivial	k	$N \rtimes T$	$\rho > 0$ and a sign	yes
$\{0\}$	0	$\mathrm{SL}(2,\mathbb{C})$	not computed here	yes
Y_κ	ℓ_κ	$\mathrm{SU}(1,1)$	the list of Lecture 23	no
X_m^-, C^-	$-k_m$, $-k$	as for X_m^+, C^+	as for X_m^+, C^+	no

In physics language, X_m^+ with $n = 0$, 1, 2 describes massive particles such as the pion, the electron and the W boson, while C^+ with $\nu = \pm 2$ describes the photon and $\nu = \pm 4$ the graviton, and no known particle has continuous spin.

25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let D_n be the irreducible unitary representation of $\mathrm{SU}(2)$ on V_n. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure μ_m on X_m^+ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

The factor $1/\omega$ makes this measure invariant. For a boost along x_3 this is visible at once. Put $p_\perp = (p_1, p_2)$ and write $p_3 = \sqrt{m^2 + |p_\perp|^2}\,\sinh\eta$. Then $\omega = \sqrt{m^2 + |p_\perp|^2}\,\cosh\eta$ and $dp_3/\omega = d\eta$. A boost along x_3 fixes $p_\perp$ and shifts the rapidity η. Exercise 25.4 proves invariance under all of $\mathrm{SL}(2,\mathbb{C})$.

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p(mI)h_p^* = \Phi(p)$, so $\Lambda(h_p)k_m = p$. The *Wigner rotation* is

$$W(g,p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

Put $q = \Lambda(g)^{-1}p$. Read from the right, the three factors move the base point k_m to q, then q to p, then p back to k_m. Figure 25.2 shows this loop for the exact data of Exercise 25.5(e).

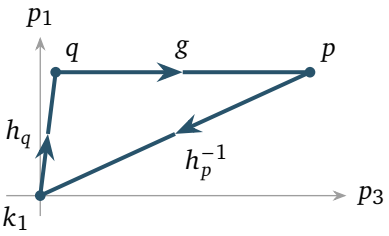


Figure 25.2: The loop of the Wigner rotation for $m = 1$ and $g = \mathrm{diag}(2, \frac{1}{2})$. The slice $p_2 = 0$ of X_1^+ is drawn as a graph over the (p_3, p_1)-plane, with $q = (\frac{713}{128}, \frac{87}{16}, 0, \frac{87}{128})$ and $p = (\frac{6713}{512}, \frac{87}{16}, 0, \frac{6087}{512})$. The boost h_q moves k_1 to q, the boost g moves q to p, and the boost h_p^{-1} moves p back to k_1. The product fixes k_1, so it lies in $\mathrm{SU}(2)$. It acts on the spin fibre V_n, which the drawing of the base does not show.

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

Proposition 25.15. *(a)* $W(g,p) \in \mathrm{SU}(2)$.
(b) $W(gg',p) = W(g,p)\,W(g',\Lambda(g)^{-1}p)$.
(c) $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
(d) If μ_m is Λ-invariant, then each $U(a,g)$ is unitary.
(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.

Proof. (a) Put $q = \Lambda(g)^{-1}p$. Then h_q maps k_m to q, g maps q to p, and h_p^{-1} maps p to k_m. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With q as above, $W(g,p)W(g',q) = h_p^{-1}gh_q\,h_q^{-1}g'h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q \cdot a' = p \cdot \Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and D_n is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2\, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary g, the matrix gh_qg^{-1} is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

On translations, $W(I,p) = I$, so $U(a,I)$ is multiplication by $e^{-ip\cdot a}$. Hence, as in Example 25.5, the measure E of this model has $E(S)$ equal to multiplication by the indicator function of $S \cap X_m^+$. So $E(X_m^+) = I$, and $E(\{p\}) = 0$ for every point p. Formula (25.6) also has no square root of a Jacobian, unlike the induced representations of Lecture 17, because μ_m is invariant.

By the rotation case of the Wigner rotation, Proposition 25.15(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum k_m the little group acts through D_n, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by t acts by multiplication by e^{-itp_0}. Its self-adjoint generator P_0 is multiplication by p_0, with domain

$$\mathrm{Dom}(P_0) = \{\psi \in \mathcal{H}_{m,n} : p_0\psi \in \mathcal{H}_{m,n}\}.$$

Since $p_0 \geq m$ on X_m^+, we get $P_0 \geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of μ_m.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$. Rotations fix p_0 and act orthogonally on $\vec{p}$. The boost d_s needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported ψ, the formula is jointly continuous in (a,g,p). The positive square root h_p is continuous by matrix functional calculus. For (a,g) in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all ψ.
3. *Identification with induction.* The section map is

   $$f \longmapsto \psi_f, \qquad \psi_f(p) = f(0, h_p).$$

   It carries the induced representation from $A \rtimes \mathrm{SU}(2)$, with inducing data $\chi_{k_m} \otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.10(b), shows that the model is irreducible.

Remark 25.16. The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. Restricted to $\mathrm{SU}(2)$, the components of a field in (j_L, j_R) carry the spins $|j_L - j_R|, \dots, j_L + j_R$ of Lecture 24. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

Self-checks

1. Is $(-1,0,0,0)$ in the orbit of k_1?
   Answer. No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of H_k on $\mathbb{C}^2$ preserve some inner product?
   Answer. No. By Proposition 25.13, every finite-dimensional continuous unitary representation of H_k is trivial on N. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   Answer. $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.15(e).

Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by Λ. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. The projection $E(S)$ selects the part of the space with momentum in S, and in the massive model it is multiplication by an indicator function. Mackey's theorem reduces the classification to little groups. It already gives the irreducible unitary representations of the $ax+b$ group and of the Heisenberg group. Up to unitary equivalence, irreducible unitary representations with E concentrated on X_m^+ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.

Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Lecture 12 induced from a character of a torus and obtained infinitely many irreducible representations. In this lecture a complex structure selects one of them. Induced representations act on equivariant functions, and the same functions can be read as sections of a vector bundle. We compute the holomorphic sections completely for the projective line. We then state the general Borel–Weil theorem, work the case* SU(3) *by hand, and prove the theorem from Peter–Weyl.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting SU(2) representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention, and write explicit sections for SU(3).
- Prove Borel–Weil from Peter–Weyl, and name the inputs that stay cited: $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. So it has a smooth local section near every point of G/H, by the local-section theorem (Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26). We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

These transition functions make $E = G \times_H V$ a smooth complex vector bundle, and this structure does not depend on the chosen family of local sections. The tool is the vector bundle chart lemma (Lee, Lemma 10.6). The section on sources at the end of this lecture checks its hypotheses.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h)f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. Moreover, s is smooth exactly when f is smooth, and s is continuous exactly when f is continuous. The section on sources gives the proof.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

We first compute everything on one example, and we start with the holomorphic language it needs. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense above.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

Since $w = 1/z$, the plane of z with one point added at infinity is all of $\mathbb{CP}^1$, so $\mathbb{CP}^1$ is a sphere. Write $E_0 = (e_0^*)^m$ and $E_1 = (e_1^*)^m$ for the two frames of $\mathcal{O}(m)$. They satisfy $E_1 = z^m E_0$ on the overlap. A section value over a point has one coefficient in each frame. If the frame grows by z^m, then the coefficient shrinks by z^{-m}. Figure 26.1 shows this at one point.

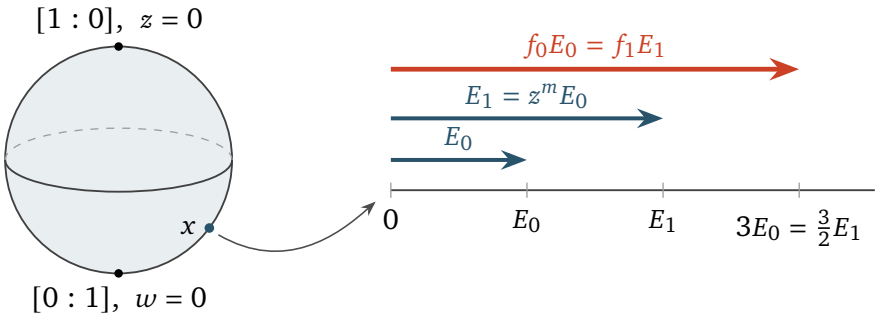


Figure 26.1: The sphere $\mathbb{CP}^1$ with the point x where $z = 2$ and $w = \frac{1}{2}$, and the fibre of $O(1)$ over x, drawn as a real line. On this sphere the chart z omits the south pole $[0 : 1]$, and the chart w omits the north pole $[1 : 0]$. Here $E_1 = 2E_0$, and the section value $3E_0$ has coefficient $f_0 = 3$ in the frame E_0 and coefficient $f_1 = \frac{3}{2}$ in the frame E_1.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from Stein and Shakarchi, *Complex Analysis* (2003), Chapter 2, Section 4. The section on sources records the exact statements. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1,z) = f_0(z)$ and $F(w,1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k} v_1^k$,

$$\Pi_m(t_\theta) F_k = e^{i(2k-m)\theta} F_k, \qquad 0 \leq k \leq m.$$

The weights are $-m, -m+2, \ldots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis $(v_0^2, v_0 v_1, v_1^2)$, expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\, v_0^2 + (|a|^2 - |b|^2) v_0 v_1 + a\bar{b}\, v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \geq 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta} f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section.

Next we identify the whole bundle. Let m be any integer. At the line $\mathbb{C}(1,0)$, the element t_θ acts on the tautological fibre by $e^{i\theta}$. It acts on the mth dual power of that fibre by $\chi_{-m}(t_\theta) = e^{-im\theta}$. Write $\mathbb{C}_{-m}$ for $\mathbb{C}$ with T acting through the character χ_{-m}.

Let ε be the value at $z = 0$ of the frame $(e_0^*)^m$. For $m < 0$, this frame means $e_0^{\otimes(-m)}$, consistently with $e_1^* = ze_0^*$ and $e_1 = z^{-1}e_0$. The map $[g,u] \mapsto g \cdot (u\varepsilon)$ from $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$ to $\mathcal{O}(m)$ is well defined, because t_θ multiplies ε by $\chi_{-m}(t_\theta)$. It is $\mathrm{SU}(2)$-equivariant and linear on each fibre. Since $\mathrm{SU}(2)$ acts transitively on $\mathbb{CP}^1$ with stabiliser T, it is bijective. We now show that its coefficients in the local trivialisations are smooth and nowhere zero.

Over U_0, use the smooth local section $s(z) = g_v$ of $\mathrm{SU}(2) \to \mathrm{SU}(2)/T$, where $v = (1,z)/\sqrt{1+|z|^2}$. Since $g_v(1,0)^T = v$, we have

$$s(z)^{-1}(1,z)^T = \sqrt{1+|z|^2}\,(1,0)^T.$$

The element $s(z)$ sends a functional ϕ on $\mathbb{C}(1,0)$ to $\phi \circ s(z)^{-1}$ on the line through $(1,z)$. Evaluating at $e_0(z) = (1,z)$ gives $s(z) \cdot e_0^*(0) = \sqrt{1+|z|^2}\, e_0^*(z)$. For vectors, the same display gives $s(z)e_0(0) = (1+|z|^2)^{-1/2}e_0(z)$. Taking mth or $(-m)$th tensor powers, we obtain for every integer m

$$s(z) \cdot (u\varepsilon) = u\,(1+|z|^2)^{m/2}\,(e_0^*)^m(z).$$

Over U_1, the section g_v with $v = (w,1)/\sqrt{1+|w|^2}$ gives the coefficient $u(1+|w|^2)^{m/2}$ with respect to $(e_1^*)^m$, by the same computation.

Thus, in both local trivialisations, the map is $u \mapsto u(1+|x|^2)^{m/2}$, where x is the chart coordinate. This coefficient is smooth. Its factor $(1+|x|^2)^{m/2}$ is nowhere zero, so the inverse $c \mapsto c(1+|x|^2)^{-m/2}$ is smooth too. Hence the map is an isomorphism of smooth line bundles

$$\mathcal{O}(m) \cong \mathrm{SU}(2) \times_T \mathbb{C}_{-m}.$$

A section in the equivariant-function description therefore obeys $f(gt_\theta) = e^{im\theta} f(g)$. The positive exponent belongs to the function covariance. The fibre itself has the negative character. For $m \geq 0$ and a homogeneous polynomial F of degree m, the vector $g \cdot (u\varepsilon)$ takes the value u at $g(1,0)^T$, by the definition of ε. With $u = f_F(g) = F(g(1,0)^T)$, it agrees with F at that vector, so the two descriptions of the section agree.

Now we can say what holomorphy does. The Hilbert completion of the continuous induced representation of Lecture 12 is the space of all L^2 sections of $\mathcal{O}(m)$. Let $m \geq 0$. By Frobenius

reciprocity, V_d occurs in this space once when $e^{-im\theta}$ is a weight of V_d, that is, for $d = m, m+2, m+4, \ldots$. Figure 26.2 shows this list. By Theorem 26.3, the holomorphic sections form exactly the first term V_m. For $m < 0$ the same reciprocity gives $V_{|m|}, V_{|m|+2}, \ldots$ while holomorphy keeps nothing.

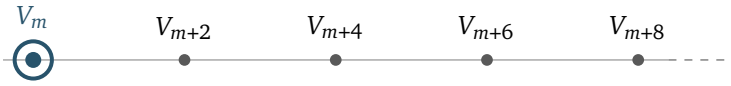


Figure 26.2: The irreducible representations of SU(2) in the L^2 sections of $O(m)$, for $m \geq 0$. Each dot occurs once. The ringed dot V_m is the space of holomorphic sections.

In physics language, $O(m)$ is the bundle of a charged particle on a sphere around a magnetic monopole, its two charts are two gauge patches, and the holomorphic sections are the lowest Landau level.

26.5 The general Borel–Weil theorem

We now replace $\mathrm{SL}(2,\mathbb{C})$ and its upper triangular subgroup by a complex group G and a Borel subgroup B. The SU(2) case at the end of this section shows that the calculation above is the first instance.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Holomorphic bundles. Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that π has holomorphic local sections s_i whose domains U_i cover G/H. Then $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps. So the local trivialisations given by the sections s_i make E a complex manifold. In this case E is a *holomorphic* vector bundle. These trivialisations are its *holomorphic bundle charts*, and the complex structure of E does not depend on the family (s_i); the section on sources checks these points. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

A section is holomorphic exactly when its function $f : G \to V$ is holomorphic. Indeed, if f is holomorphic, then so is $f \circ s_i$. Conversely, the map $q(g) = s_i(\pi(g))^{-1} g$ on $\pi^{-1}(U_i)$ is holomorphic into H by the slice-chart property. The covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big),$$

and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G.

The groups. Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14, and it is connected because K is. Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra.

We cite two facts about these groups. First, B is closed, and the inclusion of K in G induces a diffeomorphism from K/T onto the *flag manifold* G/B (Sepanski, *Compact Lie Groups*, Lemma 7.49 and Theorem 7.50). In particular,

$$G = KB.$$

Second, B is a closed complex subgroup, and G/B is a complex manifold with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections (Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 2.11 and its proof). So the paragraph on holomorphic bundles applies to G, B and G/B. The section on sources identifies Sepanski's matrix model of G with the group of Lecture 14, and it proves that B is a complex subgroup.

The character. Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. Each $b \in B$ is uniquely a product $t\exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$, and these factors depend smoothly on b (Sepanski, Lemma 7.49(d)). Our character of weight λ on T is $\exp H \mapsto e^{i\lambda(H)}$. Define $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp\mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

The bundle. Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with

$$f(gb) = \chi_\lambda(b) f(g). \tag{26.4}$$

A section is holomorphic exactly when its function $f : G \to \mathbb{C}$ is holomorphic. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

The proof below uses the cited facts about B, χ_λ and G/B above, in particular $G = KB$. It also uses the holomorphic extension of Lecture 14 and the highest-weight theorem of the prerequisite course. The rest comes from the Peter–Weyl theorem.

The dual V_λ^* has highest weight $-w_0\lambda$. Here w_0 is the longest element of the Weyl group, the element that sends the positive roots to the negative roots. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels. For $\mathrm{SU}(2)$ we have $-w_0\lambda = \lambda$, so the dual is invisible there.

The case $\mathrm{SU}(3)$ by hand

Here $G = \mathrm{SL}(3,\mathbb{C})$, and B is the subgroup of upper triangular matrices. A *flag* is a line ℓ inside a plane P in $\mathbb{C}^3$. Figure 26.3 shows one. The group G moves flags, and B is the stabiliser of the standard flag $\mathbb{C}e_1 \subset \mathbb{C}e_1 \oplus \mathbb{C}e_2$. The group $K = \mathrm{SU}(3)$ already moves flags transitively, with stabiliser the diagonal torus T. So $K/T = G/B$ is the space of flags.

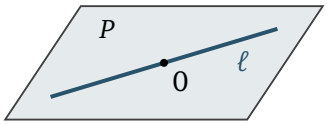


Figure 26.3: A flag $\ell \subset P$ in $\mathbb{C}^3$, drawn schematically: the complex line ℓ and the complex plane P appear as a real line in a real plane.

Example 26.5 (Sections for $\mathrm{SU}(3)$)**.** Let $\lambda = a\omega_1 + b\omega_2$ with integers $a, b \geq 0$, and write β for an upper triangular matrix in G. The character is trivial on the unipotent matrices. On the torus, ω_1 is the first diagonal entry and ω_2 is the product of the first two. So

$$\chi_\lambda(\beta) = \beta_{11}^{a+b}\beta_{22}^{b}.$$

Take a linear functional Φ on $\mathbb{C}^3$ and put $f(g) = \Phi(ge_1)$. Since $(g\beta)e_1 = \beta_{11}ge_1$, we get $f(g\beta) = \beta_{11}f(g)$. So f is a section for $(a,b) = (1,0)$. These sections form $(\mathbb{C}^3)^*$, the dual of $V_{(1,0)} = \mathbb{C}^3$, as the theorem says.

Next take a linear functional Ψ on $\Lambda^2\mathbb{C}^3$ and put $f(g) = \Psi(ge_1 \wedge ge_2)$. Since $(g\beta)e_1 \wedge (g\beta)e_2 = \beta_{11}\beta_{22}\, ge_1 \wedge ge_2$, this f is a section for $(0,1)$. These sections form the dual of $V_{(0,1)} = \Lambda^2\mathbb{C}^3$.

A product of a sections for $(1,0)$ and b sections for $(0,1)$ is a section for (a,b). These products need not be independent. For $(1,1)$, the nine products of coordinates of ge_1 with coordinates of $ge_1 \wedge ge_2$ satisfy one relation, since $ge_1 \wedge (ge_1 \wedge ge_2) = 0$. They span the eight-dimensional dual of the adjoint module.

Proof of the Borel–Weil theorem

Proof. We build sections from V_λ, and then we show that there are no others. Throughout, R_X denotes the derivative of a function φ on K along the right translation by $\exp tX$:

$$(R_X\varphi)(k) = \frac{d}{dt}\bigg|_0 \varphi(k \exp tX), \qquad X \in \mathfrak{k}.$$

We extend it complex-linearly to $Z \in \mathfrak{k}_\mathbb{C} = \mathfrak{g}$, with $R_{X+iY} = R_X + iR_Y$.

Step 1: sections from V_λ. Let λ be dominant. A module V_λ of highest weight λ exists by the highest-weight theorem of the prerequisite course and the integration theorem of Lecture 4. It extends to a holomorphic representation of G by Lecture 14. Let v_λ be a highest weight vector. Write Π_λ for this representation. On $\mathfrak{b}$, the operator $d\Pi_\lambda(Z)$ acts on v_λ by the scalar $d\chi_\lambda(Z)$: this is $i\lambda(H)$ for $H \in \mathfrak{t}$ and zero on the positive root spaces. So $b \mapsto \Pi_\lambda(b)v_\lambda$ and $b \mapsto \chi_\lambda(b)v_\lambda$ agree on $\exp \mathfrak{b}$, and hence on the connected group B:

$$\Pi_\lambda(b)v_\lambda = \chi_\lambda(b)v_\lambda.$$

For $\xi \in V_\lambda^*$ put $f_\xi(g) = \xi(\Pi_\lambda(g)v_\lambda)$. This function is holomorphic, and

$$f_\xi(gb) = \xi(\Pi_\lambda(g)\Pi_\lambda(b)v_\lambda) = \chi_\lambda(b)f_\xi(g).$$

So f_ξ is a holomorphic section of L_λ.

Step 2: the dual. Left translation by a gives

$$f_\xi(a^{-1}g) = \xi(\Pi_\lambda(a)^{-1}\Pi_\lambda(g)v_\lambda) = f_{\xi\circ\Pi_\lambda(a)^{-1}}(g).$$

So $\xi \mapsto f_\xi$ intertwines the dual representation with the section action. This is the reason for the dual: the section is linear in the functional ξ, and left translation acts on ξ. The map is injective. Indeed, if $f_\xi = 0$, then ξ vanishes on all $\Pi_\lambda(g)v_\lambda$. These vectors span V_λ because it is irreducible, so $\xi = 0$.

Step 3: the equations on K. Now let λ be any integral weight, and let f be holomorphic on G with the covariance (26.4). Let $\varphi = f|_K$, a smooth function on K. For $Z \in \mathfrak{g}$,

$$(R_Z\varphi)(k) = \left.\frac{d}{dt}\right|_0 f(k\exp tZ),$$

since the differential of the holomorphic function f is $\mathbb{C}$-linear. This is the point where holomorphy is used. A smooth function on K that is only covariant under T would not satisfy the second equation below. For $Z \in \mathfrak{b}$, the covariance gives $f(k\exp tZ) = e^{t\,d\chi_\lambda(Z)}f(k)$. Hence

$$R_H\varphi = i\lambda(H)\varphi \quad (H \in \mathfrak{t}), \qquad R_E\varphi = 0 \quad (E \text{ in a positive root space}). \tag{26.5}$$

Step 4: one block at a time. Let Π be an irreducible unitary representation of K, and let P_Π be the isotypic projector of Lecture 11. It is an integral of left translations:

$$(P_\Pi\varphi)(x) = d_\Pi \int_K \chi_\Pi(y)\,\varphi(y^{-1}x)\,dy.$$

Here χ_Π is the character of Π. The derivative $R_X\varphi$ is continuous on the compact group K, so we may differentiate under this integral, and $R_ZP_\Pi\varphi = P_\Pi R_Z\varphi$ for $Z \in \mathfrak{g}$. So $P_\Pi\varphi$ also satisfies (26.5).

The function $P_\Pi\varphi$ lies in the coefficient block M_Π. Let (ξ_i) be a basis of V_Π^*. Then

$$(P_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)v_i), \qquad v_i \in V_\Pi,$$

and differentiating gives $(R_ZP_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)d\Pi(Z)v_i)$. By Schur orthogonality (Lecture 5), the matrix coefficients are linearly independent: a sum $\sum_i \xi_i(\Pi(k)w_i)$ vanishes for all k only if every $w_i = 0$. So (26.5) gives

$$d\Pi(H)v_i = i\lambda(H)v_i, \qquad d\Pi(E)v_i = 0.$$

Thus each nonzero v_i is a highest weight vector of weight λ in the irreducible module V_Π. By the highest-weight theorem of the prerequisite course, if some $v_i \neq 0$, then λ is dominant and $\Pi \cong V_\lambda$. Its highest weight line is one-dimensional, so $v_i = c_i v_\lambda$, and

$$(P_\Pi \varphi)(k) = \xi(\Pi(k) v_\lambda), \qquad \xi = \sum_i c_i \xi_i .$$

For every other Π, we get $P_\Pi \varphi = 0$.

Step 5: completeness. We have not assumed that φ is a finite sum of coefficients. The Peter–Weyl theorem, in the form of the Plancherel theorem of Lecture 11, says that the sums of $P_\Pi \varphi$ converge to φ in $L^2(K)$. By Step 4 at most one term is nonzero. So φ equals that term in $L^2(K)$, and both functions are continuous, so they agree on K. If λ is not dominant, then $\varphi = 0$. If λ is dominant, then $\varphi = f_\xi|_K$ for some ξ.

Finally, $G = KB$, and the covariance determines a function on G from its restriction to K. So $f = 0$ when λ is not dominant, and $f = f_\xi$ when λ is dominant. With Step 2, this gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$. The dual of an irreducible module is irreducible. ∎

The case SU(2)

For $K = \mathrm{SU}(2)$ the group G is $\mathrm{SL}(2, \mathbb{C})$ (Lecture 14), and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}. The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

We now compare the two section spaces directly. Let f be holomorphic on G with $f(gb) = \chi_m(b) f(g)$. The character is trivial on unipotent b, so f depends only on the first column $v = g(1,0)^T$. Put $F(v) = f(g)$. On $\{v_0 \neq 0\}$ and on $\{v_1 \neq 0\}$, the local sections

$$v \longmapsto \begin{pmatrix} v_0 & 0 \\ v_1 & 1/v_0 \end{pmatrix}, \qquad v \longmapsto \begin{pmatrix} v_0 & -1/v_1 \\ v_1 & 0 \end{pmatrix}$$

have determinant one and holomorphic entries. So F is holomorphic on $\mathbb{C}^2 \setminus \{0\}$. Also $F(cv) = c^m F(v)$ for $c \neq 0$, by the character on $\mathrm{diag}(c, c^{-1})$. Hence $f_0(z) = F(1, z)$ and $f_1(w) = F(w, 1)$ are entire, and they satisfy the transition rule (26.2). By Theorem 26.3, for $m \geq 0$ there is a homogeneous polynomial, again called F, of degree m with these two coefficients, so that $f = f_F$. For $m < 0$, the same theorem gives $f_0 = f_1 = 0$, so $f = 0$. Conversely, each $f_F(g) = F(g(1,0)^T)$ is holomorphic and covariant.

So $F \mapsto f_F$ is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$, and for $m < 0$ both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil. In Step 1 of the proof, $V_m = \mathrm{Sym}^m \mathbb{C}^2$ has highest weight vector $e_1^{\otimes m}$, with $e_1 = (1,0)^T$. So $f_\xi(g) = \xi((g e_1)^{\otimes m})$, which is the function f_F for the polynomial $F(v) = \xi(v^{\otimes m})$.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$, where w_0 is the longest element of the Weyl group. For $\mathrm{SU}(2)$ these weights coincide. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels, so for $\mathrm{SU}(3)$ the bundle $L_{(1,0)}$ gives highest weight $(0, 1)$.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Compact induction contains $V_m, V_{m+2}, \dots$, and holomorphy keeps only V_m. For compact, connected, simply connected, semisimple K, the Borel–Weil theorem gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$ for dominant integral λ, and zero otherwise. We proved it from Peter–Weyl. Covariance and holomorphy make the vectors in each coefficient block highest weight vectors of weight λ, so only one block survives. The facts $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem stay cited. For $\mathrm{SU}(3)$, first-column and wedge coordinates give explicit sections.

Sources and hypotheses

This section collects what the main line cites, with the checks of hypotheses and the matching of conventions. Nothing later depends on the order in which it is read.

Smooth structure of associated bundles. We show that $E = G \times_H V$ is a smooth complex vector bundle. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1} s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

Regularity of covariant functions. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1}g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

Holomorphic inverse functions. Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*.

Holomorphic bundles. We work in the setting of the paragraph on holomorphic bundles in the main line. If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1}s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family.

Complex analysis. [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \leq k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k.

The matrix model of $K_{\mathbb{C}}$. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G. We identify the two groups. A faithful representation ρ of K exists (Lecture 10; also [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]). Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N, \mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N, \mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

The Borel subgroup and the quotient. By Sepanski, Definition 7.48, B is the connected Lie subgroup of G with Lie algebra $\mathfrak{b}$. By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X, Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0, 0)$ is the complex-linear isomorphism $(X, Y) \mapsto b(X + Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Weights and the source of the theorem. Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)). In our labels, Sepanski, Definition 7.52, defines the character χ_λ of the main line. Sepanski's Theorem 7.58 states the Borel–Weil theorem for the groups G and B above. His Definition 7.54 calls a section holomorphic when its function on G is holomorphic, and by the equivalence in the main line his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, so his theorem gives the same V_λ^*. Our proof does not use his theorem.

The group $\mathrm{SL}(2, \mathbb{C})$ as $\mathrm{SU}(2)_{\mathbb{C}}$. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2, \mathbb{C})$ (Lecture 14). We give $\mathrm{SL}(2, \mathbb{C})$ its complex structure as the subset $\{\det = 1\}$ of $\mathbb{C}^4$. This subset is a complex submanifold, with explicit slice charts. On the open set $\{g_{11} \neq 0\} \subset \mathbb{C}^4$, put

$$\varphi(g_{11}, g_{12}, g_{21}, g_{22}) = (g_{11}, g_{12}, g_{21},\ g_{11}g_{22} - g_{12}g_{21} - 1).$$

The map φ is holomorphic onto the open set $\{(a, b, c, t) : a \neq 0\}$, with holomorphic inverse

$$(a, b, c, t) \longmapsto \big(a, b, c, (t + 1 + bc)/a\big).$$

In this chart, $\mathrm{SL}(2, \mathbb{C})$ is the set $t = 0$. On $\{g_{12} \neq 0\}$, $\{g_{21} \neq 0\}$ and $\{g_{22} \neq 0\}$, solve $\det g = 1 + t$ in the same way for the entry that multiplies the nonzero one in $\det g$. These four open sets cover $\mathrm{SL}(2, \mathbb{C})$, because $\det g = 1$ forces some entry of g to be nonzero.

By the slice-chart property, a map into $\mathrm{SL}(2, \mathbb{C})$ with holomorphic entries is holomorphic. On $\mathrm{SL}(2, \mathbb{C})$ the inverse is

$$g^{-1} = \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

Hence multiplication and inversion have polynomial entries, and $\mathrm{SL}(2, \mathbb{C})$ is a complex Lie group.

We now compare this structure with the complex structure of $\mathrm{SU}(2)_{\mathbb{C}} = \mathcal{X}(\mathrm{SU}(2))$ from Lecture 14. The example for $\mathrm{SU}(2)$ in Lecture 14 shows that the four matrix entries D_{ij} of

the standard representation generate the representative functions of SU(2). It also gives the bijection

$$P : \mathrm{SU}(2)_{\mathbb{C}} \longrightarrow \mathrm{SL}(2,\mathbb{C}), \qquad P(\psi) = \big(\psi(D_{ij})\big)_{i,j=1,2}.$$

The formula $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$ gives $P(\psi_1\psi_2) = P(\psi_1)P(\psi_2)$, so P is a group isomorphism. Also $P(\mathrm{ev}_g) = g$ for $g \in \mathrm{SU}(2)$. In the notation of Bröcker and tom Dieck, P is the map $r_{\mathbb{C}}$ for the standard representation r of SU(2).

Cited result. We apply [C: Bröcker and tom Dieck, *Representations of Compact Lie Groups*, Section III.8, Proposition 8.2 and the paragraph after it] to the standard representation r, whose entries generate the representative functions. Proposition 8.2 states that $r_{\mathbb{C}} = P$ maps $\mathrm{SU}(2)_{\mathbb{C}}$ bijectively onto an affine variety in $\mathbb{C}^4$. This variety is the image $\mathrm{SL}(2,\mathbb{C})$ of P. The paragraph after it states that the topology of $\mathrm{SU}(2)_{\mathbb{C}}$ is the subspace topology induced by P. It also states that the variety is an analytic submanifold of $\mathbb{C}^4$, on which multiplication and inversion are holomorphic. With this structure, P is a biholomorphic group isomorphism from $\mathrm{SU}(2)_{\mathbb{C}}$ onto $\mathrm{SL}(2,\mathbb{C})$ that is the identity on SU(2).

This analytic-submanifold structure on $\mathrm{SL}(2,\mathbb{C})$ coincides with the slice-chart structure above. Indeed, the inclusion of $\mathrm{SL}(2,\mathbb{C})$ into $\mathbb{C}^4$ is holomorphic for each structure. By the slice-chart property of the other structure, the identity map of $\mathrm{SL}(2,\mathbb{C})$ is holomorphic in each direction.

Here we use the standard representation r. We identify $\mathrm{SU}(2)_{\mathbb{C}}$ with $G = \mathrm{SL}(2,\mathbb{C})$ through P, which is the identity on SU(2). By the polynomial argument above, the complex structure of Lecture 14 does not depend on the generating representation. So it is the structure transported by P, which is $r_{\mathbb{C}}$ for the standard representation. By the cited result and the preceding paragraph, this structure coincides with the submanifold structure on G.

Consider the maps

$$s_0(z) = \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}, \qquad s_1(w) = \begin{pmatrix} w & -1 \\ 1 & 0 \end{pmatrix}.$$

Both matrices have determinant one. Their first columns are the tautological frames $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. Identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$, as in the main line. Then s_0 and s_1 are local sections of $G \to G/B$ over U_0 and U_1. Their entries are polynomials, so they are holomorphic maps into G.

At a point of the overlap, $w = 1/z$ and

$$h_{01} = s_0(z)^{-1} s_1(1/z) = \begin{pmatrix} 1 & 0 \\ -z & 1 \end{pmatrix} \begin{pmatrix} z^{-1} & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} z^{-1} & -1 \\ 0 & z \end{pmatrix} \in B.$$

Hence $\chi_m(h_{01}) = z^{-m}$.

Let $f : G \to \mathbb{C}$ satisfy $f(gb) = \chi_m(b) f(g)$, and put $f_0 = f \circ s_0$ and $f_1 = f \circ s_1$. Since $s_1(1/z) = s_0(z) h_{01}$, covariance gives

$$f_1(1/z) = f\big(s_0(z) h_{01}\big) = z^{-m} f_0(z) \qquad (z \neq 0).$$

With $w = 1/z$, this is the transition rule $f_1(w) = w^m f_0(1/w)$ of $\mathcal{O}(m)$.

Next we compare the two complex structures on G/B. Write $\pi : G \to G/B$ for the projection, and identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$. The vector $g(1,0)^T$ is the first column

$(g_{11}, g_{21})^T$ of g. Hence $\pi^{-1}(U_0) = \{g_{11} \neq 0\}$ and $\pi^{-1}(U_1) = \{g_{21} \neq 0\}$ are open in G. In the charts,

$$z \circ \pi(g) = \frac{g_{21}}{g_{11}} \quad \text{on } \pi^{-1}(U_0), \qquad w \circ \pi(g) = \frac{g_{11}}{g_{21}} \quad \text{on } \pi^{-1}(U_1).$$

These are quotients of holomorphic functions with nonvanishing denominators. Hence π is holomorphic for the chart structure. The sections s_0 and s_1 are holomorphic for the chart structure, because their entries are polynomials. By the comparison of the two group structures above, the cited structure on G/B also makes π holomorphic for the submanifold structure on G, and it has holomorphic local sections.

Let s be a holomorphic local section for one of the two structures. Then $\pi \circ s$ is the identity map on the domain of s. This composition is holomorphic into the other structure, because π is holomorphic for that structure. Applying this in both directions shows that the identity map of G/B is holomorphic with holomorphic inverse. Hence the two complex structures coincide.

Let s' be another holomorphic local section over an open subset of U_0. Then $q = s_0^{-1}s'$ takes values in B, and it is holomorphic into B by the slice-chart property. Covariance gives $f \circ s' = \chi_m(q) f_0$. The factor $\chi_m(q)$ is holomorphic and nowhere zero, so it does not change holomorphy. The same holds over U_1. Hence a section of L_m is holomorphic exactly when f_0 and f_1 are entire.

This is the agreement of the two notions of holomorphic section that the projective-line calculation uses.

Reading

- M. R. Sepanski, *Compact Lie Groups* (2007), Section 7.4: the Borel–Weil theorem and the groups G and B.
- J. M. Lee, *Introduction to Smooth Manifolds*, second edition (2013), Chapters 4, 5 and 10: local sections, embedded submanifolds and the vector bundle chart lemma.
- T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (1985), Sections III.4 and III.8: faithful representations and complexification.

Lecture 27

Compact orbits and geometric quantisation

Objectives. *A compact coadjoint orbit has a symplectic form. Not every such orbit supplies a representation of the chosen global group. We calculate the obstruction on a sphere. The symplectic area of the sphere divided by $2\pi\hbar$ must be integral. An equivariant lift imposes a further condition when the group is* $\mathrm{SO}(3)$. *Holomorphic sections then connect the orbit to the representations of the previous lecture.*

By the end of this lecture you will be able to:

- Compute the coadjoint-orbit form of $\mathrm{SU}(2)$ with its radius normalisation.
- Define a unitary connection locally and prove the sphere integrality condition.
- Distinguish an ordinary prequantum bundle from an equivariant one.
- Recover spin $m/2$ by holomorphic sections and keep this construction separate from shifted character formulas.

27.1 The orbit sphere and its area

Use the basis of Lecture 4,

$$X_j = -\frac{i}{2}\sigma_j, \qquad [X_1, X_2] = X_3$$

with cyclic versions of the bracket. The Pauli matrices satisfy

$$\sigma_i\sigma_j = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k.$$

Identify $\mathfrak{su}(2)^*$ with $\mathbb{R}^3$ by $x_j = \ell(X_j)$. For $u = (u_1, u_2, u_3)$ put $X_u = \sum_j u_j X_j$. Then $[X_u, X_v] = X_{u\times v}$.

The adjoint action is the rotation cover from Lecture 4. Its dual action is the same rotation action in these orthonormal coordinates. Thus its orbits are the spheres

$$S_r^2 = \{x : |x| = r\} \quad (r > 0)$$

and the origin. The infinitesimal action is $\xi_u(x) = u \times x$. Indeed, pairing it with v gives $(u \times x)\cdot v = -x \cdot (u \times v)$, which is exactly the dual derivative from Lecture 19.

Proposition 27.1. *Let $r > 0$. The orbit form at $x \in S^2_r$ is*

$$\omega_x(v, w) = \frac{x \cdot (v \times w)}{r^2}, \qquad v, w \in T_x S^2_r.$$

If Ω is the outward-oriented area form of the sphere, then $\omega = \Omega/r$. In spherical coordinates,

$$\omega = r \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2_r} \omega = 4\pi r. \tag{27.1}$$

Proof. The KKS formula of Lecture 19 requires

$$\omega_x(u \times x, v \times x) = x \cdot (u \times v).$$

At $x = re_3$, the two tangent vectors are $r(u_2, -u_1, 0)$ and $r(v_2, -v_1, 0)$. Substitution in the stated formula gives $r(u_1 v_2 - u_2 v_1)$, the required value. Rotation invariance gives the formula at every x.

The outward area form is $\Omega_x(v, w) = x \cdot (v \times w)/r$. Thus $\omega = \Omega/r$. The usual sphere parametrisation has area form $r^2 \sin\theta \, d\theta \wedge d\phi$, giving the spherical formula (27.1). Integration over $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$ gives $4\pi r$. ∎

Write $\xi_j = \xi_{e_j}$ for the standard basis vectors e_j. At the north pole, $\xi_1 = -r\partial_{x_2}$ and $\xi_2 = r\partial_{x_1}$, and $\omega(\xi_1, \xi_2) = r$. This is a convenient sign check. The origin must be treated separately: it is a point orbit, and none of the formulas dividing by r applies there.

Fix a positive constant $\hbar$. It sets the scale at which we compare symplectic area with phases. Keeping it explicit will make the integrality condition easier to track.

27.2 Connections described in local frames

A Hermitian line bundle is a complex line bundle with a smoothly varying positive inner product on its fibres. A local *unitary frame* is a section of norm one that never vanishes. In such a frame, a section is described by a complex coefficient function u.

Definition 27.2. A *unitary connection* differentiates local coefficients by

$$\nabla u = du + Au,$$

where A is an imaginary-valued 1-form. Let unitary frames e_i on U_i and e_j on U_j have connection forms A_i and A_j. If $e_j = g_{ij} e_i$, with $g_{ij} : U_i \cap U_j \to \mathrm{U}(1)$, the connection forms are required to satisfy

$$A_j = A_i + g_{ij}^{-1} dg_{ij}.$$

The *curvature* F_∇ is the imaginary-valued 2-form given on U_i by $F_\nabla = dA_i$.

Here du is the ordinary differential, evaluated on tangent vectors. A section s has coefficients u_i and u_j in the two frames. Since $e_j = g_{ij} e_i$, these coefficients satisfy $u_i = g_{ij} u_j$, or equivalently $u_j = g_{ij}^{-1} u_i$. Differentiating $u_i = g_{ij} u_j$ gives

$$du_i + A_i u_i = g_{ij}\big(du_j + (A_i + g_{ij}^{-1} dg_{ij}) u_j\big).$$

The left side is the coefficient of ∇s in the frame e_i. In the frame e_j this coefficient is $du_j + A_j u_j$, and coefficients change by the factor g_{ij}. Hence frame independence of ∇s for every section is exactly the rule for A_j in the definition.

Since g_{ij} is scalar, $d(g_{ij}^{-1} dg_{ij}) = 0$ locally. Indeed, write $g_{ij} = e^{i\alpha}$ with real α, so this form is $i\,d\alpha$. Hence $dA_i = dA_j$ and the curvature is global.

The same connection can be written in any nonvanishing frame. The change-of-frame formulas remain valid for $g_{ij} \in \mathbb{C}^\times$, with complex-valued A_i. Only unitary frames require imaginary-valued forms.

Along a path, the equation for parallel transport is $du + Au = 0$. In one frame its solution multiplies u by $\exp(-\int A)$. This is a phase because A is imaginary. Curvature measures the resulting phase around small loops. In physics language, A is a gauge potential and ∇ is a covariant derivative.

Definition 27.3. A *prequantum line bundle* for (M, ω) at scale $\hbar$ is a Hermitian line bundle with unitary connection satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega. \tag{27.2}$$

The sign is part of our convention. A prequantum bundle is an initial geometric input. Its space of all smooth sections is usually far larger than the desired representation. A polarisation, such as holomorphicity, supplies a further selection.

We cite the general prequantisation theorem. A closed real 2-form ω on a smooth manifold admits a prequantum line bundle at scale $\hbar$ exactly when its periods divided by $2\pi\hbar$ are integers. A *period* of ω is the integral of ω over a closed oriented surface mapped smoothly into the manifold, or over an integer combination of such surfaces. Equivalently, $[\omega/(2\pi\hbar)]$ is an integral cohomology class. See Kirillov, Chapter 1, Section 2.4.

We prove the sphere case directly.

27.3 The integral sphere bundle

Put

$$m = \frac{2r}{\hbar}.$$

Take a northern chart excluding the south pole and a southern chart excluding the north pole. Define

$$A_N = -\frac{im}{2}(1 - \cos\theta)d\phi, \qquad A_S = \frac{im}{2}(1 + \cos\theta)d\phi. \tag{27.3}$$

Although $d\phi$ is singular at each pole, A_N is smooth at the north pole and A_S at the south pole. To see this we use a complex coordinate, and we recall its rules. For $z = x + iy$ put

$$dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy, \qquad \partial_z = \tfrac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

Then $df = \partial_z f\, dz + \partial_{\bar{z}} f\, d\bar{z}$ for a function f, and $dz \wedge d\bar{z} = -2i\, dx \wedge dy$. Writing the 1-form $a\,dz + b\,d\bar{z}$ in dx and dy and using the rule of Lecture 19 gives

$$d(a\,dz + b\,d\bar{z}) = (\partial_z b - \partial_{\bar{z}} a)\, dz \wedge d\bar{z}.$$

In the northern complex coordinate $z = \tan(\theta/2)e^{i\phi}$,

$$A_N = \frac{m}{2}\frac{z\,d\bar{z} - \bar{z}\,dz}{1+|z|^2}.$$

This expression is smooth at $z = 0$. The southern formula has the same property in its own coordinate. In physics language, these are the potentials of a magnetic monopole on two patches.

Differentiation gives

$$dA_N = dA_S = -\frac{im}{2}\sin\theta\,d\theta\wedge d\phi = -\frac{i}{\hbar}\omega.$$

On the overlap, $A_S - A_N = im\,d\phi$. Thus A_N and A_S satisfy the change-of-frame formula if the frame transition is

$$e_S = e^{im\phi}e_N.$$

This is a single-valued function on the overlap exactly when $m \in \mathbb{Z}$. For positive radius this means m is a positive integer. In physics language, $m \in \mathbb{Z}$ is Dirac's quantisation condition.

The idea of the necessity proof below is one loop with two caps. Transport around a loop multiplies a fibre by a phase. In a frame on a disk D bounded by the loop, Stokes' theorem makes this phase $\exp(-\int_D F_\nabla)$. In the plane of Lecture 19, a loop bounds one disk, and the Weyl relation of Lecture 18 has a phase of the same kind, a constant times an area. No condition arises there. On a sphere, a loop bounds two disks, with opposite orientations, as Fig. 27.1 shows. The two disks must give the same phase. Hence $\exp(-\int_{S^2} F_\nabla) = 1$, which says that $\int_{S^2}\omega \in 2\pi\hbar\mathbb{Z}$. We now write this with frames. In physics language, the latitude phase of Exercise 27.2 is the Berry phase of a spin.



Figure 27.1: One latitude loop, oriented by increasing ϕ, on two copies of the sphere. Its arrow points the same way in both. In the first copy, the shaded north cap D_N has this loop as its oriented boundary. In the second copy, the shaded south cap D_S has the reversed loop as its oriented boundary, so its phase has the opposite sign in the exponent.

Proposition 27.4. *The sphere S^2_r, $r > 0$, admits a prequantum line bundle if and only if $2r/\hbar \in \mathbb{Z}$. For $m = 2r/\hbar$, the preceding construction supplies one.*

Proof. Sufficiency is the explicit construction. For necessity, suppose that S^2_r carries a prequantum line bundle.

Step 1 (unitary frames on the two disks). Use the two open disk charts obtained by deleting the opposite poles. On each chart, parallel transport a unit vector at its origin along radial

segments. Cover each radial segment by finitely many local unitary frames. The change-of-frame law makes transport agree on overlaps. The resulting finite composition of ODE solutions depends smoothly on the endpoint and gives a smooth unitary frame. Each chart contains a neighbourhood of its closed hemisphere, so these frames are valid up to and beyond the equator. Call them e_N and e_S, with connection forms A_N and A_S.

Step 2 (the transition function and its lift). Write $e_S = g e_N$ near the equator, with $g : S^1 \to \mathrm{U}(1)$. Orient the equator by increasing ϕ, as the boundary of the northern disk.

Lift $g(e^{i\phi})$ to a real phase $\alpha(\phi)$ for $0 \le \phi \le 2\pi$, using the circle covering from Lecture 4. Since the endpoints give the same value, $\alpha(2\pi) - \alpha(0) = 2\pi k$ for an integer k.

Step 3 (Stokes). By the change-of-frame law, $A_S = A_N + g^{-1}dg$. Moreover $g^{-1}dg = i\,d\alpha$, since $g = e^{i\alpha}$ along the equator. Let D_N and D_S be the closed northern and southern hemispheres. We apply the calculus Stokes formula for a 1-form on each oriented disk. The southern disk induces the opposite orientation on the equator. Hence

$$\int_{S^2} F_\nabla = \int_{D_N} dA_N + \int_{D_S} dA_S = \int_{S^1} (A_N - A_S) = -\int_{S^1} g^{-1}dg = -2\pi i k.$$

By the prequantum condition (27.2) and the spherical formula (27.1),

$$k = \frac{i}{2\pi}\int F_\nabla = \frac{1}{2\pi\hbar}\int \omega = \frac{2r}{\hbar}.$$

Hence $2r/\hbar \in \mathbb{Z}$. ∎

Since $m > 0$, the curvature has no global primitive on the sphere. If $F = dA$ globally, Stokes' theorem would give $\int_{S^2} F = 0$. The period is nonzero because $m \neq 0$. The two local potentials are therefore essential.

27.4 Holomorphic sections and equivariance

First we show that $O(m)$, with one metric, carries the prequantum connection. Its holomorphic sections are then the quantisation. Finally we show that the natural SU(2) action is the equivariant lift, and that SO(3) needs even m.

The metric and the connection

Identify the sphere with $\mathbb{CP}^1$ by

$$z \longmapsto \frac{r}{1+|z|^2}\big(2\operatorname{Re} z, 2\operatorname{Im} z, 1-|z|^2\big).$$

The north pole is $z = 0$, and $z = \tan(\theta/2)e^{i\phi}$ is the northern coordinate used above. Recall from Lecture 26 that SU(2) acts on lines by $g\cdot[v] = [gv]$. The identification map is equivariant for this action and the coadjoint action. Indeed, the line of $v = (1, z)$ corresponds to the traceless Hermitian matrix proportional to

$$vv^*/\|v\|^2 - I/2.$$

Conjugation by g sends this matrix to the matrix for the line of gv, and it rotates the Pauli coordinates of the matrix. With $|z| = \tan(\theta/2)$, we have $d|z| = \frac{1}{2}(1+|z|^2)\,d\theta$ and $r\sin\theta = 2r|z|/(1+|z|^2)$. Substitution in the spherical formula (27.1) gives the sphere form

$$\omega = \frac{2ir\,dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Since $i\,dz\wedge d\bar z = 2\,d(\operatorname{Re} z)\wedge d(\operatorname{Im} z)$, this form has the same positive orientation as the spherical formula (27.1).

The holomorphic frame transition of $O(m)$ in Lecture 26 is $e_S^{\mathrm{hol}} = z^m e_N^{\mathrm{hol}}$. The Hermitian metric with

$$\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-m}$$

gives unitary frames $e_N = (1+|z|^2)^{m/2}e_N^{\mathrm{hol}}$ and e_S whose transition is $(z/|z|)^m = e^{im\phi}$. This is the transition of the integral sphere bundle, so the potentials (27.3) define a unitary connection on $O(m)$.

The metric is SU(2)-invariant. To see this, we describe it fibre by fibre. A point f of the fibre over a line $[v]$ is a function on that line with $f(\lambda v) = \lambda^m f(v)$ for $\lambda\in\mathbb{C}^\times$. In this model, the holomorphic sections of Lecture 26 are the homogeneous polynomials of degree m, restricted to each line. On the fibre over $[v]$ put

$$\|f\|^2 = \frac{|f(v)|^2}{\|v\|^{2m}}.$$

Replacing v by λv multiplies the numerator and the denominator by $|\lambda|^{2m}$. Hence the norm does not depend on the chosen vector v. An element $g\in\mathrm{SU}(2)$ sends f to the point gf over $[gv]$ with $(gf)(gv) = f(v)$. Since g is unitary, $\|gv\| = \|v\|$, and so $\|gf\| = \|f\|$. The northern frame e_N^{hol} takes the value 1 at $v = (1,z)$, as the polynomial v_0^m does. Since $\|(1,z)\|^2 = 1+|z|^2$, this norm agrees with the metric displayed above.

A connection is *compatible* with the holomorphic structure when its form in a holomorphic frame has no $d\bar z$ term. Apply the change-of-frame law to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-m/2}e_N$ and use the coordinate formula for A_N above. The form of this connection in the frame e_N^{hol} is then

$$A_N^{\mathrm{hol}} = A_N - \frac{m}{2}\,d\log(1+|z|^2) = -m\frac{\bar z\,dz}{1+|z|^2}.$$

Exercise 27.4 carries out this frame change in coordinates. The form A_N^{hol} has only a dz term, and its curvature is

$$dA_N^{\mathrm{hol}} = m\frac{dz\wedge d\bar z}{(1+|z|^2)^2} = -\frac{i}{\hbar}\omega.$$

Thus the prequantum bundle can be taken to be $O(m)$, with its usual holomorphic structure.

For a given metric, the unitary connection compatible with the holomorphic structure is unique. Let e be a holomorphic frame and put $h = \|e\|^2$. Let A be the connection form in the frame e. The unitary frame $h^{-1/2}e$ has an imaginary-valued connection form A_u. The frame e is $h^{1/2}$ times this unitary frame, so the change-of-frame law gives $A = A_u + \frac{1}{2}d\log h$. Moreover $\bar A_u = -A_u$, and $\log h$ is real. Hence

$$A + \bar A = d\log h.$$

For a function f, write $\partial f = (\partial f/\partial z)\,dz$ and $\bar\partial f = (\partial f/\partial \bar z)\,d\bar z$, so that $df = \partial f + \bar\partial f$. Suppose that A has no $d\bar z$ term. Then $\bar A$ has no dz term. Comparing the dz terms on the two sides gives

$$A = \partial \log h.$$

Thus the metric determines the compatible unitary connection. For $h = (1+|z|^2)^{-m}$ this formula gives $-m\bar z\,dz/(1+|z|^2)$, which is A_N^{hol} again.

The quantisation

When $m = 2r/\hbar$ is a positive integer, the *holomorphic quantisation* of S^2_r is the section space

$$Q(S^2_r) = H^0(\mathbb{CP}^1, O(m)) \cong V_m.$$

By the section computation of Lecture 26, this space has dimension $m+1$ and spin $j = m/2 = r/\hbar$. An invariant positive inner product comes from compact averaging. Explicitly, the formula

$$\|s\|^2 = \frac{1}{4\pi r}\int_{S^2_r} \|s\|^2_{\mathrm{fibre}}\,\omega$$

is the norm of Exercise 26.4. Indeed, in the coordinate z, the section z^k of $O(m)$ has squared fibre norm $|z|^{2k}(1+|z|^2)^{-m}$, and with $u = |z|^2$ this integral is

$$\int_0^\infty u^k(1+u)^{-m-2}\,du = \frac{k!(m-k)!}{(m+1)!}.$$

The point orbit uses the trivial line and gives the trivial representation, without a radius denominator.

The equivariant lift

Definition 27.5. Let K be a Lie group with Lie algebra $\mathfrak{k}$, and let $M \subset \mathfrak{k}^*$ be a coadjoint orbit with its KKS form ω. An *equivariant prequantum bundle* for K is a prequantum line bundle for (M,ω) at scale $\hbar$ with a smooth K-action on its total space. The action must cover the coadjoint action on M, be linear and unitary on fibres, preserve the connection, and satisfy the *orbit normalisation*. This normalisation requires that, at each $\ell \in M$, the stabiliser K_ℓ act on the fibre over ℓ with differential

$$X \mapsto i\ell(X)/\hbar, \qquad X \in \mathrm{Lie}(K_\ell).$$

The orbit normalisation fixes the infinitesimal lift. Without it, merely asking for a bundle action would not specify its relation to the orbit functional.

At the north pole, the stabiliser in $\mathrm{SU}(2)$ is generated by $\exp(tX_3)$, of period 4π. In an equivariant prequantum bundle, this stabiliser acts on the fibre by the character

$$\exp(tX_3) \longmapsto e^{irt/\hbar} = e^{imt/2}.$$

This is a character exactly when $m \in \mathbb{Z}$. It is the action on the fibre of $O(m)$ from Lecture 26: $t_\theta = \exp(-2\theta X_3)$ acts there by $e^{-im\theta}$. The natural $\mathrm{SU}(2)$ action is linear on fibres and preserves the invariant metric. Each $g \in \mathrm{SU}(2)$ acts by a holomorphic bundle map. It sends

holomorphic frames to holomorphic frames. Its holomorphic coordinate change sends forms with only a dz term to forms of the same kind. Hence g transports the connection to a unitary connection compatible with the holomorphic structure. By uniqueness, the transported connection is the original one. Therefore the natural action supplies the required lift.

For SO(3) the same stabiliser rotation has period 2π. Its required character is single-valued only when $e^{2\pi i r/\hbar} = 1$, or m is even. Conversely, for even m the centre $-I$ acts trivially on every fibre of $O(m)$, so the action descends. Therefore

ordinary prequantum bundle	$m \in \mathbb{Z}$
SU(2)-equivariant prequantum bundle	$m \in \mathbb{Z}$
SO(3)-equivariant prequantum bundle	$m \in 2\mathbb{Z}$

for the positive-radius sphere, with $m > 0$. An ordinary integral sphere with odd m does exist. It does not quantise equivariantly for the global rotation group SO(3). Figure 27.2 shows the first integral spheres.

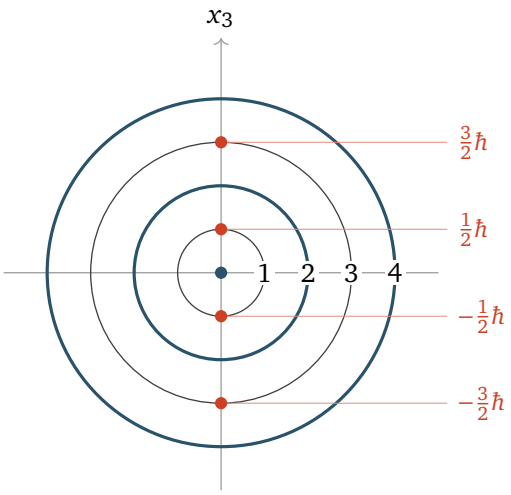


Figure 27.2: Cross-sections of the integral spheres, labelled by $m = 2r/\hbar$ with radius $r = \hbar m/2$, and the point orbit at the origin. Heavy circles have even m and admit an SO(3)-equivariant lift. The marks on the axis give the four heights $x_3 = \frac{\hbar}{2}k$, for $k = 3, 1, -1, -3$, associated with the weights of V_3, the quantisation of the sphere $m = 3$. Only the two extreme marks are points of that sphere on the axis.

Phase changes and the odd-degree obstruction

One might try to change the SU(2) lift by phases so that $-I$ acts trivially. We show that, for this sphere and its fixed connection, every such phase change is trivial.

Throughout this subsection, an action on $O(m)$ means a smooth action that is linear and unitary on fibres, as in the definition of an equivariant prequantum bundle. Suppose two such SU(2) actions cover the same sphere action and preserve the connection. For fixed g, compare their fibre maps. Their ratio is a bundle automorphism covering the identity. It is linear and unitary on each fibre, and each fibre is a complex line. Hence the ratio

is multiplication by a smooth function $a_g : S^2 \to \mathrm{U}(1)$. Since both actions preserve the connection,

$$a_g^{-1} da_g = 0.$$

The sphere is connected, so a_g is constant. Call this constant $\chi(g)$. The group-action law then implies

$$\chi(gh) = \chi(g)\chi(h).$$

Thus changing the lift while preserving the connection can only twist it by a continuous character of SU(2).

Every such character is trivial. Its differential is a real Lie algebra map from $\mathfrak{su}(2)$ to the abelian algebra $i\mathbb{R}$, so it vanishes on brackets. Since

$$[\mathfrak{su}(2), \mathfrak{su}(2)] = \mathfrak{su}(2),$$

the differential vanishes. Since the connected group SU(2) is generated by exponentials (Lecture 2), the character is identically one. Consequently a phase change of this kind cannot alter the action of the centre. Hence the natural action of Lecture 26 is the only such SU(2) action on $O(m)$ that covers the sphere action and preserves the connection.

Now suppose that SO(3) acted on $O(m)$ in this way. Composition with the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ would give such an SU(2) action on which $-I$ acts trivially. By uniqueness, that action would be the natural one, where $-I$ acts on every fibre by $(-1)^m$. Hence for odd m no SO(3) action on $O(m)$ that is linear and unitary on fibres covers the rotation action and preserves the connection. This conclusion does not use the orbit normalisation.

This is the sign of Lecture 13 on a fibre. The lift to SU(2) of the rotation through 2π about the north pole is $-I$, and it acts on the fibre by $(-1)^m$. For odd m, the calculation above shows that no change of phases removes this sign. So SO(3) acts only projectively on the sections of odd degree, and its cover SU(2) acts as a representation.

A spin-one comparison with functions on the sphere

Functions on the sphere contain every integer spin, by Lecture 12, and spin one is their linear part. Holomorphic sections of $O(2)$ contain spin one alone. We compare the two copies. For $r = \hbar$, the integer is $m = 2$. The quantisation is the three-dimensional space of homogeneous quadratic polynomials in (v_0, v_1)

$$F(v_0, v_1) = Av_0^2 + Bv_0v_1 + Cv_1^2.$$

There is another spin-one model in Lecture 12: the linear coordinate functions on the sphere. The two models are isomorphic as representations, although a bundle section and a scalar function are different geometric objects.

We now identify the two models explicitly. Write $F(v) = v^T Q v$ and set

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T(Q) = \epsilon^{-1} Q = \begin{pmatrix} -B/2 & -C \\ A & B/2 \end{pmatrix}.$$

This is an invertible linear map from symmetric matrices to traceless matrices. The polynomial action sends Q to $g^{-T} Q g^{-1}$. Since $g^T \epsilon g = \epsilon$ for $g \in \mathrm{SL}(2, \mathbb{C})$,

$$T(g^{-T} Q g^{-1}) = g T(Q) g^{-1}.$$

Thus T intertwines the quadratic section representation with the complexified adjoint representation.

For $Y \in \mathfrak{su}(2)_{\mathbb{C}}$, let ℓ_x be the functional with coordinates $x_j = \ell_x(X_j)$ and define the linear function $f_Y(x) = \ell_x(Y)$ on the sphere, extending ℓ_x complex-linearly. Under pullback by g^{-1},

$$f_Y(g^{-1} \cdot x) = \ell_x(\mathrm{Ad}(g)Y) = f_{\mathrm{Ad}(g)Y}(x).$$

Hence the map sending Y to f_Y is another intertwiner. The three basis elements X_j give the independent functions x_j, so this map is an isomorphism onto their span. Since $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, the map $Q \mapsto f_{T(Q)}$ identifies the two spin-one models explicitly. The normalised period is still $m = 2$, the dimension is three, and the action descends to $\mathrm{SO}(3)$.

27.5 The compact construction and the shift

This section compares the construction with the recipe of Lecture 19. There a subalgebra $\mathfrak{m}$ with $\ell([\mathfrak{m},\mathfrak{m}]) = 0$ gave a character χ_ℓ of the group of $\mathfrak{m}$, and we induced from it. In $\mathfrak{su}(2)$ this fails for a nonzero ℓ. A real polarisation would have dimension two, and $\mathfrak{su}(2)$ has no two-dimensional subalgebra, because the bracket of two independent vectors is perpendicular to their plane. So we pass to the complexified algebra and take the positive Borel subalgebra.

The sphere illustrates the general compact picture. Let K be compact, connected, simply connected and semisimple. Let λ be a dominant integral weight in the real convention $d\chi = i\lambda$. Recall the positive Borel subalgebra fixed in Lecture 26. Our sign conventions follow that choice. Extend λ by zero on the root directions, and take the orbit through $\ell = -\hbar\lambda$. By the orbit normalisation, the stabiliser of ℓ acts on the fibre over ℓ by a character with differential $-i\lambda$. Extended by zero on the root directions, ℓ vanishes on $[\mathfrak{b},\mathfrak{b}]$, which is the sum of the positive root spaces. The character of B with differential $i\ell/\hbar$ is therefore χ_λ^{-1}. So the rule $f(gm) = \chi_\ell(m)^{-1} f(g)$ of Lecture 19 becomes $f(gb) = \chi_\lambda(b) f(g)$, which is the rule for the sections of L_λ in Lecture 26.

For strictly dominant λ, the stabiliser is a maximal torus T. The orbit is K/T and the bundle is L_λ of Lecture 26. Choose its invariant complex structure from the positive Borel subalgebra. The Borel–Weil theorem of Lecture 26 then gives V_λ^*. The sign in $\ell = -\hbar\lambda$ and the dual cancel. Let w_0 be the longest element of the Weyl group, and put $\mu = -w_0\lambda$, the highest weight of V_λ^*. Since w_0 is represented in K, the orbit through $-\hbar\lambda$ contains $\hbar\mu$. So the irreducible representation of highest weight μ belongs to the orbit through $\hbar\mu$, and the dominant integral weights label integral coadjoint orbits. For $\mathrm{SU}(2)$, the weight m has $\lambda(X_3) = -m/2$, so $-\hbar\lambda$ is the north pole of the sphere of radius $\hbar m/2$. For a weight on a chamber wall, the stabiliser K_ℓ is larger. The orbit is then the partial flag manifold K/K_ℓ. The analogous line bundle and holomorphic-section result are cited from Kirillov, Chapter 5, Sections 2–3. We do not prove that general geometric construction here.

For a different global form of K, the stabiliser character must exist for the actual stabiliser group. This is the equivariant integrality test. It reproduces the character-lattice restriction of Lecture 8. An ordinary integral period condition alone does not encode the global form.

Kirillov's character formula, which we do not prove, integrates over the orbit through $\lambda+\rho$ (Kirillov, Chapter 5). On the sphere, $r^2 = j^2\hbar^2$, while the Casimir is $j(j+1)\hbar^2 = \big((j+\frac{1}{2})^2-\frac{1}{4}\big)\hbar^2$. So the shift appears in quantum formulas and not in the orbit.

No ρ shift was used in this construction. On the sphere, the orbit label is m, and the uncorrected section space has dimension $m+1$. The character nevertheless contains $m+1$:

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin((m+1)\theta)}{\sin\theta} \qquad (\sin\theta \neq 0).$$

This is the Weyl character formula, with its familiar $\lambda+\rho$ numerator. It does not say that the prequantum orbit above had label $m+1$. Half-form corrections are another construction and also require a convention. Optional Exercise 27.6 shows one such choice on the sphere, and nothing in this lecture depends on it.

Self-checks

1. If $r = 3\hbar/2$, find the normalised period and the quantised representation.

Answer. The period divided by $2\pi\hbar$ is $2r/\hbar = 3$. The bundle is $O(3)$ and its sections give V_3, of dimension four and spin $3/2$.

2. Does that sphere admit an SO(3)-equivariant prequantum lift with the orbit normalisation?

Answer. No. Its stabiliser character would be $e^{3it/2}$ on a rotation of period 2π. It takes the value -1 at $t = 2\pi$.

3. Why can the connection have local potentials although no global potential exists?

Answer. The difference of two local potentials is the logarithmic differential of a transition function. A nonzero curvature period prevents a single global primitive but is consistent with these local potentials.

Summary

The SU(2) orbit of radius $r > 0$ has symplectic area $4\pi r$. Prequantisation requires $m = 2r/\hbar$ to be integral. Holomorphic sections of $O(m)$ give spin $m/2$. Equivariance for SO(3) further requires even m. The ordinary period, the global equivariant lift and the holomorphic selection are separate steps. The orbit through $-\hbar\lambda$ and the dual V_λ^* of Borel–Weil combine to attach V_μ to the orbit through $\hbar\mu$. For SU(2) this attaches V_m to the sphere of radius $\hbar m/2$.

Reading

Kirillov, *Lectures on the Orbit Method*, Chapter 1, Section 2.4, for integrality, and Chapter 5, Sections 2–3, for compact orbits, holomorphic constructions and character formulas. Sepanski, *Compact Lie Groups*, Section 7.4, states the Borel–Weil theorem proved in Lecture 26.

Lecture 28

Reconstruction and the boundary of quantum groups

Objectives. *The group, its coefficient algebra and its tensor representations encode the same global information in different forms. Differentiation connects these descriptions to the enveloping algebra. It loses information, however, unless the allowed group representations are retained. We make that connection explicit and identify the structures that a later course on quantum groups will deform.*

By the end of this lecture you will be able to:

- Construct the pairing between $U(\mathfrak{k}_{\mathbb{C}})$ and representative functions, with the correct multiplication order.
- For connected K, explain why representative functions form a restricted part of an algebraic dual, and show that for simply connected K they fill the finite dual.
- Recover compact and complex groups from the appropriate tensor automorphisms.
- Use the circle and the $\mathrm{SU}(2)/\mathrm{SO}(3)$ pair to identify what global data survive differentiation.

28.1 The two algebras

Let K be a compact Lie group and $\mathfrak{k}$ its real Lie algebra. As in Lecture 14, $\mathcal{R}(K)$ is the span of the coefficients

$$c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)v), \qquad \lambda \in V^*,\ v \in V,$$

of finite-dimensional continuous complex representations. Every sum of coefficients is one coefficient of a direct sum. Its Hopf structure is

$$\Delta_{\mathcal{R}} f(g,h) = f(gh), \qquad \varepsilon_{\mathcal{R}} f = f(e), \qquad S_{\mathcal{R}} f(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

These formulas and their targets were proved in Lecture 14. We use the ordinary algebraic tensor product. Thus $\Delta_{\mathcal{R}} f$ is a finite sum of tensors, not an infinite expansion.

On the other side, put $U = U(\mathfrak{k}_{\mathbb{C}})$, the universal enveloping algebra from Lie Algebras. It is generated by $\mathfrak{k}_{\mathbb{C}}$ with $XY - YX = [X,Y]$. Its standard Hopf maps are determined by

$$\Delta_U X = X \otimes 1 + 1 \otimes X, \qquad \varepsilon_U X = 0, \qquad S_U X = -X. \tag{28.1}$$

In physics language, the first rule is the rule for adding spins. Here Δ_U and ε_U extend as algebra homomorphisms. The antipode S_U extends so that it reverses products. This

extension is well defined because reversing the defining relation sends $XY - YX - [X,Y]$ to $YX - XY + [X,Y]$, which is again zero in U. We check that Δ_U respects the defining relation:

$$[\Delta_U X, \Delta_U Y] = [X,Y] \otimes 1 + 1 \otimes [X,Y] = \Delta_U [X,Y].$$

Mixed tensor factors commute, which is why the cross terms vanish. Coassociativity and the counit identities follow by checking the generators, since the maps are algebra homomorphisms.

The antipode identities hold on X because $-X + X = 0$. They extend to products. Suppose that they hold on u and v. Then

$$\begin{aligned}\sum S_U(u_{(1)}v_{(1)})u_{(2)}v_{(2)} &= \sum S_U(v_{(1)})S_U(u_{(1)})u_{(2)}v_{(2)} \\ &= \varepsilon_U(u) \sum S_U(v_{(1)})v_{(2)} = \varepsilon_U(u)\varepsilon_U(v)1.\end{aligned}$$

The first equality uses that S_U reverses products. The second uses the identity for u, and the third uses the identity for v. The other antipode identity is proved the same way.

The two algebras have opposite kinds of commutativity. The coefficient algebra is commutative. It is cocommutative exactly when K is abelian, because coefficients separate points. The enveloping algebra is cocommutative, as the generator formulas (28.1) show, but is generally noncommutative.

The compact real form also defines an involution on U by $X^* = -X$ for real $X \in \mathfrak{k}$, extended antilinearly with $(uv)^* = v^* u^*$. For real X, Y, applying $*$ to the defining relation gives $YX - XY + [X,Y]$, which is again zero. Thus $*$ is well defined. Applying it twice fixes the generators, so it is an involution.

In every finite-dimensional unitary representation,

$$d\Pi(u^*) = d\Pi(u)^*.$$

This follows first for real generators, whose operators are skew-adjoint, then for sums and products. The involution remembers the compact real form inside its complexification.

28.2 Differentiating coefficients

For $X \in \mathfrak{k}$, let

$$(R_X f)(g) = \left.\frac{d}{dt}\right|_0 f(g \exp(tX)).$$

This is the left-invariant vector field X^L of Lecture 1, acting on functions. The letter R records multiplication on the right in the argument. It is distinct from the right regular group action, although differentiating that action gives this operator. By Lecture 1, $[R_X, R_Y] = R_{[X,Y]}$. Hence, by the universal property of U, these operators extend to R_u for $u \in U$.

Definition 28.1. The *coefficient pairing* is

$$\langle u, f \rangle = (R_u f)(e), \qquad u \in U,\ f \in \mathcal{R}(K).$$

The coefficient pairing is complex bilinear. Its angle brackets denote a pairing, not a Hermitian inner product. For real X, the value $\langle X^k, f \rangle$ is the kth derivative at 0 of $t \mapsto$

$f(\exp tX)$. These pairings record the Taylor derivatives of f at e. For a general u, the pairing is a finite linear combination of such derivatives. For a coefficient, direct differentiation gives

$$R_X c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)d\Pi_V(X)v), \qquad \langle u, c^V_{\lambda,v}\rangle = \lambda(d\Pi_V(u)v). \tag{28.2}$$

The second identity follows for products of generators and then by linearity. Since the pairing was defined on functions by differential operators, it does not depend on the chosen coefficient expression.

The order matters. For two real generators $X, Y \in \mathfrak{k}$,

$$\langle XY, f\rangle = \partial_t\partial_s|_0 f(\exp(tX)\exp(sY)).$$

Indeed, for $f = c^V_{\lambda,v}$ we have $R_X R_Y f(g) = \lambda(\Pi_V(g)d\Pi_V(X)d\Pi_V(Y)v)$. Interchanging X and Y would reverse the matrix product. Subtracting the pairing of YX with f from the pairing of XY with f gives $\langle [X,Y], f\rangle$.

Proposition 28.2 (Hopf pairing identities). *Write finite coproducts as* $\Delta_U u = \sum u_{(1)} \otimes u_{(2)}$ *and* $\Delta_{\mathcal{R}} f = \sum f_{(1)} \otimes f_{(2)}$. *Then*

$$\begin{aligned}
\langle ab, f\rangle &= \sum \langle a, f_{(1)}\rangle\langle b, f_{(2)}\rangle,\\
\langle u, fh\rangle &= \sum \langle u_{(1)}, f\rangle\langle u_{(2)}, h\rangle,\\
\langle 1, f\rangle &= \varepsilon_{\mathcal{R}} f, \qquad \langle u, 1\rangle = \varepsilon_U u,\\
\langle S_U u, f\rangle &= \langle u, S_{\mathcal{R}} f\rangle.
\end{aligned}$$

Proof. Choose a basis (e_j) of V with dual (e^j). For $f = c^V_{\lambda,v}$,

$$\Delta_{\mathcal{R}} f = \sum_j c^V_{\lambda,e_j} \otimes c^V_{e^j,v}.$$

Applying the right side of the first identity gives

$$\sum_j \lambda(d\Pi(a)e_j)\, e^j(d\Pi(b)v) = \lambda(d\Pi(a)d\Pi(b)v).$$

This is $\langle ab, f\rangle$ by the coefficient formula (28.2).

For the second identity, $R_X(fh) = (R_X f)h + f(R_X h)$ is the ordinary Leibniz rule. Iterating it gives exactly the coproduct (28.1). For instance,

$$\Delta_U(XY) = XY \otimes 1 + X \otimes Y + Y \otimes X + 1 \otimes XY.$$

Thus evaluation at e proves the claim for every product and hence for every u. The identity $\langle 1, f\rangle = \varepsilon_{\mathcal{R}} f$ follows because R_1 is the identity. Applying derivatives to constants gives $\langle u, 1\rangle = \varepsilon_U u$.

Finally, write $\Pi^\vee$ for the dual representation on V^*, given by $\Pi^\vee(g)\mu = \mu \circ \Pi_V(g)^{-1}$. The function $S_{\mathcal{R}} c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)^{-1}v)$ is the coefficient of $\Pi^\vee$ with vector λ and with the functional given by evaluation at v. On V^*, a real generator acts by $d\Pi^\vee(X)\mu = -\mu \circ d\Pi(X)$. For every $u \in U$,

$$d\Pi^\vee(u)\mu = \mu \circ d\Pi(S_U u).$$

This holds for generators and extends multiplicatively because S_U reverses products. Apply the coefficient formula (28.2) to this coefficient of $\Pi^\vee$, and then to $c^V_{\lambda,v}$. Both sides of the last identity equal $\lambda(d\Pi(S_U u)v)$, which proves it. ∎

Hence the pairing makes the product of each algebra dual to the coproduct of the other. In this algebraic form, it relates the infinitesimal action to group multiplication. The proposition does not assert that one of these algebras is the entire linear dual of the other.

A noncommutative calculation

Take $K = \mathrm{SU}(2)$, so $\mathfrak{k}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. We test the order identity in its defining representation. Put

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are complexified generators. The derivative is extended complex-linearly; we are not claiming that $\exp(tE)$ belongs to $\mathrm{SU}(2)$ for real t. Let $D_{ij}(g) = g_{ij}$ be coefficients of its defining representation. Then

$$EF = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad FE = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows that $\langle EF, D_{11}\rangle = 1$, whereas $\langle FE, D_{11}\rangle = 0$. Their difference is $\langle H, D_{11}\rangle = 1$. The pairing therefore distinguishes EF from FE in U, even though its other argument belongs to a commutative function algebra.

The coefficient coproduct gives the same answer:

$$\Delta_{\mathcal{R}} D_{11} = D_{11} \otimes D_{11} + D_{12} \otimes D_{21}.$$

Pairing the first tensor factor with E and the second with F yields

$$E_{11}F_{11} + E_{12}F_{21} = 0 + 1 = 1.$$

Pairing the first factor with F and the second with E yields zero. Thus a commutative algebra can have a coproduct that encodes a noncommutative group law. Commutativity of the product does not by itself decide cocommutativity of the coproduct.

The regular actions on coefficients

Let $L_g f(x) = f(g^{-1}x)$ and $R_g f(x) = f(xg)$, as in the compact harmonic-analysis lectures. Here R_g denotes the group action. The symbols R_X and R_u denote differential operators. The pairings of the translates of a coefficient are

$$\langle u, L_g c_{\lambda,v}\rangle = \lambda\big(\Pi(g^{-1})d\Pi(u)v\big),$$

$$\langle u, R_g c_{\lambda,v}\rangle = \lambda\big(d\Pi(u)\Pi(g)v\big).$$

For the first formula, $L_g c$ is the coefficient with functional $\lambda \circ \Pi(g^{-1})$. For the second, $R_g c$ has vector $\Pi(g)v$. Inserting these into the coefficient formula (28.2) proves both identities without any operator-domain question.

Notice the positions of the group matrix and the derivative matrix. Left translation changes the dual variable. Right translation changes the vector variable. This is the same tensor order $V^* \otimes V$ that appeared in the Peter–Weyl theorem. The enveloping-algebra pairing continues that convention rather than introducing a second one.

28.3 What the dual contains

For any unital algebra A, its *finite dual* A° consists of the functionals that vanish on a two-sided ideal of finite codimension. Equivalently they are matrix coefficients of finite-dimensional A-modules. Here is the elementary equivalence.

A finite-dimensional module gives a homomorphism $A \to \mathrm{End}(V)$ with finite-codimensional kernel; every coefficient vanishes on that kernel. Conversely, if φ vanishes on such an ideal I, use the finite-dimensional regular module A/I, vector $1 + I$ and functional $\bar{\varphi}$. Then $\varphi(a) = \bar{\varphi}(a(1 + I))$.

By the coefficient formula (28.2), the pairing therefore maps $\mathcal{R}(K)$ into U°. Only the modules that integrate to the chosen compact group occur in this image. A finite-dimensional representation of $\mathfrak{k}_{\mathbb{C}}$ need not respect the periods of K or the restrictions imposed by a central quotient.

> **Proposition 28.3** (Injectivity for connected groups). *If K is connected, the map $\mathcal{R}(K) \to U^\circ$ from the pairing is injective.*

Proof. Suppose $\langle u, f\rangle = 0$ for every u. Since every sum of coefficients is one coefficient of a direct sum, we may write $f = c^V_{\lambda,v}$. For any finite list $X_1, \dots, X_s \in \mathfrak{k}$, expand the finite-dimensional matrix exponentials in

$$f(\exp(t_1X_1)\cdots\exp(t_sX_s)) = \lambda\big(e^{t_1 d\Pi(X_1)}\cdots e^{t_s d\Pi(X_s)}v\big).$$

These power series converge absolutely for every real or complex tuple $(t_1, \dots, t_s)$. The Taylor coefficient of $t_1^{a_1}\cdots t_s^{a_s}$ is

$$\frac{\langle X_1^{a_1}\cdots X_s^{a_s}, f\rangle}{a_1!\cdots a_s!} = 0.$$

Thus the displayed function vanishes for all real tuples. A connected Lie group is generated by its exponentials (Lecture 2), so $f = 0$ on K. ∎

For disconnected K, the same argument detects only the identity component. A coefficient may vanish on that component and be nonzero elsewhere. Thus connectedness cannot simply be removed.

Example 28.4 (The circle). Let $K = \mathrm{U}(1)$ and choose X with $\exp(tX) = e^{it}$. Then

$$U = \mathbb{C}[X], \qquad \mathcal{R}(K) = \mathbb{C}[z, z^{-1}], \qquad \langle X^k, z^n\rangle = (in)^k.$$

The image consists of finite linear combinations of the evaluation functionals $p(X) \mapsto p(in)$, $n \in \mathbb{Z}$. The finite dual is larger. Evaluation at $X = 1$ belongs to U° but is not in this image. If it were a combination supported on $n_1, \dots, n_s$, the polynomial $p(X) = \prod_j (X - in_j)$ would have zero value under that combination, but $p(1) \neq 0$.

The full linear dual is larger still. It contains the functional L given by $L(X^k) = k!$. By the equivalence above, every element of U° is a matrix coefficient of a finite-dimensional module. Suppose that X acts on that module by the operator A and that the coefficient uses the vector v and the functional λ. Then the coefficient takes the value $\lambda(A^k v)$ on X^k. Choose a norm on the module, with the corresponding operator and dual norms. This value is

then bounded in absolute value by $C\|A\|^k$, where $C = \|\lambda\|\|v\|$. The sequence $k!$ has no such exponential bound. Thus L is not even in U°.

Figure 28.1 draws these functionals on $U = \mathbb{C}[X]$ in the plane of the variable X. A dot at a point X_0 means the evaluation $p(X) \mapsto p(X_0)$.

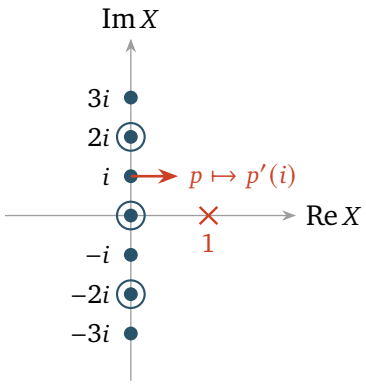


Figure 28.1: Functionals on $\mathbb{C}[X]$. The filled dots at $X_0 = in$ with $n \in \mathbb{Z}$ span the image of $\mathcal{R}(\mathrm{U}(1))$. The ringed dots at $X_0 = 2in$ span the image for the quotient of the circle by μ_2, treated later in this lecture. The cross at $X_0 = 1$ is in U° but not in the image. The arrow at i marks the functional $p \mapsto p'(i)$, which comes from the Jordan block of Exercise 28.3. It lies in U°, and it is a different functional from the evaluation at i.

28.4 Global forms and tensor reconstruction

The groups SU(2) and SO(3) have isomorphic real Lie algebras, hence the same complex enveloping algebra after this identification. Their coefficient algebras retain different representations. Pullback through $p : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives

$$p^*\mathcal{R}(\mathrm{SO}(3)) = \{f \in \mathcal{R}(\mathrm{SU}(2)) : f(-g) = f(g)\}. \tag{28.3}$$

Indeed, a representation descends exactly when $-I$ acts trivially. On V_m it acts by $(-1)^m$. The algebraic Peter–Weyl decomposition therefore identifies the right side with precisely the coefficient blocks for even m. Odd blocks have been excluded, not annihilated by differentiation. Both coefficient algebras inject into the same finite dual, with different images.

For SU(2) the image fills the finite dual. More generally, let K be compact, connected and simply connected. A finite-dimensional module of U restricts to a Lie algebra map from $\mathfrak{k}$ to $\mathfrak{gl}(V)$. By the integration theorem of Lecture 4, this map is the differential of a representation of K. So every element of U° is a coefficient of a representation of K. With Proposition 28.3, the pairing identifies $\mathcal{R}(K)$ with U°. Thus U° gives $K_{\mathbb{C}}$, whose points are the characters of $\mathcal{R}(K)$ by Lecture 14, and the involution of the first section selects K. The involution is needed: $U(\mathfrak{sl}(2,\mathbb{C}))$ is also the enveloping algebra of the complexification of $\mathfrak{su}(1,1)$. Tori and central quotients are where the coefficient algebra adds information. The circle example shows a torus, and (28.3) shows a quotient.

To recover the group, keep the category of finite-dimensional continuous complex representations of K, all its intertwiners, and its forgetful functor ω. This functor sends each representation to its underlying vector space and each intertwiner to its linear map.

A *tensor automorphism* of ω is a family of maps $\theta_V \in \mathrm{GL}(V)$, one for each representation V, such that

$$\theta_W T = T\theta_V \quad \text{for every intertwiner } T\colon V \to W, \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

Here $\mathbb{C}$ is the trivial representation. A tensor automorphism is *unitary* if every θ_V is unitary for every invariant inner product on V. Write $\mathrm{Aut}_{\otimes}(\omega)$ for the group of tensor automorphisms and $\mathrm{Aut}^{u}_{\otimes}(\omega)$ for its subgroup of unitary tensor automorphisms.

Each $g \in K$ supplies the unitary tensor automorphism $\theta_V = \Pi_V(g)$. A tensor automorphism θ corresponds to the algebra homomorphism $\mathcal{R}(K) \to \mathbb{C}$ given by $c^{V}_{\lambda,\upsilon} \mapsto \lambda(\theta_V \upsilon)$. Unitary tensor automorphisms correspond to $*$-homomorphisms. Exercises 14.4–14.5 prove this correspondence.

Let $K_{\mathbb{C}}$ be the complexification of K introduced in Lecture 14. It is a complex Lie group, and its identity component is reductive. The reconstruction theorem is cited from Bröcker–tom Dieck, Sections III.7–8. It identifies

$$K \cong \mathrm{Aut}^{u}_{\otimes}(\omega), \qquad K_{\mathbb{C}} \cong \mathrm{Aut}_{\otimes}(\omega).$$

The compact group has the topology of pointwise convergence on the representation spaces. In words, every unitary tensor automorphism is $\Pi_V(g)$ for a unique $g \in K$. Every tensor automorphism is given by a unique element of $K_{\mathbb{C}}$, acting through the holomorphic extensions of the representations Π_V to $K_{\mathbb{C}}$. The unitary condition distinguishes the compact target from its complexification.

For the circle, write $\mathbb{C}_n$ for the one-dimensional representation $z \mapsto z^n$ of $\mathrm{U}(1)$, where $n \in \mathbb{Z}$. Every finite-dimensional representation of $\mathrm{U}(1)$ is a direct sum of copies of these representations. Naturality makes θ act on every copy of $\mathbb{C}_n$ by one scalar $a_n \in \mathbb{C}^{\times}$. The tensor rule, the isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \cong \mathbb{C}_{m+n}$ and the unit rule give $a_{m+n} = a_m a_n$ and $a_0 = 1$. Therefore $a_n = a^n$ with $a = a_1 \in \mathbb{C}^{\times}$. Every such choice of a defines a tensor automorphism. This tensor automorphism is unitary exactly when $|a| = 1$. This recovers $\mathbb{C}^{\times}$ and $\mathrm{U}(1)$ respectively.

For $\mathrm{SU}(2)$, let $V = \mathbb{C}^2$ be the defining representation. The alternating tensor $e_1 \otimes e_2 - e_2 \otimes e_1$ is invariant. Since $\theta_{V\otimes V} = \theta_V \otimes \theta_V$ and $\theta_{\mathbb{C}} = 1$, naturality with the intertwiner from $\mathbb{C}$ given by this tensor forces $\theta_V \otimes \theta_V$ to preserve it. Because $\theta_V \otimes \theta_V$ multiplies this tensor by $\det \theta_V$, we get $\det \theta_V = 1$. Every irreducible representation is a symmetric power of V, and direct sums and intertwiners determine the rest of the tensor automorphism.

Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ acts on these tensor constructions and gives a tensor automorphism, using the complexification from Lecture 14. Hence the complex target is $\mathrm{SL}(2,\mathbb{C})$. Imposing unitarity on V gives $\mathrm{SU}(2)$. For the even-weight category the central element $-I$ acts as the identity tensor automorphism, so reconstruction instead yields the quotient global group.

A circle quotient and its representation lattice

There is a smaller example of global information in a coefficient subalgebra. Let $\mu_N \subset \mathrm{U}(1)$ be the Nth roots of unity. The quotient is again a circle, but keep the quotient map and the identification of Lie algebras fixed. A character z^n descends through this map exactly when

$$\zeta^n = 1 \quad \text{for every } \zeta \in \mu_N, \qquad \text{or equivalently } N \mid n.$$

Thus the pulled-back coefficient algebra is

$$\mathbb{C}[z^N, z^{-N}] \subset \mathbb{C}[z, z^{-1}].$$

Under the same infinitesimal pairing, its evaluation points are $iN\mathbb{Z}$, instead of $i\mathbb{Z}$. The real Lie algebras of these two circles are isomorphic. The quotient map, however, has changed which periods a representation must respect. Rescaling an abstract Lie algebra isomorphism does not erase the specified quotient map.

The subalgebra remains closed under the Hopf operations. For its generator $w = z^N$,

$$\Delta w = w \otimes w, \qquad \varepsilon w = 1, \qquad Sw = w^{-1}, \qquad w^* = w^{-1}.$$

Reconstruction from this algebra and its unitary structure recovers the quotient circle together with its representation lattice. The same calculation with even highest weights gives the SU(2) to SO(3) distinction. In both cases, keeping a Lie algebra while changing the allowed coefficient subalgebra changes the global representation problem.

28.5 What a quantum-group course changes

The structures above give a concrete starting point for deformation. The coefficient algebra encodes the multiplication of group elements in its coproduct. The enveloping algebra encodes the infinitesimal bracket in its associative product. The pairing between them relates these two operations. A deformation can change these operations while retaining the Hopf identities.

Classical object	Product	Coproduct
$\mathcal{R}(K)$	commutative	usually not cocommutative
$U(\mathfrak{k}_{\mathbb{C}})$	usually not commutative	cocommutative

A noncommutative coefficient algebra is no longer an algebra of scalar-valued functions on an ordinary group. A noncocommutative coproduct changes how tensor products of representations carry an action. The order of tensor factors can then matter. Associativity of the tensor product of representations still follows from coassociativity.

Here is one such deformation, as an algebraic calculation only.

Example 28.5 (Quantum SL_2)**.** On SU(2), the coefficients of the defining representation are $a = D_{11}$, $b = D_{12}$, $c = D_{21}$ and $d = D_{22}$. Keep the coproduct $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$. Change the product. For a nonzero complex number q, let a, b, c, d generate an associative algebra with the relations

$$ab = qba, \quad ac = qca, \quad bd = qdb, \quad cd = qdc, \quad bc = cb,$$
$$ad - da = (q - q^{-1})bc, \quad ad - qbc = 1.$$

At $q = 1$ these relations say that the generators commute and that the determinant is one. For other q this algebra is not commutative, so it is not an algebra of functions on a group. The coproduct formula still respects the relations. For example, using $ab = qba$, $cd = qdc$ and $ba = q^{-1}ab$,

$$\Delta(a)\Delta(b) - q\Delta(b)\Delta(a) = ab \otimes \big(ad - da - (q - q^{-1})bc\big) = 0.$$

The other relations follow in the same way, and $ad - qbc$ is sent to $(ad - qbc) \otimes (ad - qbc)$. These relations do not give a compact quantum group for every q. The involution of the classical case does not carry over unchanged.

A braiding is additional structure, not an automatic property of a Hopf algebra. In suitable quasitriangular examples on the enveloping side, an R-matrix supplies an intertwiner between the two tensor orders. The future course will define these notions and prove their compatibility identities. They are not needed for any theorem here.

The $*$-structure also remains substantive. An arbitrary Hopf $*$-algebra is not automatically a compact quantum group; positivity and analytic completion require further hypotheses.

The useful preparation is therefore the classical bookkeeping developed in this course: distinguish an algebraic coefficient space from a completion, track coproduct order, keep the real or unitary structure, and retain the global representation lattice. These are the data a deformation must replace or preserve. The theory of deformation parameters, $U_q(\mathfrak{g})$ and roots of unity begins in the later course.

Looking back

We asked what a group determines that its Lie algebra does not. One answer runs through the course. A connected Lie group is a quotient of its simply connected covering group by a discrete central subgroup. In an irreducible representation of the cover that is finite-dimensional or unitary, a central element acts by a scalar. So a global form chooses which central characters are allowed. We used this for SU(2) and SO(3) in Lecture 4 and for lattices in Lecture 7 and Lecture 8. It separates integer and half-integer spin in Lecture 13, in the Lorentz modules of Lecture 24, in the massive states of Lecture 25, and on the sphere of Lecture 27. The coefficient algebras of this lecture record the same choice.

Part I treated compact groups. For a compact connected group, the irreducible representations are labelled by dominant characters. For every compact Lie group, the invariant integral leads to a decomposition of $L^2(K)$ as a Hilbert sum of finite blocks.

Part II removed compactness. Generators, the modular function and induction joined the invariant integral as the working tools. We then described the unitary representations

of the Heisenberg group, of $\mathrm{SL}(2,\mathbb{R})$ and of the Poincaré group. Each description has a cited theorem at its centre. For the Heisenberg group, we proved that the Schrödinger representation is irreducible, and the Stone–von Neumann theorem says that it is the only one for each nonzero central character. For $\mathrm{SL}(2,\mathbb{R})$ we constructed the principal and discrete series and derived the positivity bounds, and the full list is cited from Bargmann and Harish-Chandra. For the Poincaré group, Mackey's theorem reduces the problem to little groups. We computed the massive and finite-helicity sectors, and we only named the others.

Orbits organise much of this. In Lecture 19 a coadjoint orbit of the Heisenberg group is a plane or a point, and it gives one representation. In Lecture 27 the nonzero coadjoint orbits of $\mathrm{SU}(2)$ are spheres. For $m \geq 1$, the sphere of area $2\pi\hbar m$ gives V_m, on holomorphic sections of a line bundle over the sphere. The point orbit gives V_0. In Lecture 17 and Lecture 25 the orbits lie instead in the dual of the translations.

We did not decompose $L^2(G)$ for noncompact G, apart from the statement cited in Lecture 23, and we did not treat general semisimple groups. These are the next subjects in the representation theory of Lie groups. A second direction deforms the Hopf structures of this lecture, and a later course on quantum groups begins there.

Self-checks

1. For the circle, compute $\langle X^2, z^{-3}\rangle$.

Answer. It is $(-3i)^2 = -9$. The pairing is bilinear, so there is no complex conjugation in this calculation.

2. Does $U(\mathfrak{sl}(2,\mathbb{C}))$ alone decide whether V_1 (spin $1/2$) is allowed?

Answer. No. It is allowed for $\mathrm{SU}(2)$ and excluded for $\mathrm{SO}(3)$. The chosen coefficient subalgebra, or equivalently the allowed global representations, records that distinction.

3. For the circle tensor automorphism $a = 2$, what are $\theta_{\mathbb{C}_2}$ and $\theta_{\mathbb{C}_{-1}}$? Is it unitary?

Answer. They are 4 and $1/2$. It is a complex tensor automorphism, but it is not unitary and does not represent an element of the compact circle.

Summary

Differentiation pairs representative functions with the enveloping algebra. For connected K, the pairing embeds them in the finite dual. There the global group selects a specific subalgebra, which is the whole finite dual when K is simply connected. Tensor reconstruction recovers K when unitary structure is kept and $K_{\mathbb{C}}$ when it is forgotten. The coefficient and enveloping Hopf structures provide the classical starting point for quantum groups.

Reading

Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Sections III.1 and III.7–8: coefficient algebras, reconstruction and complexification. The enveloping-algebra and tensor-product inputs are from the prerequisite Lie Algebras course. Lecture 14 contains the coefficient Hopf identities used here. The quantum-group paragraph is an outlook, not an additional classification theorem. For quantum SL_2, see Klimyk and Schmüdgen, *Quantum Groups and Their Representations* (1997), Section 4.1, whose conventions we use.